



Norwegian Ministry  
of Climate and Environment



*Empowered lives.  
Resilient nations.*

# CLIMATE SOLUTIONS FROM COMMUNITY FORESTS

Learning from Indigenous Peoples and Local Communities



EQUATOR  INITIATIVE



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*Back Cover:* Serranía de los Paraguas Mountains, Valle del Cauca Department, Colombia.  
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# PREFACE



Conserving the world's forests represents the most cost-effective climate solution available today. As well, actions to protect, restore, and manage forests sustainably will contribute to inclusive growth, poverty eradication, fighting corruption, achieving food security, and maintaining biodiversity, among many other environmental and development benefits.

Efforts by the international community to address deforestation have progressed significantly in recent years. But it is more important than ever to acknowledge the role of indigenous peoples and local communities on the forefront of forest protection. Their rights need to be recognized and respected.

The Sustainable Development Goals and the Paris Climate Agreement offer new global frameworks for decisive action on forests. Both will need to be fortified by commitments to secure and protect local rights to communal lands, territories, and natural resources. They offer the international community an unprecedented opportunity to build on the New York Declaration on Forests, launched in September 2014, which saw more than 180 governments, companies, indigenous peoples groups, and civil society organizations commit to halving deforestation by 2020 and ending natural forest loss by 2030.

This book is a UNDP contribution to the literature on indigenous peoples and land rights, forest protection, climate change mitigation and adaptation, and inclusive development. It profiles more than twenty recipients of the Equator Prize – a flagship partnership for UNDP. It honours innovative leadership at the grassroots level – and draws lessons from those leaders for community-based forest management. The aim of the book is to provide practitioners and policymakers with insights into what is and what is not working on the ground, and to have these experiences inform how governments, companies, industry, and other stakeholders engage, empower, and partner with indigenous peoples and local communities on forest protection and management.

To put the new climate and sustainable development frameworks into action, we must find partnerships which can support governments to deliver on the commitments they have made. Indigenous peoples and local communities are indispensable partners. Investing in their rights, entitlements, and empowerment will help to ensure a more sustainable and equitable future.

On behalf of UNDP, I express our deepest appreciation to the Government of Norway for its continued leadership in support for indigenous peoples and local communities, particularly in the context of forests, and for their ongoing support to the Equator Prize.

**Helen Clark**  
Administrator  
United Nations Development Programme

# FOREWORD



Forests sustain over one billion livelihoods, provide drinking water to one third of the world's largest cities and are a critical safety net for some of the world's poorest and most vulnerable communities. Halting tropical deforestation – and expanding the protection and sustainable management of forests globally – will be an essential part of meeting the Sustainable Development Goals and the targets set out in the Paris Climate Agreement.

Norway continues to support action on forests as one of the most economical and effective ways to address climate change. Since 2008 the Norwegian International Climate and Forest Initiative (NICFI) has invested 17 billion Norwegian kroner to reduce greenhouse gas emissions from deforestation and forest degradation.

At the same time, Norway recognizes that indigenous peoples and local communities are among the best managers and defenders of the world's tropical forests. Strengthening and expanding community rights to forests is one of the most viable, cost-effective and socially responsible ways of meeting global climate and development goals. Norway is committed to supporting indigenous peoples and forest communities in their efforts to protect forests in a way that promotes sustainable growth.

For over a decade, Norway has been a proud supporter of the work that UNDP and the Equator Initiative are doing to identify and profile outstanding community-based initiatives that show the power and relevance of local action to meeting global challenges. The Equator Prize recipients whose initiatives appear in this publication show pioneering work on the protection, restoration and sustainable management of forests. They illustrate how community-based action can deliver food and water security, generate income, protect biodiversity - all while providing global climate benefits.

Protecting the world's forests is a great responsibility that must be shared among a diverse range of stakeholders. The international community, national governments and the private sector have major roles to play through Reducing Emissions from Deforestation and Forest Degradation (REDD+) and other initiatives. Indigenous peoples and local communities are a critical piece of the puzzle and are our indispensable partners in the push towards reaching the new global goals on climate and sustainable development.

A handwritten signature in dark ink, reading 'Vidar Helgesen'.

**Vidar Helgesen**

Minister of Climate and the Environment  
Government of Norway

# EXECUTIVE SUMMARY

## A ROLE FOR COMMUNITIES IN CLIMATE ACTION

Indigenous peoples and local communities have a vital part to play in the global effort to reduce greenhouse gas emissions and address climate change through their role as custodians of many of the world's remaining standing forests. Community-based forest management—an approach that encompasses a wide range of forest protection, restoration, and sustainable use activities—can effectively reduce rates of deforestation, restore degraded forests, and allow local people to better adapt to climate change. It can achieve these global climate benefits even as it supports local livelihoods and indigenous cultures, conserves biodiversity, and increases ecosystem resilience and stability.

This book brings together case material from 22 examples of successful community-based forest management. These stories showcase the wide range of initiatives that indigenous peoples and other forest-dependent communities take to sustainably use, protect, and restore local forests through their collective action. Such community-based action has a well-documented record of delivering an array of local benefits. These include job creation and economic gains, greater social cohesion and community self-reliance, political and social empowerment, food security and water access, conflict resolution, wildlife protection, and ecosystem restoration.

While these local development benefits are amply demonstrated in the cases presented here, the case materials also demonstrate the global climate benefits that flow from these local actions. In short, community-managed forests offer an effective method to maintain and increase forest canopy, suppress illegal forest activities that degrade forests, and—through education and first-hand experience—build a lasting commitment among community members to sustainable forest use. All of these actions maintain and enhance the ability of the world's forests to sequester CO<sub>2</sub> from the atmosphere and to moderate the effects of this build-up on ecosystems and people.

Many of these local initiatives successfully scale up to become influential voices in the governance of entire ecosystems, forests, wildlife corridors, and agricultural landscapes, refuting the common but mistaken view that local solutions invariably remain small in scope and impact. They are often the foundation on which national progress towards climate and development goals are built.

## LESSONS FROM INDIGENOUS AND COMMUNITY-BASED FOREST MANAGEMENT

The community-based forest management models profiled in these case studies—and many more taking place in rural communities around the world—represent an underutilized tool to meet global climate goals while serving rural development needs. As the cases show, local indigenous peoples groups, women's groups, and other community-based organizations, when empowered to manage their ecosystems and natural resources, can become innovators in local development and climate action.

As the international community begins to implement the climate agreement adopted in December 2015 in Paris, it is an opportune time to reflect on the conditions that make community-based forest management successful, how such initiatives can be empowered and enabled, what climate and human development benefits are produced, and what role community-managed forests can play in a global framework to address climate change.

It is also crucial to ask how these local initiatives align with and support the aspirations of indigenous peoples and local communities to secure their land and resource rights and to pursue their own development goals. Aligning the global climate agenda with the movement to empower indigenous peoples and local communities will invariably strengthen both efforts.



## THEME 1: SUSTAINABLE FOREST LIVELIHOODS

Case studies in this section profile community-based forest management initiatives that have evolved to maximize and maintain the local forest as a substantive community asset, contributing to local livelihoods through subsistence and cash income, and generating revenue streams that fund community institutions, public service delivery and local infrastructure. This includes cases where communities have established substantial commercial forest enterprises, as well as cases where forest management has yielded more broad-based community uses and public benefits. The following lessons emerge from cases in this section:

- **Climate Potential:** Community-based forest management can be effective over substantial areas and in situations where forest pressures are intense.
- **Community Motivations:** The desire for new livelihood options that offer sustained economic benefits is an important factor motivating community forest management. But self-determination and social and political empowerment are just as important to many indigenous forest communities.
- **Forest Ownership:** Secure, enforceable community land rights increase the chances for successful community-based forest management.
- **Enterprise Development:** Community-based forest enterprises can be highly successful, but require considerable attention to developing business capacities and tackling market, transport, and regulatory obstacles, often with the help of commercial partners.
- **Local Knowledge and Innovation:** Local knowledge is a key resource that makes community-based forest initiatives relevant and adaptive, while local innovation bridges between traditional practices and modern forest management.
- **Scalability and Sustainability:** Successful community-based forest initiatives have shown high potential to scale up through a combination of local advocacy, effective outreach, and the help of intermediary support organizations. Inclusive governance, consistent benefits, and continuous public education promote sustainability.

## THEME 2: FOREST-FRIENDLY AGRICULTURE

Demand for agricultural land is a leading cause of deforestation, while poor farming practices can be a significant source of forest degradation. Forest-friendly farming practices, on the other hand, can maintain and enhance forest cover, food security, and local livelihoods. Cases in this section display the many ways in which community-managed forests have incorporated agriculture within their land management plans. The following lessons emerge from cases in this section:

- **Food Security and Forests:** Smallholder agriculture is a bulwark of local food security, but also a primary source of pressure on local forests. Successful community-based forest initiatives enable local people to overcome this essential forest-farm tension.
- **Forest-Friendly Options:** Agricultural intensification, agroforestry, and adoption of alternative crops and markets can increase the productivity of local agriculture and integrate farming into the forest landscape, easing the pressure for forest conversion and overuse.
- **Education and Training:** Locally adapted training programs, demonstration sites, and community education efforts are essential to build the capacity and willingness to change local agricultural practice.
- **Integrated Landscape Management:** Sustaining local forests can be helped by adopting an integrated approach to landscape management. Community-managed forests show a particular aptitude for balancing the interdependence of local agriculture, forestry, and other land uses.



## THEME 3: FOREST RESTORATION

This section profiles communities who have successfully restored degraded forests and deforested land as a pathway to ecosystem regeneration and sustainable livelihoods. Targeted efforts to improve forest quality in degraded local forests or to reforest cleared areas are often part of larger efforts to protect and improve watershed functions and regain ecosystem services that local forests once provided. Forest restoration is also an important element of community-based adaptation to climate change. The following lessons emerge from cases in this section:

- **Motivation and Results:** Communities undertake forest restoration primarily to support local livelihood goals or regain lost services such as water regulation or coastal storm protection. But restoration has social benefits and teaching value as well, inspiring local participation in forest activities.
- **Mitigation Potential:** Community-led forest restoration has considerable global mitigation potential because it is widespread, is especially suited to restore degraded forests, is applicable in areas where large-scale reforestation projects are not, and is sustainable due to community buy-in.
- **Restoration and Adaptation:** Forest restoration is a powerful climate adaptation tool. It rebuilds landscape resilience, supports a transition to climate-adapted livelihoods, and restores protective functions such as flood control, erosion prevention, and coastal buffering.

## THEME 4: FOREST PROTECTION

Cases in this section feature community-led initiatives to protect forests through community conserved areas, sanctuaries, or other explicitly protected zones. This has become a potent strategy to maintain forest cover and its associated climate and livelihood benefits, as well as to maintain and even rejuvenate local cultural identity. The following lessons emerge from cases in this section:

- **Conservation and Local Culture:** Community-protected forests reflect local cultures, express local conservation values, and respond to local environmental pressures. Their strength is that they are designed with the community in mind, making them more effective and sustainable than state-managed protected areas.
- **Effectiveness:** The conservation value of community-protected forests is high. Because they are embedded in local culture and arise from local demand, community compliance with use restrictions within the community-protected forests is high and local participation in their management is strong, boosting results.
- **Beyond the Local Level:** Community-protected forests can be a powerful complement to state-designated protected areas, increasing connectivity across conservation zones, wildlife corridors and landscapes. They often occur in otherwise neglected areas, which states bypass. Linking them in dynamic networks can magnify local benefits and scale up biodiversity and climate benefits.



## FINDINGS: A COMMUNITY ROUTE TO GLOBAL CLIMATE GOALS

The case studies in this book provide an overview of both the local development benefits and the global climate benefits of community-based forest management, and the link between the two. The case materials also reveal what motivates communities to organize and carry out local forest initiatives, and the enabling conditions—the resource rights, partnerships, and capacities—that help them to succeed. Below are the principal findings that emanate from the cases and the associated analysis.

### 1. **The scope of climate benefits from community-based forest management is large.**

These global climate benefits are significant for several reasons:

**Significant in area.** Community-based forest initiatives work at a local level, village by village and forest by forest, but their aggregate impacts are much larger in scope.

**Significant in the quality and sustainability of benefits.** Successful community-based forest management generates mitigation and adaptation benefits of high quality. These are notable for their sustainability in a rural environment where conversion and extraction pressures on forests can be intense.

**Significant for where it can be applied.** Community-based forest management is marked by its wide applicability and ability to generate local benefits and regenerate vital forests in situations where other interventions fail. It can be effective in areas where the pressure to overuse or clear local forests is strong, as well as in areas of degraded forest.

**Significant for the change in local thinking it engenders.** Community-based interventions work because they change the way communities think about and act toward their local forest resources.

This change in thinking is often a prerequisite for community willingness to adopt and even innovate new management practices, establish new sustainable forest enterprises, and persist through obstacles and over years.

### 2. **To generate global climate benefits, community-based forest initiatives must deliver on local development priorities.**

Communities are motivated more by local benefits and empowerment than by global climate goals. Indeed, respecting these local motivations is the only route to achieving the significant climate benefits associated with community-based forest initiatives.

### 3. **Empowerment begins with secure land rights.**

No single factor provides a greater motivating force for the adoption of community-based forest management, or plays a more central role in its success, than secure land rights. Many of the cases show the power of combining advocacy for indigenous lands rights with sustainable forest management practices to safeguard community forests.

### 4. **An integrated, landscape approach maximizes the potential of local forest management.**

Adopting an integrated approach to forest management, in which a variety of forest uses—including agriculture—may take place within the forest landscape, is the best way to maximize local forest benefits and manage inherent trade-offs.

### 5. **Partnerships are vital.** Partners provide community groups with an array of support services—from technical and business consultations to political organizing and outreach. Supporting and participating in dynamic partnerships with communities is one of the best ways that donors, governments, NGOs, and the private sector can encourage the expansion of local forest initiatives and the scaling up of the climate benefits they produce.

## SUPPORTING COMMUNITY-BASED FOREST MANAGEMENT IN A NEW GLOBAL CLIMATE REGIME

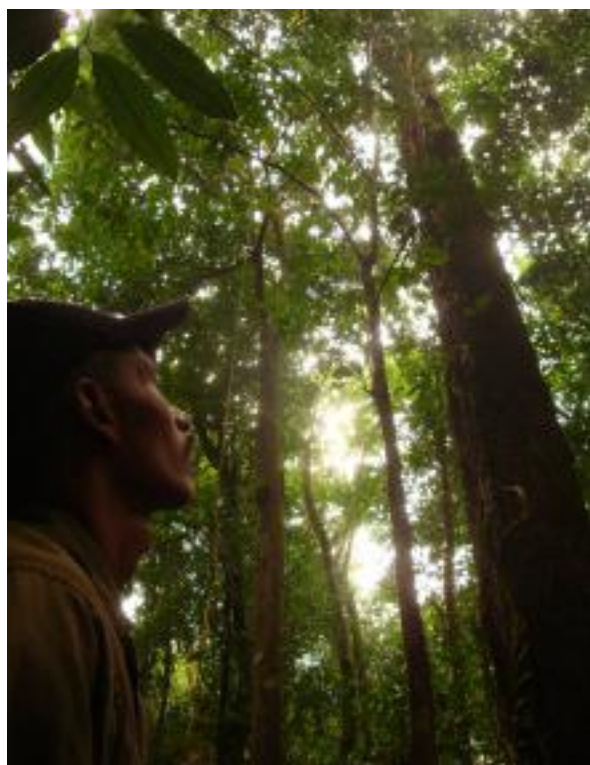
Indigenous and local communities already play an important, although largely underappreciated, role in safeguarding the global climate through the sustainable management of communal forests. But they can and should play a more prominent and acknowledged role in the new international climate framework. Community-based forest initiatives undertaken by indigenous groups and local communities can make unique and timely contributions to the goals of reducing deforestation and restoring degraded forests, thereby reducing emissions. This lends credence to the international endorsement of Reducing Emissions from Deforestation and Forest Degradation (REDD+) and continuing efforts to create national programs to support community based forest initiatives. Indeed, with an enabling policy environment and the right partnerships, community efforts could be an important factor in achieving the international goal of keeping global warming to below 2°C. Likewise, community-based forest management, because of

its effectiveness in restoring degraded forests, could undoubtedly play a substantive role in achieving the forest restoration targets set by the Sustainable Development Goals and the New York Declaration on Forests. Just as importantly, community-based efforts have a record of sustainability that will be critical to maintaining progress on climate change and poverty reduction targets in decades to come.

To facilitate this, the international community and national authorities would be well-served by taking action in three areas.

1. **Acknowledge the contributions and potential of community-based forest initiatives.** A first step in encouraging the forest management efforts of indigenous peoples and local communities is for the international community to acknowledge their current achievements and potential contributions to mitigating forest emissions and adapting to climate change. The widespread implementation of REDD+ offers an opportunity to take advantage of community-level contributions.
2. **Endorse the expansion of indigenous territories and community land rights.** There is a huge unmet demand for secure land rights in forest communities. Action to greatly expand the official recognition and legal protection of community land rights is thus a first order of business. Such tenure reform can capitalize on technological improvements in mapping, demarcation, and land titling that have recently decreased the costs of legally recognizing indigenous and community lands.
3. **Provide an enabling environment of policies and support.** Land and resource rights by themselves are not enough. They are only one element of an enabling environment that can facilitate local empowerment, develop local governance and business capacities, provide financial and capacity support where needed, and remove regulatory obstacles to the growth of rural enterprises.

Addressing global climate change in a way that recognizes the sustainable development aspirations of indigenous and local communities is one of the defining rural development challenges of our day. Community-based forest management offers a route to meeting this challenge by generating a range of economic and social benefits for local people while reducing global carbon emissions, and helping rural communities to adapt to the unavoidable impacts of climate change.









A photograph of a dense tropical forest. The upper portion of the image is dominated by a thick, white mist or fog that hangs over the forest canopy. Below the mist, the forest is a dense, lush green, with various shades of green indicating different types of trees and vegetation. The trees appear to be tall and closely packed together. In the lower right corner, a small portion of a tree with large, light green leaves is visible. The overall atmosphere is serene and somewhat mysterious due to the fog.

# Introduction

# INTRODUCTION

Indigenous peoples and local communities have a vital part to play in the global effort to reduce greenhouse gas emissions and address climate change. They are the custodians of many of the world's remaining standing forests, which are essential in mitigating greenhouse gas emissions. Community-based forest management—an approach and land governance model that encompasses a wide range of forest protection, restoration, and sustainable use activities—can effectively reduce rates of deforestation, restore degraded forests, and allow local people to better adapt to climate change. It can achieve these global climate benefits even as it supports local livelihoods and indigenous cultures, conserves biodiversity, and increases ecosystem resilience and stability.

**“Community-based forest management—an approach and land governance model that encompasses a wide range of forest protection, restoration, and sustainable use activities—can effectively reduce rates of deforestation, restore degraded forests, and allow local people to better adapt to climate change.”**

In this book, we bring together case material from 22 successful community-based forest management initiatives. These initiatives showcase the wide range of approaches that indigenous peoples and other forest-dependent communities take to sustainably use, protect, and restore local forests through their collective action. Such community-based action has a well-documented record of delivering an array of local benefits. These include job creation and economic gains, greater social cohesion and community self-reliance, political and social empowerment, food security and water access, conflict resolution, wildlife protection, and ecosystem restoration.

While these local development benefits are amply demonstrated in the cases presented here, the case materials also demonstrate the global climate benefits that flow from these local actions. In short, community-managed forests offer an effective method to maintain and increase forest canopy, suppress illegal forest activities that degrade forests, and—through education and first-hand experience—build a lasting commitment among community members to sustainable forest use. All of these actions maintain and enhance the ability of the world's forests to mitigate CO<sub>2</sub> build-up in the atmosphere and to moderate the effects of this build-up on ecosystems and people.

Many of these local initiatives successfully scale up to become the predominant governing bodies for entire ecosystems, forests, wildlife corridors, and agricultural landscapes, refuting the common but mistaken view that local solutions invariably remain small in scope and impact. They are often the foundation on which national progress towards climate and development goals are built.

## A ROLE FOR COMMUNITIES IN CLIMATE ACTION?

The community-based forest management models profiled in these case studies—and many more taking place in rural communities around the world—represent an underutilized tool to meet global climate goals while serving rural development needs. As the cases show, indigenous peoples groups, women's groups, and other community-based organizations, when empowered to manage their ecosystems and natural resources, can become innovators in local development and climate action.

As the international community embarks on implementation of the new climate framework, it is an opportune time to reflect on the conditions that make community-based forest management successful, how such initiatives can be empowered and enabled, what climate and human development benefits are produced, and what role community-managed forests can play in a global agenda to address climate change.

It is also important to ask how these local initiatives align with and support the aspirations of indigenous peoples and local communities to secure their land and resource rights and to pursue their own development goals. Aligning the global climate agenda with the movement to empower indigenous peoples and local communities will invariably strengthen both efforts.

The intent of this book is twofold.

- To examine the community dynamics, enabling policies and partnerships that are behind successful community-based forest management initiatives, and demonstrate the resulting climate and sustainable development benefits.
- To show how legal and social empowerment, such as strengthening land tenure security and resource rights, can meaningfully impact the ability of indigenous peoples and local communities to mitigate and adapt to climate change.

Our hope is that the case studies and the accompanying analysis in this reader can help to clarify the important role of indigenous peoples and local communities in achieving global climate change goals. In addition, we hope the cases and findings presented here can provide the basis for practical guidance to communities coping with increasing pressures on local forests and agricultural landscapes and the growing threat of climate-related stresses; to donors and other partners wishing to catalyze,

support, and scale up these efforts, whether through Reducing Emissions from Deforestation and Forest Degradation (REDD+) or other similar initiatives; and to government officials and policymakers who want to provide an enabling environment of policies, resources, and knowledge to aid the kind of community-driven development these cases represent.

## ABOUT THE CASES

All of the community-based forest initiatives profiled in this book are recipients of the Equator Prize, awarded biennially by the United Nations Development Programme (UNDP) and its partners to recognize outstanding local solutions for people, nature, and resilient communities. Since its inception in 2002, the Equator Prize has been awarded to more than 200 indigenous peoples and local community groups from more than 70 countries around the world. Forest communities make up some 60 percent of all Equator Prize winners.

Detailed case studies have been compiled on each of the Equator Prize winners. This collection of case material comprises a unique archive of best practices in community-based ecosystem management, and it is from this archive that the cases in the book are drawn. The cases included here have been edited to reduce their length and focus reader attention on the most critical material. Full-length versions of these cases can be found in the [Equator Initiative Case Study Database](#).

## Cross-Cutting Questions

The case studies in this reader have been assembled to consider the following cross-cutting questions.

- **Motivations and Scope:** What motivated these community-based forest management initiatives and what has been the scope of climate benefits achieved? How has community-based forest management improved local resilience?
- **Climate Potential:** What role can rural forest communities play in an international regime to combat climate change? How can this role be recognized and encouraged by national governments and the international community?
- **Empowerment:** What constitutes an enabling policy environment for indigenous and local community action? What are the policies, laws, and regulations that need to be in place for local action to thrive?
- **Partnerships:** What are the partnership models that add to local capacities and support effective local action? What is the role for private sector and other partners?



*There are over 200 Equator Prize winners from more than 70 countries around the world.*

### **The Equator Initiative: A Partnership for Resilient Communities**

The Equator Initiative brings together the United Nations, governments, civil society groups, businesses, and grassroots organizations to recognize and advance local sustainable development solutions for people, nature, and resilient communities. The partnership arose from recognition that the greatest concentrations of both biodiversity and acute poverty coincide in equator belt countries, and the high potential for win-win outcomes where biological wealth could be effectively managed to create sustainable livelihoods for the world's most vulnerable and economically marginalized populations. The direct dependence of the rural poor on nature for their livelihoods means that biodiversity loss often exacerbates local poverty. But by the same token, action to sustain ecosystems and maintain or restore biodiversity can help stabilize and expand local resource-based economies and relieve poverty.

The Equator Initiative aims to recognize the success of local and indigenous initiatives, create opportunities and platforms for the sharing of knowledge and good practice, inform policy to foster an enabling environment for local and indigenous community action, and develop the capacity of local and indigenous communities to scale up their impact. The center of Equator Initiative programming is the Equator Prize, awarded biennially to recognize and advance local climate and sustainable development solutions. As local and indigenous groups across the world chart a path towards sustainable development, the Equator Prize shines a spotlight on their efforts by honoring them on an international stage. The Equator Prize is unique for recognizing group or community achievement, rather than focusing on that of individuals.



## A RANGE OF SUSTAINABLE FOREST MANAGEMENT METHODS

A review of Equator Prize-winning initiatives shows that community groups use a wide range of forest management methods to protect their forest assets from threats and increase the range and quantity of benefits that local forests produce. These methods include applying better harvesting practices for timber and non-timber products; developing new forest-based products, medicines, and specialty crops; practicing ecotourism and cultural tourism; pursuing agroforestry and other sustainable forest agriculture; restoring and regenerating cleared and degraded forest areas; and protecting local forests in community conserved areas and restricted use zones.

To organize the case material, the book is divided into four themes that reflect the major activities that community-based forest initiatives undertake.

### Theme 1: Sustainable Forest Livelihoods

Case studies in this section consider the wide variety of strategies that communities have employed to use their forests to support local livelihoods while maintaining and enhancing forest ecosystems. This includes cases where communities have established substantial commercial forest enterprises, as well as cases where forest management has yielded more broad-based community uses and public benefits.

### Theme 2: Forest-Friendly Agriculture

Demand for agricultural land is a leading cause of deforestation, while poor farming practices can be a significant source of forest degradation. Forest-friendly farming practices, on the other hand, can maintain and enhance forest cover, food security, and local livelihoods. Cases in this section display the many ways in which forest communities have incorporated agriculture within their sustainable forest management regimes.

### Theme 3: Forest Restoration

This section profiles communities that have successfully restored degraded forests and deforested land as a pathway to sustainable livelihoods and ecosystem regeneration. Targeted efforts to improve forest quality in degraded local forests or to reforest cleared areas are often part of larger efforts to protect and improve watershed functions and regain ecosystem services that local forests once provided. Forest restoration is also an important element of community-based adaptation to climate change.

### Theme 4: Forest Protection

Cases in this section feature community-led initiatives to protect forests through community conserved areas, sanctuaries, sacred sites or other explicitly protected zones. This has become a potent strategy to maintain forest cover and its associated climate and livelihood benefits, as well as to maintain and even rejuvenate local cultural identity.



## Examples of Community-Based Forest Management

### Sustainable Forest Livelihoods

- Applying sustainable harvest practices for timber and non-timber products
- Drawing up forest management plans specifying location, timing, and intensity of forest uses
- Developing and commercializing new forest-based products and medicines and establishing forest-based enterprises
- Practicing ecotourism and cultural tourism
- Mapping community forest boundaries, monitoring forest conditions, and measuring forest productivity

### Forest-Friendly Agriculture

- Pursuing agroforestry and other sustainable forest agriculture
- Replacing slash-and-burn agriculture with intensified crops systems in designated crop zones
- Better soil management through terracing, mulching, no-till planting, and other slope and forest-adjusted tillage methods
- Developing high-value specialty crops

### Forest Restoration

- Restoring and regenerating cut-over and degraded forest areas through temporary closures and reforestation
- Establishing local nurseries to raise indigenous tree species for reforestation projects and augmentation planting
- Replanting and restoring mangrove forests

### Forest Protection

- Establishing community conserved areas, reserves, and restricted-use zones within local forests
- Co-managing established parks and protected areas with state agencies
- Mixing forest conservation areas with agricultural zones and forest use zones within an integrated landscape management plan
- Forest conservation campaigns based on the protection of flagship wildlife species

## THE UNIQUE ROLE OF FOREST COMMUNITIES

For the 1.6 billion people living in or near forests, income from forest products such as wild foods, fuelwood, fodder, timber, medicines, and other non-timber forest products comprises a significant portion of total household income—an average of 28 percent, according to a Center for International Forestry Research study (Angelsen, et al, 2014). For many communities, dependence on forests for subsistence and income is much higher, particularly among the estimated 60 million indigenous peoples who live in forests. This dependence is not a matter of economy alone. Forests also provide the spiritual and cultural context for many rural communities and indigenous groups, a factor that plays an important part in motivating community action to sustainably manage forests.

**“One of the roots of local vulnerability is the widespread lack of legally recognized land rights in rural communities, meaning that many forest communities lack legal control over their local forests.”**

Unfortunately, threats to this critical resource base—from unsustainable forest management and land uses, extractive industries, land grabs and climate change—are high. Forest loss from agricultural conversion, large-scale timber removal, mining, and other extractive uses exceeds some 13 million ha per year, with an attendant loss of biodiversity. The livelihoods and cultural heritage of many forest communities are undermined, and economic and social vulnerability increases. One of the roots of local vulnerability is the widespread lack of legally recognized land rights in rural communities, meaning that many forest communities lack legal control over their forests.

Nonetheless, Equator Prize-winning communities have shown that community groups—when properly motivated—can be highly effective in meeting these threats and protecting their economic and cultural assets through collective action. Indigenous peoples and local communities have a long record of forest stewardship, informed by generations of close contact with forest ecosystems and manifest in

a variety of traditional forest management practices and accumulated ecosystem knowledge. The Intergovernmental Panel on Climate Change (IPCC) acknowledged in a recent report that the traditional knowledge systems of indigenous peoples and their holistic vision of community and environment are key resources in the effort to achieve global climate change mitigation and adaptation goals. These traditional forest practices are typically highly tailored to the local environment, friendly to biodiversity, and sustainable over time. At the same time, they have proven adaptable, with many communities augmenting their traditional practices with modern approaches such as forest monitoring and mapping techniques.

Rural forest communities also have a major incentive to manage local forests in climate-friendly ways. These communities are among those most likely to be adversely affected by—and called upon to adapt to—climate change, including severe weather, floods, and drought. These adverse conditions will likely bear directly on the health and productivity of forests, and thus on the economic and cultural well-being of forest communities. It should not be surprising then that community action is often a key catalyst to slowing deforestation, restoring degraded forests, and increasing local climate resilience.



## MAKING THE LINK: COMMUNITY-BASED FOREST MANAGEMENT AND CLIMATE BENEFITS

Rural communities are often well-placed to mitigate greenhouse gas emissions from forests because they live among and depend on these resources. As local forest users, their harvest and management practices can strongly affect whether local forests are degraded or maintained. Several lines of evidence have emerged that indicate the importance of community-based forest management in fighting global climate change.

1. **Forest sector emissions are substantial, and deforestation remains a potent risk.** Emissions from deforestation and forest degradation are responsible for some 14-21 percent of global greenhouse gas emissions, according to a recent analysis of tropical forests (ISU, 2015:19). While there has been a decrease in the rate of deforestation in some countries in the last few years, global forest loss is still substantial—about 13 million ha per year—and pressure on forests remains high. Recent projections of global deforestation from 2016-2050 run to nearly 290 million ha—more than 8 million ha per year—in the absence of new forest conservation policies and agricultural practices (Busch and Engelmann, 2015:14). As a consequence, forest management has long been part of international climate negotiations, with the need to decrease forest emissions well-accepted by the international community. The potential for forest communities to contribute to this effort has been acknowledged internationally through establishment of national REDD+ programs in which communities receive support for carrying out projects to protect, sustainably use and restore local forests.

“The forest area officially recognized by governments as indigenous or community-controlled is far smaller than the area customarily claimed and occupied by indigenous peoples and communities.”

2. **Community-based forestry solutions are effective at cutting deforestation rates.**

The case studies in this book illustrate how community-based forest initiatives are effective at reducing deforestation rates in local forests. This is corroborated by a global-level analysis of deforestation rates in community-managed forests released in 2014 by World Resources Institute and the Rights and Resources Initiative (WRI and RRI, 2014). The results showed significantly lower deforestation rates in community-owned forests compared to similar state- and privately-managed forests, particularly when the national government recognized and protected community rights to own and manage local forests. In Brazil, for example, the deforestation rate from 2000-2012 in designated indigenous community forests was 0.6 percent, compared to 7.0 percent in adjacent forests—more than 11 times lower (WRI and RRI, 2014:4). In Guatemala’s Peten region (in the Maya Biosphere Reserve), legally recognized community forests experienced a 0.02 percent deforestation rate from 1986-2007, while nearby Protected Areas of the Maya Biosphere Reserve experienced a 0.41 percent deforestation rate—that’s 20 times lower (WRI and RRI, 2014:2). The difference between the deforestation rates in community forests versus public forests can largely be attributed to the value communities place on their forest resources, the sense of ownership and empowerment they feel, and the incentive they have to actively manage their forests for productivity and police them to suppress illegal logging and forest conversion.

3. **The area of community-owned or occupied forests is substantial at a global level, making potential climate benefits of community forestry globally significant.**

Indigenous peoples and local communities have legal title or officially recognized rights to at least 513 million ha of forests worldwide—about 15.5 percent of the global forest area (RRI, 2014:2). In lower and middle-income countries where most tropical deforestation occurs, the proportion of indigenous and community-controlled forests is higher—over 30 percent—reflecting an expansion of legally recognized indigenous and community forests in these countries since the 1990s. These community forests contain some 38 billion tons of carbon—about 29



times the annual emissions from all passenger vehicles in the world (RRI, 2014:3). Thus, the climate advantages of community-based forest management, when taken together, are significant at a global level. Moreover, there is much room for expanding the area of recognized indigenous and community forests, and thus potentially expanding the global climate benefits of community-based forestry. The forest area officially recognized by governments as indigenous or community-controlled is far smaller than the area customarily claimed and occupied by indigenous peoples and communities.

4. **Community-based forest solutions offer good value.** Reducing deforestation and forest degradation are low-cost strategies for addressing climate change. Recent research confirms that action to cut deforestation and forest degradation rates can cut global emissions at a cost that is quite low compared to other mitigation strategies such as reducing carbon emissions from industry, cars, and dwellings through greater energy efficiency or new energy technologies. Community-based forest management is a proven means to tap into this low-cost climate mitigation that can be deployed in rural areas at minimal government expense, with additional community benefits produced in the bargain (Busch and Engelmann, 2015:1, 14-18).



## ACTING ON THE NEW YORK DECLARATION ON FORESTS AND THE SDGs

In addition to supporting the new global agreement to address climate change, encouraging community-based forest management initiatives will also be essential to achieving the goals of the [New York Declaration on Forests](#) (adopted at the United Nations Climate Summit in 2014) as well as the new slate of [Sustainable Development Goals \(SDGs\)](#) adopted in September 2015 by national leaders assembled at the United Nations.

The New York Declaration on Forests, signed by a diverse group of 180 developed and developing nations, indigenous peoples organizations, private sector companies, NGOs, and other civil society organizations, calls for concrete action to halt global forest loss and restore degraded forest areas worldwide. While citing the important role forests play in regulating global climate and absorbing greenhouse gases, the New York Declaration also stresses the other crucial functions forests serve: supporting as much as 80 percent of terrestrial biodiversity, and providing food, water, fuel, and medicines for some 1.6 billion people, as well as supporting their traditional cultures and livelihoods (UN, 2014: 3). To serve its goals, the New York Declaration sets robust targets. It calls for cutting in half the current rate of loss of natural forests by 2020, and restoring 150 million ha of degraded forest. By 2030, it calls for completely halting forest loss and restoring an additional 200 million ha of forests. Without the active participation of rural forest communities, it is hard to conceive of attaining such ambitious targets. Importantly, the New York Declaration explicitly calls for empowering forest communities and recognizing the land and resource rights of indigenous peoples.

Likewise, the new suite of Sustainable Development Goals, adopted by the international community in 2015 as a follow-on and expansion of the Millennium Development Goals, will clearly require participation by rural forest communities if they are to be achieved. Sustainable Development Goal 15 specifically targets halting deforestation by 2020, restoring degraded forests, and implementing sustainable forest management, mirroring the language of the New York Declaration on Forests (UN, 2015:25).

Perhaps more important still is the ability of community-based forest initiatives to deliver on SDGs beyond those that are forest-specific. These community initiatives have shown in the past that they are effective in reducing poverty, improving food security and nutrition, and empowering women and other vulnerable and marginalized groups.

Through their support of public infrastructure such as water systems, schools, and health clinics, they also contribute to the provision of safe drinking water, improved education, and greater access to health care. In essence, they act as efficient delivery platforms for a suite of services essential to meeting the new SDGs in rural forest communities.

## Action Targets of the New York Declaration on Forests

### Deforestation

- By 2020, cut in half the rate of loss of natural forests
- By 2030, strive to end natural forest loss

### Forest Restoration

- By 2020, restore 150 million ha of degraded forests
- By 2030, restore an additional 200 million ha

### Forest Governance

- Strengthen forest governance, transparency and the rule of law, while also empowering communities and recognizing the rights of indigenous peoples, especially those pertaining to their lands and resources

## Sustainable Development Goal 15

Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

### SDG Target 15.2

- By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests, and substantially increase afforestation and reforestation globally.







# Theme 1

## Sustainable Forest Livelihoods





**T**he cases in this section profile community-based forest management intended to maximize and maintain the local forest as a substantive community asset, contributing to local livelihoods through subsistence and cash income, and generating revenue to fund community institutions and infrastructure. The first four cases emphasize a mixed-use approach, where subsistence and small-scale enterprises predominate, and broad-based public benefits are favored. The next three cases involve substantial commercial ventures that communities have developed from forest resources, hoping to evolve greater revenues while maintaining forest cover and related non-cash forest benefits. In all of the cases, the nature of the community's forest access, ownership, and management rights plays an important role, as does the community's market access and the capacity building support received from partners. In all cases, substantial climate benefits result from maintenance of forest cover and improvement in forest conditions due to well-articulated community management goals, often in circumstances where deforestation pressures are high.

## CASES IN THIS SECTION

- **Ekuri Initiative**, Nigeria
- **Mosquitia Pawisa Apiska (MOPAWI, Agency for the Development of the Mosquitia)**, Honduras
- **Fikambanan'ny Terak'i Manambolo (FITEMA, Association of Manambolo Natives)**, Madagascar
- **Asociación para la Investigación y el Desarrollo Integral (AIDER, Association for Research and Integrated Development)**, Peru
- **Associação Sociocultural Yawanawá (Sociocultural Association of Yawanawá)**, Brazil
- **Comunidad Nativa Ese'ejá de Infierno (Ese'ejá Native Community)**, Peru
- **Comunidad Indígena de Nuevo San Juan Parangaricutiro (Indigenous Community of Nuevo San Juan Parangaricutiro)**, Mexico

## UNDERSTANDING THE CASES

To understand the structure, challenges, impacts, and implications of the forest initiatives in this section, readers should consider the following questions as they read the cases.

- **Climate Potential:** What is the potential climate benefit of these initiatives and how is it generated?
- **Community Motivations:** What factors inspired these initiatives? How was this motivation channeled?
- **Forest Ownership:** What role does forest tenure play in the viability and success of community forest initiatives?
- **Enterprise Development:** What factors are behind successful development of local forest enterprises? What are the obstacles to rural enterprise and how have successful ventures navigated them?
- **Local Knowledge and Innovation:** How do local knowledge and innovation play into the success of community-based forest initiatives?
- **Scalability and Sustainability:** Can the impact of these initiatives be up-scaled? Are their experiences transferable to other forest communities? Are they sustainable over time?



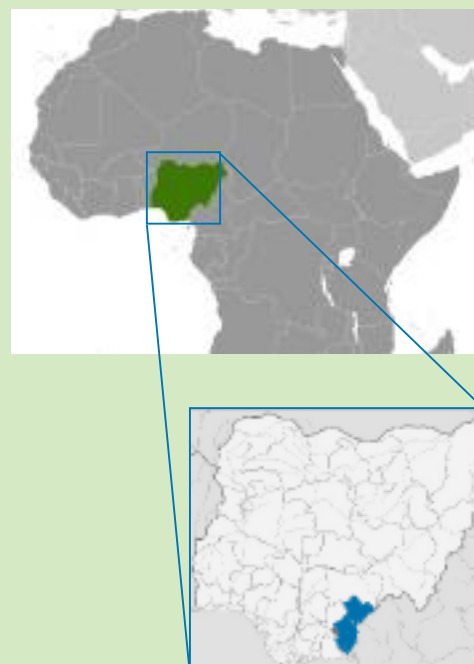


# EKURI INITIATIVE

## Nigeria

### PROJECT SUMMARY

Located in Nigeria's Cross River State, the Ekuri community manages a 33,600 ha community forest adjacent to the Cross River National Park. Community forest management began in the 1980s, when the villages of Old Ekuri and New Ekuri united in response to the proposed logging of their forest. The project would have included the construction of a road linking the villages to local market centers. Instead, the community decided to sustainably manage the forest as a community asset, generating income, subsistence materials and food. Levies on the sale of non-timber forest products by community members financed a road that eventually reached both villages. In addition to allowing farm and forest products to reach new markets, the road has enabled the transport of construction materials for two schools, a health center, and a civic center.



## BACKGROUND AND CONTEXT

The villages of Old Ekuri and New Ekuri, jointly known as the Ekuri Community, are located seven kilometers apart in Cross River State, eastern Nigeria and together own 33,600 ha of community forest situated in the buffer zone of the Cross River National Park. Around 95 percent of this land is covered by primary or secondary forest. The Ekuri Community speaks Lokorli, an indigenous language not spoken anywhere else. The community is rich in tradition, with strong cultural cohesion and respect for traditional chieftain institutions. The primary occupations of the 6,000 community residents are farming, producing non-timber forest products and handicrafts, and small-scale trading, hunting, and fishing.

### *A West African – and global – biodiversity hotspot*

The Cross River National Park is the largest protected area in the Nigeria-Cameroon region, forming a significant part of the Guinean Forests of West Africa, a global biodiversity hotspot. The park has a total area of 4,227 km<sup>2</sup>, most of which consists of primary tropical rainforests in the north and central parts, with



mangrove swamps on the coastal zones. The forest is largely untouched in the less-accessible areas, but around the margins it has been considerably affected by human activity including oil palm and rubber plantations as well as illegal logging.

### *A forest-dependent local economy*

Population growth in buffer zone villages is increasing human pressure on the forest ecosystem. Because the nearest markets are far away and forest and agricultural goods have to be carried by head-load, high-value forest products are prioritized over farming produce. The main items traded are bush mango fruits (*Irvingia gabonensis*) and edible forest leaves (*Gnetum* spp.), as well as bushmeat including endangered species such as chimpanzee (*Pan troglodytes*) and drill (*Mandrillus leucophaeus*).

During the 1980s, there was no road access to the Ekuri Community. To acquire staples such as salt, soap, kerosene, and clothing, community members had to walk over four hours to reach the nearest road. In 1982, Old Ekuri began negotiations with a logging company to construct a road to the Ekuri Community in exchange for logging rights. This attempt by Old Ekuri was vetoed by the Chief of New Ekuri who cited the enormous benefits the Ekuri people had historically received and continued to gain from the forest. He strongly advocated for the conservation of the forest so that these benefits might continue to sustain the community in perpetuity and proposed construction of the road through self-help efforts. His appeal was well-received by the community and marked the formal beginning of the Ekuri community's involvement in sustainable forest management.

### *Beginnings of collective action*

To raise income, the villages imposed levies on the proceeds from sales of non-timber forest products. Over four years, this levy system produced enough revenue to fund the construction of a 40 km road, reaching Old Ekuri in 1990 and New Ekuri in 1997. When the Cross River National Park was legally recognized in 1991, the Ekuri Community approached the park authority to gain formal support to manage their forest on their own. The request was approved and a Community Forest Officer was based in the Ekuri Community for two years to build the foundation of a formal community-managed forest system – the first of its kind in Nigeria.





### *A community-based response to an environmental threat*

The initiative's aim was to protect and safeguard the forest, both as an inherited responsibility from past generations and as a responsibility to future generations. This recognized the spiritual and cultural values of the forest for the Ekuri community, as well as the ecosystem services it provides. Since its inception, Ekuri has worked to develop sustainable forest-based livelihoods, reduce poverty, promote capacity building, and strengthen local leadership.

## KEY ACTIVITIES AND INNOVATIONS

### *Participatory forest zoning*

The initiative began in 1997 with a perimeter survey of the Ekuri community forest. Community land boundaries were demarcated through a process of negotiation with six neighboring communities. 32 community members were also given training

on developing timber inventories, and a preliminary land-use plan was developed. This plan delimited eight zones: an agroforestry stream buffer zone; protected area and animal corridor zones; non-timber forest product and forest management zones; farming and cash crop zones; and ecotourism zones.

### *Communal forest ownership and benefit sharing*

The entirety of the Ekuri community forest is under communal ownership, including trees on individual farm plots. Every Ekuri community member has the right to harvest trees for their own use, but not for commercial sale; communal sales take place through the Ekuri Initiative. This is substantially different from other communities, where forest areas have been claimed by individuals and sold to logging companies, accelerating the loss of forests. The system of communal ownership has been underpinned by the equitable sharing of benefits from Ekuri's various income generating activities. Two 50 ha plots have been set aside in which trees grown for sale are

harvested using a directional technique, a process that reduces damage to the forest and to the soil. The community has introduced regulations for forest and farm products, including a registration fee for products, the regulation of sale prices to different buyers, and the introduction of sales taxes and gate fees to ensure that 70 percent of weekly revenues are paid into the community treasury. In its efforts to reduce poverty, the initiative has developed microcredit schemes; given trainings and skills development workshops; provided scholarships, technical and financial support to farmers; and invested in value-added secondary processing equipment as well as the transporting and marketing of non-timber forest products.

## BIODIVERSITY IMPACTS

The Cross River National Park is mostly covered with lowland rainforest. More than 1,550 plant species have been identified in the park, of which 77 are endemic to Nigeria. In addition, over 350 bird species have been recorded and the area is one of only two locations in Nigeria where Xavier's Greenbul (*Phyllastrephus xavieri*) is found. The park is home to

950 species of butterflies and at least 75 mammal species, including the African buffalo (*Syncerus caffer*), the endangered African forest elephant (*Loxodonta africana*), chimpanzee (*Pan troglodytes*), and the highly endangered drill (*Mandrillus leucophaeus*).

### *The personal costs of conservation*

The Ekuri initiative began while the community forest was still largely intact. Protection has come against great odds, including attempts to coerce and intimidate the community into granting logging concessions. In 1989, the Chief of the Ekuri Community reneged on his initial commitment to protecting the forest, and secretly agreed to lease a forest concession to a logging company. This was not discovered until 1994, when the logging company began construction on a road from a neighboring community. The Chief was dethroned and the logging concession was blocked, but in 1996 a court ordered his reinstatement and sentenced six Ekuri leaders to two-year prison sentences for obstructing logging of the community forest. They served these sentences rather than concede the forest area, and the community was able to bring a successful civil suit against the deposed Chief, his supporters, and







the logging company. As a result of these personal sacrifices and the efforts of the community as a whole, total forest cover has increased within the Ekuri Community area to nearly 31,000 ha.

#### *Ongoing threats to wildlife*

Despite the community's efforts to protect the forest, wildlife populations have continued to decline due to overhunting for the commercial meat trade and poaching. Hunting, water pollution, and poisoning (for fishing) are severe constraints to the regeneration of wildlife species. The initiative is implementing an extensive environmental education program in the Ekuri community, and attempting to expand outreach to neighboring communities where species loss is linked with youth unemployment.

### **SOCIOECONOMIC IMPACTS**

#### *The Ekuri road – enabling improved community well-being*

Revenues from the sale of timber and non-timber forest products have funded crucial development and infrastructure projects in the community, including the construction of a road, culverts, and bridges. The road facilitates transport of forest products to markets and the supply of building material to the

community. This link between forest revenues and local development has given the community a clear understanding that if the forest is lost, the community also loses its social safety net.

#### *Changing the face of the local economy*

Market proximity guarantees higher income for food and cash crops, reducing pressure on high-value forest products. Individual families can now earn up to US\$100 per month from growing diversified cash crops. Sustainable harvesting techniques have also been promoted. For example, the community harvests afang leaves (*Gnetum africanum*) with a technique that leaves the vines undamaged to regenerate new leaves for future harvesting cycles. The group also harvests mature rattans in clumps, allowing younger plants to grow and mature. Okra and bush mangos are dried so they can be sold at better prices in periods of scarcity.

#### *Women's empowerment*

The initiative has promoted the participation of women at all levels of decision-making and in all activities to an unparalleled degree when compared to neighboring locales. Women have been included in forest governance, community development, and poverty reduction activities. Four of ten members of the initiative's board are women, while in many of

the microcredit groups, women outnumber men. The initiative also works closely with the Kristi Women's Movement, a group based in Ekuri that has organized women's cooperatives to increase agricultural production.

## POLICY IMPACTS

Implementation of the joint forest management model has strengthened links between the Ekuri Initiative and park and Forest Department authorities. In 1992, a volunteer Community Forest Officer (CFO) was assigned to the area by the national park authority. The forester assisted the Ekuri community in compiling the first forest inventory. The relationship between the villages and the Forestry Department, previously described as antagonistic, gradually improved. Park authorities also provided training for 16 local people in forest demarcation and enumeration skills.

The successes of the Ekuri Initiative inspired the Cross River State government to rewrite its forest sector strategy in 1994, enshrining the practice of community forestry as a guiding principle for the forestry sector and banning logging concessions throughout the state. The Cross River National Park has also been included as a key site in Nigeria's Reducing Emissions from Deforestation and Forest Degradation (REDD+) programme, with three REDD+ pilot projects being considered from Cross River State. One of these projects would include the Ekuri forests, along with neighboring community forests that together comprise nearly 214,000 ha.

## SUSTAINABILITY AND REPLICATION

The initiative's sustainability is largely based on ongoing awareness-raising activities, monitoring and enforcement of compliance with the land use plan, sanctions on offenders, direct community participation in forest governance, and equitable benefit sharing. Leveraging funds from partners as well as using revenues from the sale of sustainably managed forest products, the initiative has been able

to scale up activities in five neighboring communities which cover 10,000 people. The initiative's model has been shared with communities in Nigeria, Cameroon, Uganda, Mozambique, and South Africa. Within Nigeria, the Ekuri model has been replicated in nearly 40 communities in Cross River State.

## PARTNERS

- **Cross River National Park:** Community forester, vehicle, funds for health center and road maintenance
- **Cross River State Forestry Commission:** Inventory mapping, vehicles to transport building materials, training in directional felling and technical skills
- **Ford Foundation:** Forest survey and plan, microcredit scheme, registration of Ekuri Initiative, conference on community forestry
- **Forest Management Foundation:** Vehicle to transport building materials to villages and farm and non-timber forest products from villages to markets
- **Bonny Allied Industries:** Cement for four bridges
- **Crushed Rock Industries:** Material for bridges
- **Global Green Grants Fund:** Legal fees to fight illegal forest lease (US\$5,000)
- **IUCN:** Office relocation, palm oil pressers, chainsaws, microcredit for women's groups (EU€30,250)
- **European Union Micro Projects Programme:** 15 facilities: schools, markets, health centers, civic centers, water infrastructure, roads
- **UNDP/GEF Small Grants Programme:** Capacity building (US\$29,000)
- **Wheels4Life:** 20 bicycles for five communities (US\$2,000)



# MOSQUITIA PAWISA APISKA (MOPAWI, AGENCY FOR THE DEVELOPMENT OF THE MOSQUITIA) Honduras

## PROJECT SUMMARY

For more than 25 years, the Mosquitia Pawisa Apiska (MOPAWI) has worked to engage local and indigenous communities in the integrated management of the Río Plátano Biosphere Reserve and other protected areas in northeastern Honduras. Located within the Mosquitia area, the reserve contains the largest intact rainforest north of the Amazon. The organization assists communities in obtaining legal title to indigenous lands and successfully blocked the construction of a dam project in the Mesoamerican Biological Corridor. In addition to promoting community forestry, the group's activities include sustainable agriculture, micro-enterprise development, and ecotourism.





## BACKGROUND AND CONTEXT

Mosquitia Pawisa Apiska (MOPAWI) is a non-governmental organization that has been working alongside local communities for over 25 years to protect the environment and advance sustainable economic development in the Mosquitia region of northeastern Honduras. Projects focus on agroecology, community forestry, sustainable agriculture, ecotourism, community health, preservation of indigenous culture and language, and advocacy for indigenous land rights and self-determination. The organization has collaborated with indigenous groups in Honduras and Nicaragua to create a forest guard program that develops ecological guidelines and zoning for the Mesoamerican corridor, including rules for hunting, fishing, forestry, and agriculture.

### *Region and people*

The Mosquitia area of eastern Honduras is part of the Greater Mosquitia Ecosystem, one of the last great wilderness regions in Central America. Considered one of the most important and biologically-rich areas on the planet, the Mosquitia houses more than forty smaller ecosystems. Within the Mosquitia area, the Río Plátano Biosphere Reserve contains the largest intact rainforest north of the Amazon and was classified as a UNESCO World Heritage Site in 1982.

The region is home to four indigenous groups: Misquito, Tawahka, Pesch, and the African-indigenous Garifuna. The Misquitos form the dominant ethnic group in the region with a population of approximately 80,000 people. Traditional livelihoods include subsistence agriculture, fishing, hunting and gathering from the forest, and occasional wage labor.

### *Project catalysts*

The primary environmental threats in the region are posed by logging, petroleum exploration and drilling, mining, and cattle ranching. The steady expansion of commercial extractive industries in the region has resulted in extensive deforestation, loss of vegetation and wildlife, air and water pollution, and soil degradation.

Mosquitia is one of the most remote and economically marginalized regions in the state of Gracias a Dios; the area is only accessible by boat or plane. Due to its remoteness, the area receives few public services such as education, health, and energy. Weak political and institutional development in the area makes it socially and environmentally risky to open the land to road access and development.

MOPAWI was founded in 1985 as a non-profit, civil society organization. The main objectives of the organization are to promote development and well-being of local indigenous families and



conserve the rich biodiversity of surrounding forests and ecosystems. MOPAWI aims to ensure that development activities in the region are pursued on terms dictated by the indigenous population and in ways that are culturally-sensitive and economically and environmentally sustainable.

## KEY ACTIVITIES AND INNOVATIONS

### *Community forest management*

MOPAWI has worked to prevent deforestation by involving communities in local forest protection committees and by promoting community forest management. The organization has also been active in providing training to local farmers on the responsible harvest and use of non-timber forest products, including on quality standards to attain Forest Stewardship Council certification. The organization has focused on forest products such as batana oil, swa oil, tunu fiber, and achiote seed.

Batana oil, a natural ingredient used in hair products, has proven particularly lucrative. The oil is extracted from the nut of the American palm (*Elaeis Oleifera*), which is endemic to the region. MOPAWI has created market supply chains to export the oil to cosmetic companies in the United States and Canada. Participating communities have steadily increased their production of the oil, from 1,000 liters to 88,000 liters, while ensuring sustainable harvest practices that allow the forest to regenerate. Prices for the oil have grown from US\$1.50 per liter to US\$7 per liter, providing 2,000 producers from 40 communities with a significant source of income. Cultivation of this and other non-timber forest products are reducing the need of local community members to practice destructive slash and burn agriculture.

### *Organic cacao cultivation*

Another area of alternative livelihood programming for MOPAWI has been the sustainable cultivation of organic cacao (*Theobroma cacao*). MOPAWI promotes an agroforestry approach to cacao production, working with farmers to integrate the crop into ecoagriculture landscapes that also support fruits, timber trees, coconut palms, and food crops such as banana and plantain. This integrated approach to forest management has proved highly successful in improving local incomes without damaging local forests or felling trees. Through this activity, MOPAWI has helped to catalyze a cacao producers association with its own office, storage center, and processing facilities.

### *Advocacy for land rights and resource entitlements*

MOPAWI advocates for indigenous land rights, tenure security, and resource entitlements. The organization has brokered several key negotiations with government authorities to secure community land rights. As a complement to this activity, MOPAWI has trained hundreds of local and indigenous community members in forest conservation and stewardship, sustainable natural resource management practices, and the administration of forest management institutions.

MOPAWI has worked with *Muskita Asla Takanka* (MASTA), the federation of representative indigenous organizations in the Mosquitia, to define a strategy for obtaining legal title to indigenous lands. The first step in the process was to survey and document existing land uses to provide evidence and justification for the land claims. These findings were then compiled into land legislation proposals and submitted to the government. MOPAWI's work has empowered indigenous groups like MASTA to tackle land legalization processes and develop their own capacity for advocacy.

In the Mocoron zone, for instance, the group was able to negotiate with the government to gain indigenous land rights over 68,000 ha of land. The concession has now been in place for 40 years. MOPAWI also trained local communities in the region in sustainable natural resource management and developed land-use plans and zoning. Two distinctive management plans were developed: one for 3,500 ha of pine forest, the other for 14,500 ha of broadleaf forest. MOPAWI also supported local communities to create strict protected areas within the Mocoron zone.

MOPAWI employs a number of different media in its outreach on land rights and tenure security. Through radio, posters, and workshops, the organization has informed more than 500 local people living in the region's remote villages about their legal rights to traditional land as outlined in national law.

### *Ecotourism and promoting cultural integrity*

MOPAWI works with local and indigenous communities to promote low-impact ecotourism. Visitors to the region are provided with a number of community-delivered services, including: tours of local waterways in dugout canoes; swimming and snorkeling in the Caribbean Sea; ethnotourism excursions; nature walks; and visiting ancient archaeological sites.



In collaboration with the Ministry of Education, MOPAWI has also created a bilingual, intercultural education program that incorporates both the Miskito language and Spanish in order to strengthen cultural pride and improve learning in local schools. The program includes an environmental education module which emphasizes sustainable development, biodiversity conservation, and the responsible management of local forests.

## BIODIVERSITY IMPACTS

### *Rio Plátano Biosphere Reserve*

The Rio Plátano Biosphere Reserve is the largest area of virgin tropical rainforest in Honduras. Habitat types in the reserve include mangrove forests, coastal savannah, pine savannah, wetlands, and mature broadleaf rainforest. The reserve is home to 39 species of mammals, 377 species of birds, and 126 species of reptiles and amphibians. Threatened species include the giant anteater, Baird's tapir, jaguar, Caribbean manatee, American crocodile, military macaw, and great curassow.

MOPAWI workshops and trainings focus on sustainable forest management and agroforestry as viable alternatives to slash and burn agriculture. For example,

MOPAWI promotes the planting of guama plants (*Inga edulis*) as a source of firewood to limit the felling of old-growth species. Guama grows quickly and is nitrogen-fixing, which in turn improves soil quality and the growth of traditional crops like maize and beans.

### *Turtle conservation and butterfly farming*

MOPAWI works with more than 100 inhabitants of the coastal Garifuna community of Plaplaya in a protection program for leatherback and loggerhead turtles. Community members are trained to patrol more than 12 km of beach which are used by the turtles as nesting sites. Monitoring of the beaches and protection of nests is carried out until the baby turtles hatch. To date, 5,967 turtles have been born and safely released into the Caribbean Sea. A similar project of egg protection and local awareness-raising has been implemented in the area to help protect populations of the increasingly rare green iguana.

MOPAWI also operates the only butterfly farm in Honduras. Butterflies are exported to zoos and museums in the United States. The farm sells pupas, which change into butterflies within their destination exhibition houses. The butterfly farm is one of the main ecotourism attractions in the Rio Plátano Biosphere Reserve.



## SOCIOECONOMIC IMPACTS

### *Sustainable agriculture and agroforestry*

MOPAWI promotes sustainable agriculture and agroforestry in more than 48 communities, involving 700 families and 20 primary schools. MOPAWI encourages farmers to diversify their crops and to limit hunting activities. Technical training and assistance are provided for agro-industrial crops, plantains and bananas, basic grains, and for family garden plots. More than 200 indigenous farmers have been trained in organic cacao production. Expansion into organic cacao has improved local incomes and reduced dependence on harmful chemical and fertilizer inputs. Farmers have also been supported in the uptake of sustainable alley cropping techniques that increase forest cover and decrease soil erosion.

### *Ecotourism*

MOPAWI promotes ecotourism as an alternative livelihood activity to supplement the incomes of subsistence farmers. The organization has provided ecotourism-related training to over 100 indigenous people in the reserve. Training is provided in guide services, food preparation, transportation and accommodations. Ecotourism has created more than 28 full-time jobs. In the past, less than 25 percent of community families received an income or benefited from the tourism industry. Today, all of the families in Las Marias are benefiting, and average household income has doubled.

### *Women's empowerment*

Working through local banks, MOPAWI established a microenterprise program and provided loans to more than 260 women, empowering them to launch small-scale businesses which in turn generate employment, income, and family savings. Among the businesses launched are butcher shops, libraries, clothing shops, and mosquito net retailers. Owing in part to the success of the microcredit program, community banks have grown in number from eight to thirteen.

## POLICY IMPACTS

MOPAWI has been active in pursuing policy change that favors local and indigenous communities, particularly in the area of land rights and tenure security. At the legislative level, MOPAWI plays a key role in forging productive linkages between government, research organizations, NGOs, and indigenous organizations for campaigning and awareness-raising. MOPAWI has effectively leveraged these relationships and partnerships to influence government policy.

MOPAWI joined forces with the Tawahka community to organize the Committee of Land Vigilance and First Congress on Indigenous Lands in Mosquitia in 1992. The initiative mapped out which communal lands in the region were under the direct management of indigenous peoples and local communities. This mapping exercise helped lead to the creation of



the 230,000 ha Tawahka Biosphere Reserve, which is now managed by MOPAWI and other indigenous groups. The reserve links an important biodiversity corridor to protected areas in Nicaragua. Formal legal designation has helped reduce illegal trafficking of forest timber and wildlife from the region.

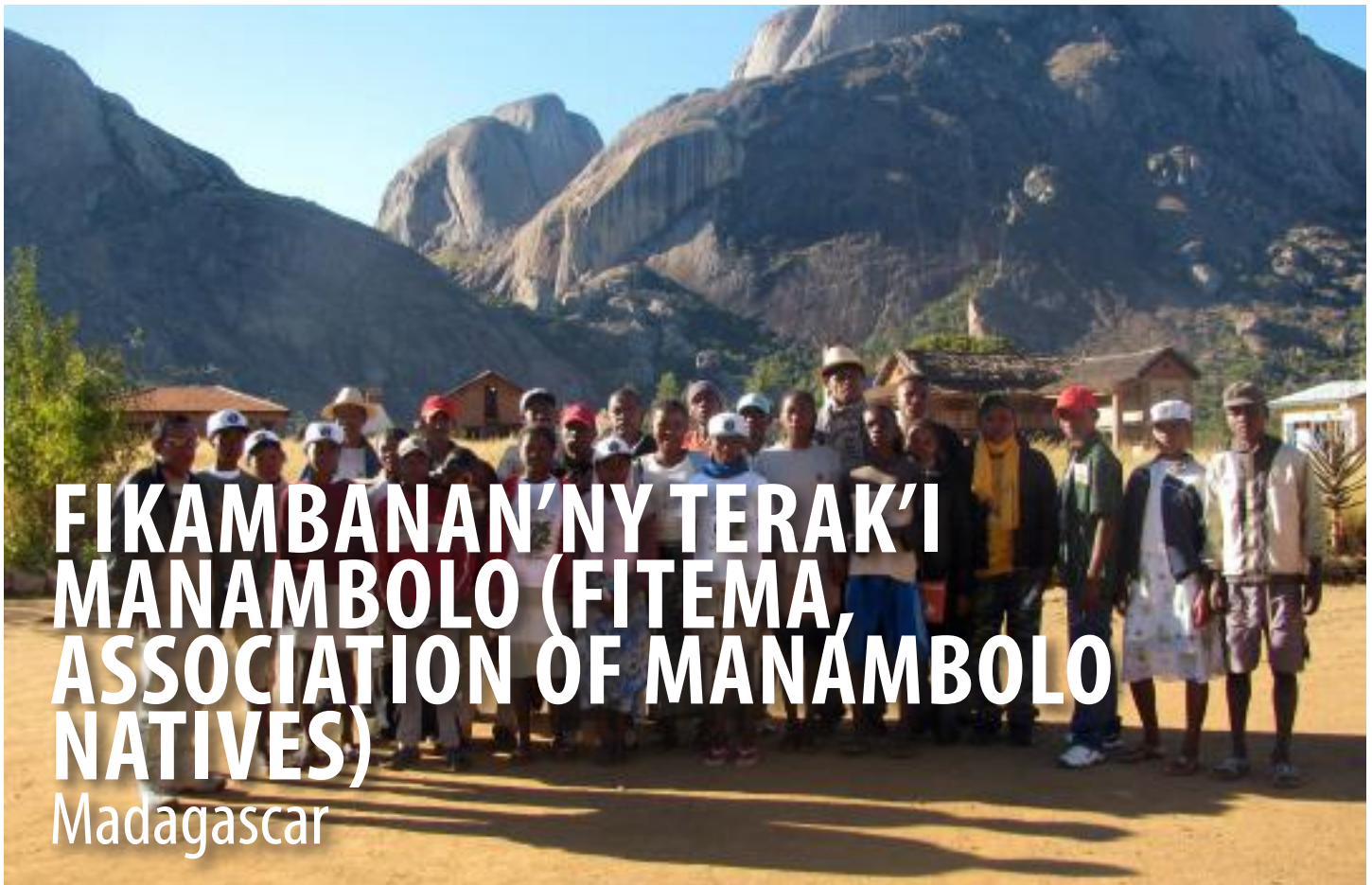
MOPAWI also partnered with the Tawahka community and other environmental conservation activists to protest construction of a dam which was to take place in the heart of the Mesoamerican Biological Corridor. As a result of this lobbying and the delivery of arguments against the economic and ecological feasibility of the dam, construction companies abandoned their plans. MOPAWI is also an active participant on the National Environmental Council, which advocates for sustainable practices and policies throughout the region.

## SUSTAINABILITY AND REPLICATION

MOPAWI has promoted community-to-community knowledge exchanges and site visits to share its experiences with groups such as the Pech and Miskito Peoples in the Río Plátano World Heritage Site, the Capapan Basin Mestizo People in the Río Patuca National Park, and the Miskito People in the Río Kruta Basin. The organization holds workshops on diverse subjects ranging from cacao production, agroforestry, alley cropping for sustainable food production, ecotourism, community mobilization and organizing, and collective land rights advocacy.

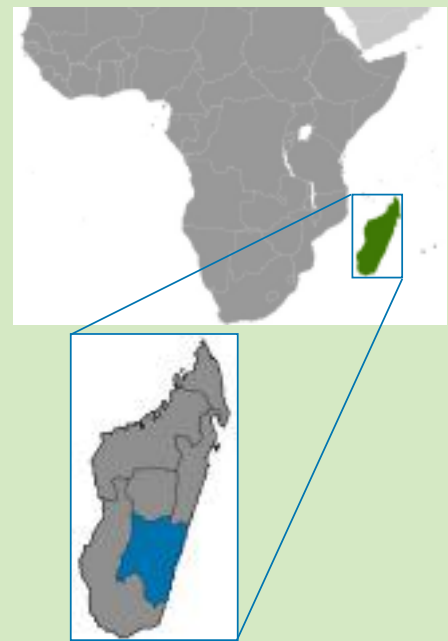
## PARTNERS

- **WWF:** Financial and technical assistance, capacity building
- **The Nature Conservancy:** Financial and technical assistance, trainings, protected area administration
- **US Department of Interior:** Financial and technical assistance, development of Plátano Reserve (turtle conservation, ecotourism)
- **The National University of Honduras:** Collaborative research on nature conservation and biodiversity
- **Ford Foundation:** Support for land titling
- **Tearfund:** Support for livelihoods, disaster risk prevention and climate change
- **Lutheran World Relief:** Support for conservation and sustainable development
- **IUCN:** Deforestation monitoring
- **The Ojon Corp – Estee Lauder Company:** Support for non-timber forest product commercialization (Social Corporation Responsibility)
- **The Methodist Church of Ireland:** Water and sanitation
- **Rainforest Alliance:** Support for forest certification (under the Forest Stewardship Council)



## PROJECT SUMMARY

Fikambanan'ny Terak'i Manambolo (FITEMA) has used the reintroduction of an indigenous land use system to help conserve forests and wetlands in the 7,500 ha Manambolo Valley – a forest corridor which joins the Andringitra and Ranomafana National Parks – while improving food security for local communities. The valley's forests are home to a high number of endemic species and also provide critical ecosystem services to the approximately 200,000 residents of five neighboring districts, including timber and non-timber forest products, water regulation, and watershed protection. The organization works on forest restoration through the establishment of nurseries with local tree species, including the native ravena palm (*Ravena madagascariensis*). The group has also constructed irrigation infrastructure and is guided in its work by a commitment to the full participation of its target communities.





## BACKGROUND AND CONTEXT

### *A hotspot for biodiversity and endemic species*

Madagascar is a micro-continent which is home to thousands of flora and fauna species, approximately 80 percent of which are unique to the island. The Malagasy moist forest eco-region is made up of forest ecosystems ranging from sea level to 2,600 m in altitude; the flora of this ecoregion comprises five of the six families endemic to Madagascar and 97 out of 209 known genera. The fauna is composed of ancient and diversified lineages, including at least one quarter of all primates in the world.

The Manambolo Valley is located between Andringitra and Ranomafana National Parks. This 7,500 ha forest corridor provides an unbroken chain of intact rainforest rich in endemic plants such as tree ferns (*Cyathea* spp.), and ebony wood (*Dalbergia* spp. and *Diospyros* spp.), as well as animals including flying mammals, birds, and lemurs. Due to deforestation, the majority of the remaining forests are located on the peaks. The lives of nearly 200,000 people depend on the sustainable management of the forests and other resources in this corridor. Most of the people living in the area are subsistence farmers who grow rice in the lowlands and food crops in the hills.

### *The dina system*

Prior to the colonization of Madagascar by the French, the use of natural resources in the Manambolo Valley was governed through the traditional *dina* system. The *dina* is a system of rules and regulations used in isolated regions of Madagascar to guide and control resource use and community behavior. These rules were traditionally passed orally from generation to generation. By establishing the Department of Water and Forests, the French colonial administration effectively replaced the *dina* system, leaving village elders no legal avenue to manage local resources that had traditionally been under their control, triggering overexploitation of forests by villagers and outsiders. In the mid-1990s, villagers approached the Worldwide Fund for Nature (WWF) seeking assistance, leading to the creation of Fikambanan'ny Terak'i Manambolo (FITEMA). WWF project personnel (who were also living in the local community) acted as intermediaries to re-open communication between village residents and the Department of Water and Forests. The project began helping local residents to re-establish and legalize the *dina* system in order to regain control

of their resources through traditional management methods, whereby elders make decisions after consulting their ancestors. This has been a crucial component in project success because Madagascar is a country where ancestors are highly venerated.

### *Linking the National Parks*

The main pressures on the region's forests are slash-and-burn agriculture, grazing, forest fires, and illegal timber and precious stone operations. These pressures are exacerbated by inadequate technical supervision by the Forest Service and subsistence harvest of forest resources. These environmental problems are widespread throughout the country and, among other things, have prompted the national government to recognize the need to provide continuity between existing protected areas.

In 2003 Madagascar announced plans to triple the country's protected areas from two million to six million hectares. WWF supported the government in identifying suitable locations for these new protected areas, which included the forest corridor between Ranomafana and Andringitra National Parks. WWF was initially responsible for managing the park, but later transferred the management of this area to Madagascar National Parks.

### *Reestablishing a traditional management system*

Malagasy law provides a legal framework for the transfer of all renewable natural resources to communities who meet conditions set out by law. WWF worked with FITEMA and government agencies to make the *dina* system legally binding. Approval was sought from relevant local authorities, finances were secured, social structures were created, and the contract was approved by the state. 1,000 ha of government forest land have now been legally transferred to local residents to manage independently, through the signing of GELOSE (Gestion Locale Sécurisée) conventions with the Department of Water and Forests. The *dina* system is now enshrined in the laws of the region and government regulations are imposed only if the *dina* is incapable of resolving conflicts.

Legal recognition of the *dina* system has given communities the power to effectively control local resources. The *dina* assures sustainability by respecting key functions of land management including timing, quantity, frequency, and rights of usage. It controls virtually all forest products including

honey, wood, eels, and crayfish. Outsiders can no longer use the forest's resources unless authorized by village elders, and a system for equitable sharing of resources within the valley has been implemented. Under the *dina* system, Manambolo residents now have a common vision for the use of the forest. Instead of being simply viewed as a source of first-come-first-serve raw material, the forest is now regarded as local heritage, to be managed for sustainable long-term use.

FITEMA's mission is to ensure the transfer of resource management to local communities, to educate local people about the importance of biodiversity, and to actively involve communities in its protection. In the beginning, the association focused on the application of the management plan (as required for resource rights transfer by the Forest Service) and establishment of an ecological restoration program. Since achieving their initial objectives, FITEMA has implemented accompanying measures meant to ensure food security in local communities. In this way, FITEMA aims to reduce the dependence of target communities on forest products through

the adoption of alternative livelihoods that provide economic benefits. Projects are implemented to increase farmers' agricultural production while discouraging detrimental practices such as slash-and-burn agriculture. Giving the communities a sense of responsibility over their natural resources has been a central goal of FITEMA. The beneficiaries of the initiative include twelve communities, or 1,300 households with an average of six persons per household, for a total of 32,000 targeted beneficiaries.

## KEY ACTIVITIES AND INNOVATIONS

FITEMA's twelve community-based associations are engaged in forest management based on a simplified management plan adhering to the *dina* system. Under this authority, 18,809 ha of the Andringitra-Ranomafana forest (20 percent of the corridor forest) is currently managed by local communities who are trained to sustainably use the land while increasing crop yields. Community management involves monitoring and patrolling the forest area by villagers independently or in collaboration with the Forest Service.







### *Food security*

FITEMA is increasing agricultural productivity by improving existing farming practices and diversifying crops. Traditionally, local farmers use slash-and-burn techniques in natural forests because their fields are either insufficient in size or unproductive due to degraded soil quality. FITEMA conducts trainings to encourage farmers to use their land more intensively by rotating crops and by restoring soil quality with cattle manure as fertilizer. FITEMA trains farmers in the use of no-till farming techniques and cover crops to control weeds, improve soil fertility, and retain soil moisture. Trained farmers pass their knowledge onto neighbor farmers. This system has been so successful that it is estimated that more than 85 percent of local non-members have adopted the improved methods through 'transversal diffusion', a process wherein people copy their neighbors' farming techniques.

The project also supports the management of 8.7 km of irrigation micro-infrastructure for the production

of rice, the main staple food of Madagascar. FITEMA also encourages the production of citrus crops and market gardening. Non-agricultural sources of income such as fish farming, artisanal products, and weaving are targeted for livelihood diversification, with the aim of reducing poverty and increasing economic opportunity.

### *Innovative responses to deforestation*

FITEMA's success in forest restoration can be traced to the method they developed using wild stock. Instead of relying on nurseries, seedlings are collected in the wild and replanted in areas where reforestation is desired. The success rate of this restoration method is above 80 percent, far higher than seed germination in nurseries.

FITEMA also encourages the planting of ravena, a palm species whose trunk begins to develop after 12 years and is usable only after 50 to 60 years. Farmers generally do not wait 50 to 60 years to use a resource, but in the case of ravena people have agreed to



plant it knowing that younger generations will reap the benefits. The reason this species is targeted is because its fibrous trunk can be used for a variety of purposes, including the construction of granary walls to protect grains since rats are unable to pierce its fibers.

## BIODIVERSITY IMPACTS

Through community-based management and monitoring of the forest, external exploitation of local resources, deforestation, forest fires, and erosion have been virtually eliminated and farmers have stopped using slash-and-burn farming techniques altogether. FITEMA's initiatives are protecting forest and water resources, including flagship species such as tree ferns, ebony, and lemurs. In the past, lemurs were rarely seen, but farmers now regularly report sightings of groups of five to seven lemurs at a time.

Nurseries are a key part of ecological restoration activities, providing fuel wood (16,000 plants), fruit trees (2,000 plants), and native plants for reintroduction to natural forests (45,000 plants). More than 16,000 seedlings of the locally valuable *ravenea* have been planted since 1998.

Participatory, multi-media (meetings, posters, and films) environmental education has raised awareness of environmental issues (e.g., slash-and-burn, lemur poaching, overharvesting of palm trees, and forest laws) within the vicinity of the forest corridor. Primary and secondary school students play a crucial role in the monitoring of lemurs and other species. After receiving lessons on the biology and ecology of lemurs, students take periodic trips to the forest to observe and record changes in lemur populations. The student environmental club boasts 491 members who are actively engaged in spreading awareness of the importance of conservation for forest-based livelihoods.

## SOCIOECONOMIC IMPACTS

Before the initiative began, farmers planted traditional crops of rice, cassava, beans, Madagascar peanut, and sweet potato for consumption. As a result of FITEMA's projects, local farmers have more than doubled their agricultural output, enabling them to sell a large percentage of their harvest. Improvements to irrigation micro-infrastructure have allowed almost 2,800 villagers in Ambalahosy to benefit from the

availability of water throughout the year and to refrain from slash-and-burn techniques that destroy the forest. The irrigation system waters 115 ha of rice fields, enabling villagers to grow enough food to feed the community for eleven months of the year (versus seven months previously).

### *Developing alternatives to deforestation*

FITEMA promotes income diversification through the introduction of non-traditional crops such as citrus and potatoes, and the establishment of fish farming and beekeeping activities. FITEMA provides technical, material, and marketing assistance to villagers; for example, in the case of beekeeping, FITEMA distributes new, improved hives and trains villagers in modern extraction techniques to increase honey output. Training is offered to select groups who agree to share their knowledge with others, with priority being given to single mothers and the elderly. FITEMA also encourages farmers to plant off-season crops (e.g. potatoes, beans, and cabbage) in fallow rice fields and to participate in market gardening as a way of adapting to climate change and improving soil fertility. Along with improved rice cultivation techniques, the planting of off-season crops has successfully reduced the lean period from seven months to three months.



### Collective action and mutual aid

FITEMA has improved food security by building community granaries in every village, providing all communities members with a facility in which to store their harvested crops. Usage rules are decided on as a group, by the members who constructed the granary. Sometimes, members buy grains to stock and later resell to members (especially during the lean season) at a lower price than asked for in markets or by dealers. The construction of granaries has also contributed to compliance with environmental regulations. Villagers who had previously been dependent on the forest for their daily needs now only visit the forest from time to time to collect fruits and mushrooms.

FITEMA has also been able to build two classrooms for the public elementary school and supply drinking water to the village (about 1,400 people). The latter helps to reduce child mortality and to improve maternal as well as general health. In addition, WWF provides support to address illiteracy, and to build local capacity in accounting and record keeping.

### Empowering women

FITEMA has emphasized the participation of women within the association. Between 2008 and 2012, the number of female members doubled. The large number of women within the association has improved their ability to form social and cooperative groups, leading to an increased sense of empowerment and improved incomes.

## POLICY IMPACTS

When the protected area was established, the FITEMA communities were actively involved in influencing changes in the boundaries of the Ranomafana-Andringitra forest corridor through public consultation, successfully lobbying to ensure that pre-existing management plans be taken into consideration. For example, zones where communities already held the right to use resources were maintained and not converted into conservation zones. In 2009 FITEMA obtained a renewal of their contract to manage the corridor forests for another ten years from the Forest Service. Two representatives of FITEMA are currently members of the Federation of Community-Based Organizations, an advisory body for decision-making regarding the management of the forest corridor between Ranomafana and Andringitra parks.

## SUSTAINABILITY AND REPLICATION

Donors to the project have often made short-term investments, creating relationship breakdowns between WWF and local communities. As a result, the project has been limited in its ability to consistently monitor changes in deforestation over time. Despite these challenges, improvements in incomes and increased food security have positively changed people's perceptions about conservation, and on the few occasions when there was a breakdown in project activities, the momentum to conserve the forest was maintained by the local people.

WWF Madagascar works with a wide range of buffer communities within the rainforest region and has been able to replicate the FITEMA model in eight other sites, bringing the total number of people reached to around 100,000. FITEMA frequently organizes farmer-to-farmer knowledge exchanges, an activity that allows communities to learn new techniques and approaches to conserve biodiversity and reduce poverty. Since 2002, twelve new grassroots community organizations have been formed in the Ranomafana-Andringitra forest corridor. In one of these cases, a neighboring community was able to receive funding for the establishment of a tourist route, thanks to FITEMA's success.

## PARTNERS

- **WWF Madagascar, WWF Switzerland, WWF Germany, WWF Sweden, WWF France, WWF US & WWF International:** Technical and financial support
- **Ministry of Environment and Forests:** Management transfer of natural resources, material and technical support
- **Regional government:** Support for regional development planning
- **Multilocal Planning Committee:** Technical assistance
- **USAID:** Technical assistance
- **Ecological Training Program:** Biological and ecological inventories





# ASOCIACIÓN PARA LA INVESTIGACIÓN Y EL DESARROLLO INTEGRAL (AIDER, ASSOCIATION FOR RESEARCH AND INTEGRATED DEVELOPMENT)

Peru

## PROJECT SUMMARY

Asociación para la Investigación y el Desarrollo Integral (AIDER) is a participatory initiative that provides capacity building and technology transfer to enable community-based conservation of forest resources across Peru. By providing technical assistance to forest-based communities in both the humid tropical forests of central eastern Peru and the tropical dry forest in the country's northern coastal region, the initiative has enhanced local capacity to improve livelihoods, protect the environment, and stop deforestation. Much of this work has benefitted indigenous communities, helping them to secure autonomous control of their resources.





## BACKGROUND AND CONTEXT

Asociación para la Investigación y el Desarrollo Integral (AIDER) works with indigenous and rural communities to implement projects that conserve the natural biological diversity of Peru's forested regions while improving community livelihoods and well-being. AIDER serves seven regions of Peru through its five regional headquarters, with a staff of more than 90 forest science professionals, technicians, administrative officials, and volunteers.

### *Facing a diversity of challenges to forest ecosystems*

AIDER takes a multidisciplinary approach to assisting communities living in diverse forest types in Peru. The organization works to protect tropical dry forests that are crucial to the livelihoods of farm families on the northern coast, but are under threat from livestock rearing, agriculture, clearing for firewood and charcoal, and desertification. To counter these threats, AIDER provides training in forest conservation, rotational grazing, and agriculture, enabling households to maintain incomes while incorporating activities such as reforestation.

In Ucayali and Madre de Dios, the challenges confronting forest-dwelling communities are different. The regions are home to approximately 12 million ha of tropical rainforests which are relatively

well-maintained, but under threat from illegal logging and gold mining, poaching, and conversion to agriculture. AIDER develops technical and managerial capacities of the forests' indigenous communities to produce non-timber forest products using low-impact methods, and establishes alliances to market their products. AIDER also helps communities to take advantage of voluntary forest certification schemes, supporting rural producers in meeting Forest Stewardship Council (FSC) standards for sustainable forest management, as well as fair trade standards.

### *Recognizing the role of forests in carbon capture*

As part of the organization's work to diversify the use of forest goods and services, AIDER promotes forest carbon trading in protected areas, private lands, and forest concessions. This strategy incorporates reforestation projects for carbon sequestration and REDD+ markets.

### *Equitably managing protected areas*

AIDER has substantial experience in helping to manage Peru's protected areas. Typically, the costs associated with managing protected areas (e.g., monitoring and research) are not fully covered by public resources or international donors, and are unsustainable. AIDER therefore advocates for the participation of civil society in the management of



protected areas, working with partners to sustainably manage several Peruvian protected areas and ensure that benefits are shared with communities.

### *Rehabilitating degraded ecosystems*

Deforestation caused by farming, mining, and infrastructure development has resulted in the loss of forest soil productivity in coastal, highland, and jungle areas of Peru, to the extent that these lands cannot recover naturally. AIDER has demonstrated the potential benefits that recovery of these degraded areas can have for conservation by planting native species that enhance soil organic matter. AIDER has also proposed afforestation for timber production and environmental services as an alternative use of these areas.

### *Utilizing secondary forests for local communities*

Secondary forests represent a significant percentage of Peru's forest estate and are increasingly present on Amazonian farms. They have economic potential for rural families and can contribute to forest conservation objectives. AIDER promotes agroforestry practices that increase the productivity of soils under cultivation as well as economic activities that use secondary forests, such as beekeeping. The organization also encourages research into timber species that generate the highest economic value for secondary forests.

### *Integrated forest research*

AIDER's mission is to "contribute to improving the quality of life of Peru's rural population in harmony with environmental conservation, through sustainable approaches that are based on applied research, policy action, the recovery of local knowledge, and establishing interinstitutional synergies." Toward this end, AIDER engages universities and research centers in research on technical and social issues that contribute to solving complex problems related to sustainable development and resource management. The organization also actively supports graduate student theses on topics related to its fields of work.

## KEY ACTIVITIES AND INNOVATIONS

In 2011, AIDER implemented eighteen different projects across its five headquarters, including FSC-certified sustainable timber harvesting, preparing forest-based communities for REDD+, and using native tree species to reforest and recover areas being affected by desertification.

### *Madre de Dios*

In the Madre de Dios region, several projects are being implemented within and around Tambopata National Reserve and Bahuaja-Sonene National Park. These include: methods for biological monitoring funded by Payments for Environmental Services (PES); strengthening participatory management of resource use regulations through the use of community policing; management of a revolving fund for local producers accessing the Nuevo América 'special use zone' of Tambopata National Reserve; and strengthening the capacity of the Madre de Dios Bureau of Environmental Services and REDD+. AIDER is also supporting a REDD+ project in the Alto Mayo Protection Forest of San Martín, a northern region of the Peruvian Amazon.

### *Ucayali*

In Ucayali, AIDER works with forest-dwelling native communities of the Shipibo-Conibo people. Much of this work focuses on developing sustainable forest-based livelihoods. This has entailed identifying market opportunities, developing value chains, and obtaining certification for sustainable timber harvesting and craft-making.

### *Piura*

In the coastal region of Piura, AIDER's work emphasizes sustainable land management strategies to combat desertification of tropical dry forests. Three projects in 2011 addressed the issues of engaging civil society organizations in identifying funding for sustainable land management projects, including Clean Development Mechanism project financing; improving regional integration for combating desertification along the coasts of Peru and Ecuador; and improving the financial sustainability of Peru's northern protected areas.

### *Technological and financial innovations*

In 2011, AIDER installed a 43 m tower in Tambopata National Reserve for long-term monitoring of carbon dioxide and methane emissions. The tower allows researchers to measure the contribution of Tambopata to the global carbon cycle and was the first attempt to record continuous measurements of CO<sub>2</sub> in the Peruvian Amazon rainforest.

AIDER has achieved notable 'firsts' in Peru. Through AIDER support, the Shipibo-Conibo community became the first FSC certified forest (35,000 ha) in Peru in 2005. AIDER also supported Peru's first forestry project

registered under the UNFCCC's Clean Development Mechanism, and achieved the validation of the Campo Verde Reforestation Project as the first forest carbon project to meet Voluntary Carbon Standards. AIDER's Madre de Dios Amazon REDD+ Project was the first in Peru to meet Climate, Community and Biodiversity standards. These achievements illustrate AIDER's role in helping communities to take advantage of innovative sources of financing, enabling them to become financially sustainable.

## BIODIVERSITY IMPACTS

### *Conservation in the Peruvian Amazon*

In Ucayali, AIDER has worked with 14 indigenous Shipibo-Conibo communities to develop community-based forest management plans in the Amazonian rainforest. Productive forests in the region cover more than 8 million ha, of which over half have been declared permanent production forests. The remaining forests fall within the territories of various native communities, reserve areas, and private landholdings.

The total area under community-based conservation covers more than 150,000 ha of forest, a critical habitat for 21 species listed as endangered by the Peruvian government due to illegal hunting and harmful commercial activities. Hunting and logging are prohibited in community reserves, reducing human pressures on the forest ecosystem. Central to this work has been certification of sustainable timber harvesting under FSC standards. AIDER is also cataloguing the properties of lesser-known primary and secondary forest timber species and distributing fact sheets to local and international buyers in an attempt to relieve pressure on remaining primary forest areas.

### *Forest conservation through protected area management*

AIDER was involved in the preparation of the Tambopata National Reserve management plan for 2011–2016. The Tambopata National Reserve covers more than 550,000 ha of pristine tropical habitat, hosts exceptional biodiversity, and is home to thousands of plant and butterfly species, and hundreds of species of birds, mammals, and reptiles. AIDER plays a lead role in the conservation and management of the park, coordinating research into its biodiversity and carbon-carrying capacity. Community-led enforcement of forest use regulations

ensures the ecological integrity of the reserve. AIDER secured innovative sources of financing for forest management in the reserve, which is one of two REDD+ project sites in Peruvian protected areas.

In San Martín, AIDER has helped the Alto Mayo Protection Forest to benefit from REDD+ financing. This protected area covers 182,000 ha of forestland along the upper Mayo River basin, protecting the Ucayali moist forest and Peruvian Yungas eco-regions, the critically endangered yellow-tailed woolly monkey (*Oreonax flavicauda*), and numerous species of orchid. AIDER has coordinated baseline studies to quantify the forest's carbon stocks, model existing deforestation, and develop a monitoring plan for the protected area.

## SOCIOECONOMIC IMPACTS

### *Micro-enterprise development in Ucayali*

In Ucayali, the population is comprised of immigrants from coastal regions and the highlands. In 2007, the population numbered 432,159, with a density of 4.3 inhabitants per km<sup>2</sup>, and 54 percent living below the poverty line. The department is home to almost 300 communities of the ethnically and linguistically diverse Arawak and Pano families. These communities have preserved their traditional rituals, beliefs, dance, music, language, clothing, livelihoods, medicines, and craftsmanship. Projects in Ucayali include the creation of 20 community-based enterprises, directly benefiting 300 families through the sale of sustainably harvested wood and non-timber forest products, including fruits and handicrafts.

### *Community-managed forest concessions in Madre de Dios*

In the Madre de Dios region, there are 26 native communities that belong to six ethnic groups, occupying a total of 414,202 ha of forestland. The region contains approximately 485,000 ha of mature Brazil nut (*Bertholletia excelsa*) forests. Peruvian forest laws grant individual concessions to communities to gather Brazil nuts for up to 40 years. Between 15,000 and 20,000 people benefit directly or indirectly from the collection, processing, and export of Brazil nuts.

AIDER has developed alternative livelihood options for the region's inhabitants, including introduction of low-impact artisanal gold mining. Gold is found in the Madre de Dios region in alluvial deposits. Although





the state has issued grants totaling almost 623,000 ha in titled mining concessions and pending mining leases, this activity is also being conducted illegally, causing widespread environmental damage.

## POLICY IMPACTS

### *Advocating for community land tenure*

In the forest sector, indigenous and local community tenure rights are governed by forestry and wildlife legislation. The Forestry and Wildlife Law of 2000 and its associated National Strategy for Forest Development aimed to facilitate sustainable production of forest products through establishment of permanent forest concessions. The law created tension between communities, indigenous peoples, and the state, due to local claims of inadequate consultation and favoritism toward commercial timber interests over community needs and concerns.

In the late 1990s, under the Ucayali Titling and Communal Reserve Project, indigenous communities in Peru worked with national and international NGOs such as AIDER to find novel ways to secure indigenous territories within the constraints of national

legislation. Using a combination of the Law of Native Communities and the existing national Forestry Law, these groups obtained land titles for 209 indigenous communities over 2.5 million ha and established the basis for access rights over 7.5 million ha of forest reserves. In Madre de Dios, indigenous people succeeded in excluding concessions from their titled indigenous lands and the territories of indigenous people in voluntary isolation.

In 2005, AIDER and partners coordinated six regional workshops on community and indigenous natural resource management across the country to develop recommendations for more effective community forest management policy in Peru. AIDER continues to play an important role in developing local awareness and capacity to adapt to changes in forest and indigenous rights legislation. In 2011, indigenous organizations from San Martín and Madre de Dios agreed not to sign REDD+ contracts until indigenous peoples' and local communities' rights are guaranteed, and the nature of REDD+ projects are clearly defined at national and international levels.

## PARTNERS

### *Government partners*

- **Regional Government of Madre de Dios:** Consultation
- **Ministry of Production:** Consultation
- **National Institute of Nature Resources (INRENA):** Funding for regional workshops
- **National Service of Protected Natural Areas (SERNANP):** Management of the Tambopata area
- **Peruvian Trust Fund for National Parks and Protected Areas (PROFONANPE):** Protected areas in northern coastal Peru

### *Regional NGOs and indigenous peoples' organizations*

- **Interethnic Association for the Development of the Peruvian Rainforest:** Consultation
- **The Regional Development Organization of Masisa:** Consultation
- **Asociación Cáritas Madre de Dios:** Low-impact artisanal mining in Madre de Dios

- **Fauna Forever:** Research and training, Tambopata area
- **Consortium for Sustainable Development in the Andean Ecoregion (CONDESAN):** Consultation
- **The Federation of Native Communities of the Iparia District (FECONADIP):** Consultation

#### Research institutes

- **Asociación Nacional de Centros de Investigación, Promoción Social y Desarrollo (ANC):** Consultation
- **National Forestry Association:** Consultation
- **Institute for the Investigation of the Peruvian Amazon:** Consultation
- **Organization for Tropical Research:** Consultation
- **Catholic University of Peru:** Research and monitoring in Tambopata National Reserve and Bahuaja-Sonene National Park
- **National University of the Madre de Dios Amazon:** Consultation
- **National University of Ucayali:** Research on lesser-known timber species
- **National Intercultural University of the Amazon:** Consultation
- **Andean Center for Rural Technology:** Consultation
- **University of Edinburgh:** Carbon dioxide monitoring project

#### Private partners

- **Bosques Amazónicos S.A.C.:** Research and management in Tambopata National Reserve and Bahuaja-Sonene National Park, reforestation in Campo Verde

#### International NGOs and donor organizations

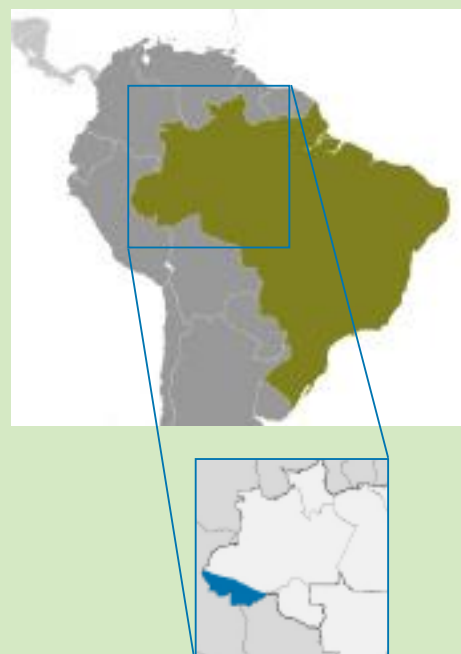
- **Agreement on the Conservation of Tropical Forests (ACBT):** Consultation
- **Belgian Technical Cooperation/Groenhart:** Business model for certified timber in the Amazon
- **Center for International Forestry Research:** Funding for regional workshops
- **Conservation International:** Forest baseline studies, monitoring plans, forest modeling, and REDD+ in San Martin
- **Critical Ecosystem Partnership Fund:** REDD+ in Madre de Dios region
- **European Union:** Sustainable environmental management and anti-desertification in Piura
- **Flemish Fund for Tropical Forests:** Sustainable forestry in Shipibo-Conibo
- **Fund for the Americas:** Management in Tambopata and Bahuaja-Sonene, sustainable production systems in buffer zone of Tambopata reserve, timber certification in Ucayali, timber management in Shipibo-Conibo
- **German Society for International Cooperation:** Restoration of lands degraded by gold mining, funding for regional workshops
- **Global Mechanism:** Capacity building in Piura
- **International Tropical Timber Organization:** Sustainable forest management in Madre de Dios, research on lesser-known timber species
- **Italo-Peruvian Fund:** Craft making in Shipibo-Conibo communities
- **SNV (The Netherlands):** Funding for regional workshops
- **WWF:** Funding for regional workshops



## PROJECT SUMMARY

Associação Sociocultural Yawanawá is a representative body of the Yawanawá indigenous people of Acre state in northern Brazil. The group works to create income-generating opportunities for its members through the conservation of the community's indigenous territory and the promotion of Yawanawá culture. Through the sustainable extraction of native agricultural products like urucum, a local plant that produces a red dye, and a partnership with an international cosmetics firm, the initiative has been able to generate revenue to invest in local infrastructure.

The community secured the revision of the boundaries of the Yawanawá's indigenous land, extending their legal control over 187,400 ha of Amazonian forest. This achievement, coupled with the initiative's innovative use of their traditional culture – for instance, through a Yawanawá clothing brand – has made the group a model for indigenous sustainable forest management in Brazil.





## BACKGROUND AND CONTEXT

### *A history of the Yawanawá in Rio Gregório*

Since the sixteenth century, the indigenous Yawanawá have traditionally worked as farmers and rubber tappers. After a sharp decline in rubber sales in 1992 due to competition from Malaysian rubber plantations, the Yawanawá lost their primary source of income. Around the same time, the state of Acre began construction of the BR-364 highway, which passed through land held by the Yawanawá in Rio Gregório. This land had first been identified in 1977, and in 1984 was officially demarcated as an area for preservation. The demarcation was based on inaccurate boundaries, however, and did not include mangroves, river sources, sacred burial grounds, lakes, and hunting sites that were of enormous cultural and socioeconomic importance to the Yawanawá.

The highlight of the organization's work has been the revision of the boundaries of their indigenous land. The association collected petitions signed by the majority of state congress and senate members, and sent a committee to Brasília to request a revision of the limits of the Yawanawá territory from the National Indian Foundation (FUNAI, the national government agency for indigenous Indian issues) and the Ministry of Justice. Anthropologists from FUNAI conducted an investigation and awarded more than 95,000 additional hectares to the Yawanawá and Katukina peoples. The new territory was approved by the Ministry of Justice in April, 2007 and was officially enacted by presidential decree in the same year. Today, the Yawanawá territory covers 187,400 ha. The

Yawanawá tribe was the first indigenous community in Brazil to successfully lobby for the revision of their recognized boundaries, a significant achievement and a major contribution to the discussion on indigenous land rights in Brazil.

### *Partnership with Aveda*

The Yawanawá initiated a commercial partnership with the North American cosmetics firm Aveda Corporation in 1993. In this, the group was initially organized as one united community under a unified leadership, overcoming a history of internal divisions and external pressures. This led to the formation of a cooperative organization, Associação Sociocultural Yawanawá, in 2003. The group oversees the communal collection and sale of urucum seeds to Aveda, and coordinates local development projects. In 2003, the Yawanawá signed three contracts with Aveda: one for the sale of urucum seeds, one for the use of the Yawanawá image, and another to support social projects within the Yawanawá community. This partnership is direct, transparent, and does not involve intermediaries.

## KEY ACTIVITIES AND INNOVATIONS

### *A win-win partnership for sustainability and well-being*

Support from Aveda allowed the Yawanawá to build a new village, named Nova Esperança ('New Hope'). This work initially targeted around 220 people. Currently, Associação Sociocultural Yawanawá encompasses six villages, with a total of nearly 750 beneficiaries





All profits arising from the sale of locally-harvested natural produce are reinvested for the welfare of the group. The Yawanawá image has been used in branding various goods that promote the group's rich cultural heritage. These products are sold at the annual Yawa Festival, and include CDs, DVDs, clothing, ceramics, and jewelry.

Revenues from the sale of urucum are no longer sufficient to meet the needs of the expanded Yawanawá target population, meaning that the social projects component of the partnership constitutes the majority of the benefit for the Yawanawá community. These projects have focused on infrastructure investments for the villages.

## BIODIVERSITY IMPACTS

The Yawanawá's struggle to preserve their indigenous territory has been successful in the face of numerous pressures and land claims. One such challenge came in 2005 when a prominent Sao Paulo-based television presenter purchased 200,000 ha of land bordering Yawanawá territory, with the intention of felling the forest and converting the land to cattle pasture. The

Yawanawá resisted the land sale, claiming 50,000 ha of this land as their indigenous territory. FUNAI vindicated the community's claim, leading to the inclusion of the land within the newly-demarcated Yawanawá territory in 2007. More recently, the same company's claim to the remaining 150,000 ha was successfully delayed after the Yawanawá petitioned the state on the grounds that an environmental impact assessment had not been carried out by the logging company.

Challenges of this kind are indicative of the pressures not only on the Yawanawá's land, but on surrounding territories and other indigenous peoples. Continued advocacy and campaigning on the part of the Yawanawá people and their international partners is protecting forest resources from such threats and is preserving the rich cultural heritage and biodiversity of the Brazilian Amazon.

### *Conservation of Amazonian species and habitats*

The biodiversity benefits of the Yawanawá's work are substantial. As the largest tract of tropical rainforest in the world, the Amazon rainforest is home to one in ten known species. The Amazon provides habitat



for around 2.5 million insect species, 40,000 plant species, nearly 1,300 bird species, over 800 species of amphibians and reptiles, and more than 400 mammals.

As the Yawanawá community has grown and become more concentrated in villages, the impact on game species in the surrounding forests and riverine habitats has increased. To reduce their impact on wildlife species, the Yawanawá introduced an animal breeding project in 2004. The community now runs a captive breeding program for alligators, fresh water turtles, wild boar, and fish, with specimens reintroduced into the wild as well as harvested for consumption.

## SOCIOECONOMIC IMPACTS

The community's partnership with Aveda has helped the Yawanawá to enhance food security and healthcare, improve education, and invest in local infrastructure.

### *Food security*

The community has worked to develop intercropping of edible plant species around village houses to improve food security. Beginning in 1992, an initial 6.5 ha of land were planted with urucum and other tropical forest products such as cassava, mango,

banana, pineapple, chestnut, pupunheira, guaraná, as well as nitrogen-rich Ingá plants and fast-growing tree species for timber. In recent years, these efforts have been supplemented by an animal husbandry program. Enclosures for pigs and a rotational dairy farm are currently under construction.

### *Health*

Health interventions have taken the form of stocking a local health clinic with the facilities to treat malaria cases, and improving sanitation in villages. 14 sanitation facilities were installed in six locations in the Yawanawá community to address increasingly common cases of diarrhea and guinea worm in infants.

### *Education*

Since 1984, more than 20 young members of the Yawanawá community have received educational training from the Pro-Indigenous Commission of Acre in partnership with the State Secretariat for Indigenous Education. These teachers work in four local schools, teaching more than 200 children. A school-parent association was mobilized to secure governmental funds for the construction and renovation of school buildings. With support from Aveda, four Yawanawá youths have been sponsored to study in state universities. The youths commit themselves to returning to their communities upon graduation.







### Community development investments

Infrastructure investments include the purchase of machinery to process urucum seeds and the construction of storage facilities. In 1999, with backing from Aveda and the state government, the Yawanawá initiated a project to produce andiroba oil. Andiroba (*Carapa guianensis*) is a tree in the mahogany family and oil obtained from its seeds is widely used as an insect repellent and as a remedy for cuts and irritated skin. By 2004, the community had prepared a business plan for the construction of an andiroba processing refinery. Because andiroba does not grow on Yawanawá lands, the seeds had to be purchased from outside the community. A lack of funding has halted the development of this enterprise, although transforming the andiroba refinery in Tarauacá into a center for processing 'oils and essences of the Amazon' remains a goal for the Yawanawá.

### Investments in sustainable energy

More recently, the Yawanawá community has attempted to generate locally-produced biogas to fuel the boats that connect their villages with the outside world. This has long been the most

important mode of transport for the community, with the introduction of on-board motors cutting travel time to the nearest town down to nine hours. The economic and environmental costs of using diesel fuel are considerable; the community spends approximately US\$9,500 per month on fuel. To overcome this problem, the Yawanawá have begun harvesting and processing the seeds of jatropha plants (*Jatropha curcas*) into biofuel. The project was a winner of the 2009 IDEAS Energy Innovation Contest for improving energy efficiency and expanding access to renewable energy.

### Improvements in communications technology

The community has facilitated contact with the outside world by opening an office in the city of Tarauacá and by providing Internet access in Nova Esperança.

### Celebration of Yawanawá cultural heritage

In partnership with Aveda, the community has promoted the Yawanawá brand within Acre and across Brazil. One enterprise involved recording traditional folk songs sung during harvest celebrations. This was the first CD of indigenous folk

music recorded in the state of Acre. The group has also produced a DVD about the history of community, which has been translated into eight languages. Dissemination of the DVD has served to increase the visibility of the Yawanawá and to promote the Yawa Festival, an annual celebration of Yawanawá culture, history and music. The Yawa Festival helped to launch a fashion project called Kurã Kene. The sale of merchandise has increased appreciation of the Yawanawá culture and includes drawings based on traditional body paintings, ceramics, and jewelry. With support from the state of Acre, Rainforest Concern, and Aveda, the initiative launched a collection of clothes with traditional Yawanawá drawings and paintings in 2004, successfully reaching a wide urban audience.

### *Women's empowerment and social equity*

Associação Sociocultural Yawanawá has helped to devolve power within the Yawanawá community. From its starting point in the village of Nova Esperança, the population now extends to six separate villages, each with its own chief. This network has changed the social dynamics of the community. Associação Sociocultural Yawanawá has also encouraged, for the first time in the history of the Yawanawá people, the initiation of two women into the rigorous training and dieting necessary to become tribal chiefs. This work represents an important victory for Yawanawá women, given the history of resistance to and mistrust of this idea among men in the community.

## POLICY IMPACTS

The conservation work of the Yawanawá community has benefitted from close ties with the state government of Acre, which has gained a reputation for its forest-friendly policies. The group has gained widespread recognition for its initiatives. In 2005, the

Kurã Kene fashion project received an award from the Government of Acre for best initiative of the year for preservation and promotion of traditional indigenous knowledge. This award highlighted the role of Yawanawá women in particular, as they were the main actors in the production of the products. The group's DVD on Yawanawá culture was honored with the 2004 Chico Mendes Award, bestowed by the state of Acre to innovative community initiatives. In recognition of the Yawanawá's work for the empowerment of women, Raimunda Putani, one of the new tribal chiefs, recently received the Bertha Lutz Award in 2007, an important distinction offered by the Federal Senate.

## PARTNERS

- **Aveda Corporation:** Funding for construction of Nova Esperança, financial and technical assistance, support for key infrastructure, educational sponsorships
- **Fundação Nacional do Índio (FUNAI):** Support for land claims
- **Committee for Democratization of Information Technology:** Support for internet access
- **Rainforest Concern UK:** Support for land claims, sanitation project, captive breeding program and clothing line
- **Pro-Indigenous Commission for Acre:** Teacher training, support for biofuel project
- **Demonstrative Project of Indigenous Populations, Pilot Program for the Conservation of Tropical Forests in Brazil:** Support for the Yawa Festival
- **Global Village Energy Partnership (GVEP):** Support for biofuel project

# COMUNIDAD NATIVA ESE'EJA DE INFIERNO (ESE'EJA NATIVE COMMUNITY OF INFIERNO)

## Peru

### PROJECT SUMMARY

Comunidad Nativa Ese'eja de Infierno was the first community to take advantage of Peru's law of native communities in the state of Madre de Dios, receiving legal title to 9,558 ha of land. As a condition of defending their lands in the 1980s, the community was obliged to set aside 3,000 ha as a reserve where hunting, logging, forestry, and any other type of resource extraction was prohibited.

In partnership with a private sector company, the community jointly manages an ecotourism lodge. From 1997 to 2007, net revenues from the lodge totaled more than US\$250,000. Profits are equally divided among the community's 500 members; in 2000, the community set aside 25 percent for investment in education, enabling the construction and operation of the only rural secondary school in the region.





## BACKGROUND AND CONTEXT

Comunidad Nativa Ese'ejá de Infierno is an indigenous group in Peru whose ancestral homeland is located on the Tambopata River in the Madre de Dios region – one of the world's biodiversity hotspots. The region is characterized by low-lying Amazonian rainforest, and contains a number of endangered species including the giant river otter. As with many regions of the Amazon, land conversion for agriculture, illegal logging, deforestation, and habitat loss are ever present challenges.

### *Native Community of Infierno*

The formation of Comunidad Nativa Ese'ejá de Infierno dates to 1974, when the Peruvian government passed the Law of Native Communities which stipulated that all indigenous peoples in the Amazonian region are entitled to form communities, have their lands demarcated, and gain legal recognition of inalienable territorial rights. Towards this end, Ese'ejá joined with other Andean and

riparian inhabitants of the Infierno region to form the Comunidad Nativa Ese'ejá de Infierno. The process of mobilization and drafting a constitution took two years before the group was officially recognized as the first 'native community' in the state of Madre de Dios and given legal title to 9,558 ha of land on both sides of the Tambopata River.

### *Beginnings of community-based ecotourism*

During the 1990s, the Madre de Dios region saw a sharp increase in tourism, due in part to the creation of the Tambopata National Reserve. When demarcating this conservation area, a section of the Native Community of Infierno territory was mistakenly added to the reserve. The community successfully disputed this decision in 1987, winning back the land. A condition of the resolution process, however, was that the area be declared a reserve. The community set aside roughly 3,000 ha of land – nearly 30 percent of their territory – as a reserve where hunting, logging, forestry, and other types of resource extraction were prohibited. The community later decided to create a low-impact ecotourism venture in the area to benefit the Infierno community.

### *Partnership with Rainforest Expeditions*

In 1996, the community agreed to a contract with a private partner, Rainforest Expeditions. The agreement established a participatory ecotourism project including an ecolodge. A 20-year timeline was established to transfer management of the lodge to the community. In the interim, community members are supported to develop organizational and managerial skills to independently run the business. While the community retains full ownership of the lodge, Rainforest Expeditions receives a 40 percent profit share until 2016 to recover the costs of investment and capacity building.

## KEY ACTIVITIES AND INNOVATIONS

Comunidad Nativa Ese'ejá de Infierno numbers 500 people, 20 percent of whom are Ese'ejás, 21 percent Andean immigrants, 23 percent local immigrants, and 34 percent mestizos. Prior to the initiative, the community was poor and marginalized, surviving on subsistence agriculture, the collection of Brazil nuts, and small-scale hunter-gathering. An overdependence on these activities, and a lack of viable alternative livelihood options, created unsustainable pressure on local ecosystems.



### *Posada Amazonas*

In partnership with Rainforest Expeditions, the community now operates Posada Amazonas, a 30-bedroom ecolodge which is adjacent to the 770,000 ha Tambopata National Reserve. The lodge was built using traditional indigenous architectural designs and materials – locally harvested wood, palm fronds, and wild cane – as well as modern, low-impact technology.

The close proximity of the lodge to the Tambopata National Reserve provides guests with access to some of the Amazon's most biologically diverse rainforest. The goal of Posada Amazonas' activities is to protect biodiversity through low-impact, educational tourist activities. The community offers a four-day program of non-motorized raft visits to the natural habitats of endemic species. Guests at the lodge are further offered rainforest hikes, ethno-botanical walks, and farm visits. A 35 m scaffolding tower provides easy canopy access for viewing toucans, parrots, and macaws. In a nearby oxbow lake, a small population of endangered giant river otters – as well as caiman, hoatzin, and horned screamers – can be observed.

The ecolodge prioritizes use of local materials and products, as long as they are of equal quality and price to comparable items on the market. This gives the broader community a chance to benefit from a new market for their products. This same principle guides the gradual introduction of ecotourism, whereby local culture will become a source of pride and revenue generation.

The emphasis on benefit-sharing and joint-decision making between a private sector partner and the local community is a noteworthy innovation. When ownership of Posada Amazonas is transferred to the community, all operational and managerial positions will be occupied by community members. This is a pioneering effort in indigenous leadership in biodiversity conservation in Peru. The end result is a management approach effective in the protection of a significant area rich in biodiversity.

## **BIODIVERSITY IMPACTS**

Conservation activities in the communal reserve have had considerable impacts on the region's biodiversity and ecosystems, protecting some of the richest habitats of flora and fauna on the planet. Community guides report the return of several species, including macaws, hummingbirds, toucans, turtles, monkeys, and increased populations of giant river otters.



### *Partnerships for monitoring, conservation incentives*

Partnerships with NGOs and research partners have improved biodiversity monitoring and evaluation, allowing the community to gather data on species within their lands and to determine threat levels. These partnerships have enabled the community to effectively and accurately determine the impact of tourism activities on wildlife species and to use this data to craft conservation strategies and land management plans.

In collaboration with Conservation International (CI), the community has set up a wildlife monitoring program wherein community members collect data on wildlife populations. CI analyzes the data and reports back to the community to inform conservation planning. Other monitoring programs are incentive-based, such as the 'Harpy Eagle Nest Watching Program'. In this program, if a community member finds an active harpy eagle nest on their land, they receive a monetary award based on the number of tourists who view it. A finder's fee is paid until the eagle chick fledges, a period of up to nine months.



Giant river otters have also benefited from the community's conservation efforts. In the past, otters were hunted for their pelts, or targeted by local fishermen because of their negative impact on fishing returns. After construction of the ecolodge, community bylaws were established to regulate fishing and access rights to the oxbow lakes that form the otter's habitat. The community implemented codes of conduct to protect reproductive sites, such as delineation of a special reserve zone that comprises half of the oxbow lake area and is off limits to tourists and community members alike.

## SOCIOECONOMIC IMPACTS

Between 1998 and 2007, the number of visitors to the Posada Amazonas lodge grew from 2,000 to more than 7,000. As of 2013, Posada Amazonas accounts for 20 percent of the tourist market within the region. This success has directly translated into substantial improvements in local incomes and livelihoods.

Most of the wage labor positions at the lodge are filled by local community members. From 1997 to 2007, the ecolodge earned US\$662,225. This sum does not include all of the community benefits and additional returns from the sale of local construction materials, handicrafts, or agricultural goods.

### *Contributions to local well-being*

Profits from the ecolodge have been invested into social support projects (e.g., an emergency health fund, elderly care and support services, schools, infrastructure, and loans and scholarships for youth) and are also split among 150 families in the community, increasing income from US\$150 to US\$805 per household between 2001 and 2007. Guides are among the biggest earners from the project, with an additional income of US\$195,894 between 1997 and 2007. This discrepancy in earning is attributable to tips given to guides based on their expertise in tracking wildlife and birds.

Looking to extend spillover success from Posada Amazonas, the community has formed committees to create strategic plans for sectors such as agriculture, education, healthcare, handicrafts, and cultural rejuvenation. In 2000, the community set aside 25 percent of ecotourism profits for investment in education. This investment led to the construction of the only rural secondary school in the Puerto Maldonado area. The handicrafts committee has also made significant strides with relatively modest amounts of investment by the community, resurrecting traditional handicraft techniques, building a workshop, and drafting a marketing strategy.



### *Strategies for sustainability*

The revenue potential of Posada Amazonas has reached its limits, forcing the community to create alternate employment opportunities tangential to or outside of the tourism sector. These efforts include an ethnobotanical center, a port, a handicraft workshop, a fruit processing center, and a fish farm. The ethnobotanical center creates annual revenues of US\$12,000 from its medical services alone.

While the initiative has had a positive impact on the community, it has also produced challenges. Primary among these has been income disparity in the community, exacerbated by racial tensions between mestizos and the Ese'ejá. Higher incomes raise concerns about shifts in emphasis from local culture to a more consumerist lifestyle, and easier access to chainsaws and rifles is associated with increases in logging and poaching.

## POLICY IMPACTS

The Ese'ejá community has lobbied against the construction of the Interoceanic Highway designed to connect the Atlantic Ocean in Brazil to the Pacific Ocean in Peru, a proposal which will drastically alter

the economic and environmental face of Madre de Dios. In response, the community has lobbied the Peruvian government for recognition of community needs in the proposed area of development and established strong partnerships with surrounding communities to mobilize a united front.

In 2003, the Ese'ejá community applied for a concession to manage lands surrounding their oxbow lake, with its resident population of giant river otters. While the concession was granted, the government also awarded a fishing permit for the same area to a private sector interest. However, the objectives of conservation and commercial fishing were incompatible. With assistance from the Peruvian Society for Environmental Law, the community fought the fishing concession, ultimately winning an ecotourism concession that covered an extra area of 2,000 ha.

The Ese'ejá community has worked with Rainforest Expeditions to create alliances with other communities upriver from the lodge to ensure that the land between Posada Amazonas and the Tambopata National Reserve remains protected. All four communities in the area – who together constitute the entire population between Infierno



and the border of Tambopata National Reserve – have been contacted. The community intends to align all communities and secure their support for the conservation of the rainforest corridor that connects them and create a functional buffer zone to the Interoceanic Highway.

## SUSTAINABILITY AND REPLICATION

As of 2013, Posada Amazonas has an occupancy rate of 57 percent, 20 percent above the break-even point. The lodge provides a 35 percent rate of return on investment, and there is room for growth in tourism-related trades such as handicrafts, food production, and retail.

Community members are trained in basic lodge staff positions, including housekeeping, cooking, accounting and administration, and guiding. Positions at the lodge can only be held for a maximum of two years per individual, ensuring access on a rotating basis to employment and the transfer of knowledge and skills throughout the community. The community has developed a train-the-trainers program, whereby staff in each position trains other community members.

To foster the replication of good practices in environmental conservation and low-impact tourism, Rainforest Expeditions has established an information exchange called Trueque Amazónico. The initiative brings together tour operators and indigenous communities from three ecotourism projects in the

Amazonian regions of Bolivia, Ecuador, and Peru. These groups met for the first time in 2003 to pool information and draw up a list of best practices that have been shared with other communities facing similar challenges.

## PARTNERS

- **Rainforest Expeditions:** Financial and technical support for ecotourism
- **American Bird Conservancy:** Support to ecolodge
- **Conservation International:** Biodiversity monitoring project
- **MacArthur Foundation:** Grant (US\$50,000) for ecolodge construction
- **Critical Ecosystem Partnership Fund:** Support to Trueque Amazónico initiative
- **Peru-Canada Bilateral Fund:** Loan (US\$350,000) to build lodge, training
- **Peruvian Society for Environmental Law:** Support for litigation and ecotourism application
- **Frankfurt Zoological Society:** Codes of conduct for viewing giant river otters
- **World Bank:** Support for ecolodge and artisans (US\$50,000)

# COMUNIDAD INDÍGENA DE NUEVO SAN JUAN PARANGARICUTIRO (INDIGENOUS COMMUNITY OF NUEVO SAN JUAN PARANGARICUTIRO) Mexico

## PROJECT SUMMARY

The town of Nuevo San Juan Parangaricutiro is located in the western part of the Mexican state of Michoacán. Since 1982, indigenous Purépecha community members have engaged in sustainable timber and non-timber forest extraction and processing from the town's local pine forests. In 1991, a landmark national resolution led to the legal transfer of ownership of 18,138 ha of communal land to the 1,254 community members engaged in the project. In 1999, the project gained Forest Stewardship Council (FSC) certification for its forest practices. In 2013, training had been offered to more than 450 local people in technical aspects of sustainable forestry. The enterprise is active in more than twenty areas of production, the majority of which involve non-timber forest products, and generates an average of 900 permanent and 300 temporary jobs each year.





## BACKGROUND AND CONTEXT

Comunidad Indígena de Nuevo San Juan Parangaricutiro is located in the western part of the Mexican state of Michoacán. The local climate is temperate-humid with abundant rains in the summer. The predominant ecosystem is pine-oak forest, a vegetation type typical of Mexico's mountainous areas, and ranges in altitude from 1,000 to 2,800 m.

### *A forest-dependent economy*

The indigenous Purépecha community owns nearly two thirds of the municipal territory - 18,000 ha of land - of which 11,000 ha are forested. The indigenous group is made up of 7,500 *comuneros*, or 'community members', about half of the residents of Nuevo San Juan Parangaricutiro. The rest are private property owners and members of surrounding *ejidos* and agrarian communities. These two designations - *ejidos* and agrarian communities - are the two constitutionally established types of communal land tenure in Mexico: whereas *ejidos* can be formed by peoples without previous cultural or other connections, agrarian communities are constituted by pre-existent groups of peoples.

More than 35 percent of the local population is engaged in logging and timber transport, resin collection, and fruit cultivation. A smaller number of residents are involved in agriculture, such as avocado and maize farming, and the majority of crops are grown for community consumption. Other economic activities include tourism, animal husbandry, handicrafts, and trade.

### *Historical roots of communal ownership*

The roots of the indigenous Purépecha community date back to 1715 when, by order of the Spanish Crown, they were granted ownership of the land that they continue to occupy today. In the early 20th century, they formally constituted themselves as a communal enterprise in response to intensive logging by private companies that returned few benefits to the community and produced widespread environmental degradation. In the late 1970s, the community resolved to advance a new model of management that would restrict forest access and use to community members and prioritize sustainable resource harvest for job creation and poverty reduction.

The Purépecha indigenous group was one of the few rural communities in Mexico that held official documents dating back to 1715 which enabled them to secure tenure rights. The eruption of the Parícutin volcano in 1943, however, destroyed the community and roughly 1,500 ha of surrounding forest, forcing residents to move outside the boundaries of their communal land. As a result, when the *comuneros* began collectively harvesting and selling forest resources in 1982, the community did not have legally-recognized property rights. A breakthrough came in the 1980s when the state of Michoacán authorized *ejidos* and communities - some of whom lacked official documentation of their property rights - to use and manage natural resources. In 1991, community land rights were nationally recognized by a landmark presidential resolution that legally transferred ownership of 18,138 ha of communal land to the 1,254 *comuneros* and their families.



### *Early years of community forestry*

In 1982, the community began to independently harvest and sell wood from local forests on a small scale. The initial experiment proved so successful that the community bought an industrial sawmill, enabling larger scale timber management while ensuring regeneration and healthy forests. In 1984, the community received the National Forestry Merit Prize for its exemplary efforts in community organization and sustainable forest management. In 1986, the community signed the Convention on the Coordination, Consultation and Shared Stewardship of Forest Technical Services, which provides an organizational framework that guides its operation to this day. In 1999, the community was accredited by the Forest Stewardship Council, an independent organization that encourages sustainable forest management. The FSC ensures that forest management practices adhere to strict environmental and socially responsible standards and requires an annual certification audit.

Comunidad Indígena de Nuevo San Juan Parangaricutiro is the only communally-owned enterprise in the State of Michoacán that has its own Department of Technical Forest Services, which ensures the enterprise is able to develop forest use and management plans that are aligned and harmonized with Mexican Forest Law. The community enterprise has offered training to more than 450 indigenous people in the technical management of forests, agronomy, organizational management, and wood harvesting and processing.

## KEY ACTIVITIES AND INNOVATIONS

### *The forest beyond the trees*

The community forests are managed by forest technicians who implement a forest management program that provides a framework for forest use including: the collection of resin, seeds, and medicinal and ornamental plants; the cultivation of seedlings for reforestation; ecotourism; wildlife management; and agroforestry. All raw materials harvested from the forests are processed in a facility owned and operated by the community and all profits are reinvested into the enterprise and community projects.

Communal land is zoned for different purposes: over 10,880 ha are allocated for forestry, 1,200 ha for forest plantations and nurseries, 1,931 ha for agriculture, 2,122 ha for fruit orchards, and 35 ha for livestock pasture.

Two leading objectives of the community are: to increase economic benefits through sustainable natural resource management, ensuring that benefits are distributed fairly; and to create jobs for the local population in order to prevent out-migration. Partners are enlisted to provide farmers with alternative livelihood options and to offer technical support in agriculture, ranching, fruit tree cultivation, and marketing and sales. A community 'laboratory' is used for training purposes as well as for soil analysis, as a distribution center for fertilizers, and for packaging peaches, avocados, and other crops.

### *Forest management planning*

The community has received national and international recognition for its vertical integration of forest production, its scale of operations, and its innovative management approach. In partnership with university researchers, the *comuneros* have been able to classify their land according to landscape type and vegetation features. This information is used to develop science-based forest management plans that systematically regulate timber extraction and diversify production activities to ensure long-term resource sustainability. This data is complemented by the wealth of traditional knowledge on forest management that *comuneros* have acquired over generations of interacting with surrounding forests.

## BIODIVERSITY IMPACTS

### *Reforestation and conservation zoning*

Forestry activities are implemented in a manner that guarantees the long-term sustainability of trees, soil, water, vegetation, and wildlife. Ranching and agricultural activities are restricted to delimited zones to reduce pressure on standing forests. Reforestation activities are undertaken in the rainy season in areas where regeneration fails to take place after timber harvest. Ongoing reforestation efforts are also taking place in the area of land damaged by eruption of the Parícutin volcano. To protect water quality and prevent erosion, the community enforces 'no cut' buffer zones along streams and riverbanks.

Land use plans are supported by the innovative use of technology. Aerial photographs are used to provide the community with spatial data of where previous logging has taken place and where new logging can begin. Boundaries are established by painting perimeter trees. In 2007, the community reforested approximately 125



ha of land with more than 314,000 trees. In 2010, an area of 180 ha was reforested using more than 450,000 trees. A further 1,050,000 trees that were raised in community nurseries were given to the government for reforestation efforts across the region.

### *Biological monitoring*

The community conducts field research and takes inventory of flora and fauna in local forests every five years, cross-referencing findings with national inventory lists. The research has concluded that there are no species in the communal territory that are rare, threatened, or endangered. The community has enlisted the support of several universities in the development of its monitoring and evaluation system.

Hunting was a traditional activity of the *comuneros* of Nuevo San Juan. White-tailed deer in particular were hunted for celebrations of traditional holidays, causing a decline in the population. In 1996 the community developed a plan to address deer management which includes a semi-captive breeding program, the sale of individual animals to neighboring communities to repopulate the region, and economic incentives for conservation, such as ecotourism.

Forest fires are a consistent problem in the region and pose a major environmental threat to forests. In response, the community built a forest fire watchtower on the Cerro de Pario, a high point in the territory. In addition, the community produces and distributes educational materials on preventing forests fires.

## SOCIOECONOMIC IMPACTS

### *Job creation, income-generation, and service delivery*

The organization has 20 different production areas with total annual sales of US\$11 million. It has created more than 1,400 jobs, of which a little less than half are full time and the rest are seasonal. Another 89 people are employed in the enterprise's non-timber forest product 'adjunct areas.' The 600 full time employees receive employment benefits including retirement plans, life insurance, vacation time, and social security. Most of the seasonal or part-time employees work for family-run enterprises and cooperatives in transportation, sawmills, and resin collection. Other employment areas include avocado orchards, water purification and bottling, ecotourism, and furniture making. Cooperative members are also supported to connect to market supply chains which link them with national and international markets. Average household incomes across the community now stand at US\$516 per month.

Economic development has reduced poverty and all community members now report an ability to meet basic needs. In addition, community members report substantial improvements in service provision, notably in the areas of potable water access, sewage and sanitation systems, and access to electricity. A vast majority of families now have access to health services, which did not exist previously. In addition, the community enterprise provides a social safety net for widows, the elderly, or those with physical



handicaps which hinder their ability to work. These improvements have reduced out-migration and caused an influx of people from neighboring areas looking for employment.

### **Education and training**

Education is a priority for the community enterprise. The organization helps to finance local preschools, elementary schools, and high schools. The community recently inaugurated a bilingual primary school that teaches children the indigenous Purépecha language and aims to foster a sense of cultural pride. Between 2009 and 2010, the school went from 6 to 58 students. On occasion, the community enterprise supports *comuneros* to pursue higher education.

## **POLICY IMPACTS**

### **Advocacy for communal tenure and forest conservation**

An ongoing challenge for the community is acquiring the 4,354 ha of land that was previously identified by the national government as communal property. The Presidential Resolution of 1991 that granted communal land rights to the *comuneros* also acknowledged the claim to private land ownership by several families who were opposed to communal ownership of their land, creating a situation of legal pluralism. To date, the *comuneros* have recovered more than 1,000 ha of land through bilateral agreements with private land owners and other legal measures.

As a member of the Alianza de Ejidos y Comunidades Forestales Certificados de México, an alliance of FSC-certified cooperatives and communities, the community advocates for the conservation and responsible stewardship of forests across Mexico. In the future, the community aims to expand its influence to international indigenous peoples' forums.

## **SUSTAINABILITY AND REPLICATION**

In its 30-year history as a self-governing producer of sustainable forest products, the organization has proven its ability to overcome obstacles (e.g., low levels of education and capacity) and adapt to new market conditions. In its early years, land

for construction of the community enterprise headquarters was donated by a neighboring ejido and local families provided equipment (e.g., chainsaws) and transportation to assist the enterprise. These actions built a strong sense of community identity, trust, and solidarity, laying the cornerstone for the organization's self-sufficiency.

Comunidad Indígena de Nuevo San Juan Parangaricutiro has hosted many seminars, workshops, site visits, and exchange visits with indigenous groups from across Mexico. *Comuneros* have also given training programs at academic institutions in Mexico and for rural communities across Central America. Their community forestry model is currently being replicated in the states of Oaxaca, Guerrero, and Jalisco.

## **PARTNERS**

- **Servicio de Extracciones Forestales:** Training in harvest of timber, road infrastructure
- **Celulosa y Papel de Michoacán:** Harvest of cellulose (funds used to invest in infrastructure and equipment)
- **Santander Serfin Bank:** Loans to invest in road equipment, infrastructure, and operations
- **Autonomous University of Mexico:** Technical support for forest management planning, inventorying, and monitoring
- **Autonomous University of Chihuahua:** Support for inventorying and monitoring
- **University of Michoacana de San Nicholas de Hidalgo:** Support for inventorying and monitoring
- **Mexico's Forest Sub-secretariat and Rural Development Department:** Technical support and assistance with permits
- **Rigoberta Menchu Foundation:** Support for exchange visits
- **World Bank:** Funding to support forestry trainings by *comuneros* to communities across Central America

# WHAT DO THE CASES TELL US?

These cases represent only a small fraction of the Equator Prize communities practicing sustainable forest management, but they reveal many of the common motivations, community processes, achievements, and obstacles found in the larger archive of cases in the Equator Initiative database.

Keeping in mind the questions posed at the beginning of Theme 1, the following lessons emerge.

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**Climate Potential:** Community-based forest management can work over substantial areas and in situations where forest pressures are intense.

Although community-based forest initiatives vary widely in size, they can and often do encompass large areas of primary and secondary forest with considerable carbon storage, making them highly relevant to both the goals outlined in the New York Declaration on Forests and the Sustainable Development Goals. The seven cases in this section, for example, together cover more than one million ha of forest area. When communities succeed in their sustainable management goals, they maintain these community lands as working forests, with large areas of intact forest cover and high forest quality, and thus high potential to yield climate benefits.

Just as important as the area of forest under community management is the location and vulnerability of these forests. Most of these initiatives work in areas where pressures to remove or overuse the forest are intense. Studies of deforestation and forest degradation make it plain that without clear and attractive alternatives, such areas will eventually be deforested or substantially degraded. Community-based initiatives provide one of the only functional counterweights to these deforestation and degradation forces.

Indeed, the fact that there are any climate benefits at all associated with the vulnerable lands in these case studies is testament to the efficacy and necessity of these initiatives. Moreover, projects that address pressing community needs, as these examples do, have high potential for sustainability over time. One of the key factors in their success is the mix of

uses that these initiatives allow, and the alternative livelihoods that they work hard to support. This distinguishes them from large-scale afforestation projects, state-owned forests, or state-administered protected areas, which are often ineffective in countering deforestation and forest degradation because they can lack community support, and offer communities neither the legal standing nor economic incentive to engage in forest protection.

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**Community Motivations:** The desire for alternative livelihoods that offer sustained economic benefits is an important factor motivating community forest management. But self-determination and social and political empowerment are just as important to many indigenous forest communities.

Community initiatives usually spring from a mix of motivations. They often involve an imminent environmental or economic threat, such as rapid loss of local forests and the cash and subsistence income they provide. The prospect of maintaining or increasing these economic benefits by conserving intact forest is certainly a prime motivating factor of community forest initiatives and a strong correlate of their success. Initiatives that offer well-developed alternative livelihood programs and revenues that can fund community infrastructure offer a powerful inducement for collective action.

**“Just as important as the area of forest under community management is the location and vulnerability of these forests. Most of these initiatives work in areas where pressures to remove or overuse the forest are intense.”**

The Ekuri Initiative, for example, involves a mix of different forest uses, each in a designated forest zone. Together, income from several non-timber and timber products, cash crops, and food crops provides a diversified income

stream to families. A system of fees and taxes imposed by the community on the sale of forest products results in considerable income to the community treasury, which in turn has funded a road providing access to the community, as well as two schools, a health center, and a community center. Community drinking water systems, irrigation systems, public granaries, and processing facilities for local forest products are other examples of community infrastructure investments from other initiatives. In most instances, these direct investments in community well-being would not have been made in the absence of these community-generated initiatives. Government services and public expenditures are often minimal in the rural areas in which most community-based forest initiatives take place.

### **“No factor is more central to successful community-based forestry than strong community land rights.”**

As important as these economic benefits are, however, they are not the only factor motivating community forestry, and in many cases they are not the prime factor. Other social and political benefits offer as much or more inducement for community action, particularly for many indigenous peoples. Initiatives that empower communities to manage their resource base through local ordinances and rules, demarcate and control their forest borders, and reap the bounty of their stewardship through harvest and use rights offer a strong rationale for community action.

For many indigenous peoples, the desire to restore or retain their ancestral territory and revitalize their indigenous culture is a strong motivating force. Forest initiatives offer an effective way to channel this desire into both political action and actual forest control on the ground. Nearly every case in this section involves advocacy for recognition or expansion of the historical forest rights of the local indigenous groups involved. In the Yawanawá case, for example, advocacy for the expansion of the Yawanawá's legally recognized ancestral lands formed the foundation of the initiative. Meanwhile, the desire for local development that would support and renew traditional Yawanawá culture determined the shape of their commercial partnership with Aveda and the use they made of the resulting revenue.

### **Forest Ownership: Strong community land rights increase the chances for successful community-based forest management.**

No factor is more central to successful community-based forestry than strong community land rights. Such rights allow the community to determine forest management goals, adopt and enforce forest use rules, and exclude those who flaunt these rules. Without this kind of secure and legally recognized forest tenure, a community's investments in sustainable forestry can be undermined by illegal forest users, commercial land grabs, or the granting of government forest concessions.

In every one of the cases in this section, issues of forest ownership play an important part. In Ekuri, the community's decision to forego commercial logging and establish its own sustainable forest regime was only possible because of the status of the local forest as community-owned. Indeed, one of the initiative's first actions in 1997 was a careful perimeter survey to document the community forest boundaries.

For many local initiatives, advocating for indigenous land rights and resource entitlements is an essential step that unifies and empowers the community and enables it to pursue forest interventions and commercial enterprises on the ground. For example, the commercial successes of Comunidad Indígena de Nuevo San Juan Parangaricutiro, Comunidad Nativa Ese'eja de Infierno, and Associação Sociocultural Yawanawá were only possible after their indigenous land claims had been resolved. In Honduras, Mosquitia Pawisa Apiska (MOPAWI) understands this dynamic and for the last 25 years has brokered negotiations with the government to secure community land rights, helping indigenous communities survey their native lands, document traditional land uses, and file land claims. One notable result was the eventual recognition of indigenous land rights over 68,000 ha of tropical forest in Honduras' Mocerón region that is now under sustainable management.

Although holding clear title to forest land is optimal for many communities, it is not the only route to secure forest tenure. Many Equator Prize-winning forest initiatives take place on public forest lands where forest management authority has been officially granted to the local community through a variety of devolution and co-management arrangements. For example, in the case of Fikambanan'ny Terak'i Manambolo (FITEMA) in Madagascar, the government has devolved forest



management and use rights over some 19,000 ha of forest to community-based associations in each of the 12 communities in which the initiative functions. This allows community members to participate in forest planning, monitoring, rulemaking, and patrolling. The sense of ownership this provides has translated into community stewardship, even while the forest remains in the public domain.

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**Enterprise Development:** Community-based forest enterprises can be highly successful, but require considerable attention to developing business capacities and tackling market, transport, and regulatory obstacles, often with the help of commercial partners.

Forest enterprises are a prominent feature of most community forest management efforts—the vehicle for generating local employment and turning forest resources into local income. Such local enterprises not only provide the economic rationale for sustainable forest management, they also provide a center of gravity to the community’s collective work and a platform for building social capital among community members and developing professional and organizational skills that will be useful beyond the enterprise itself.

Depending on the forest type and the community’s business acumen and access to markets, a wide variety of enterprises can be found, from marketing non-timber forest products like rattan, forest fruits and essential oils, cacao, and coffee, to the manufacture of local crafts and finished timber products. Service-based enterprises such as ecotourism, cultural tourism, and guide services are also increasingly common. While most communal forest enterprises remain fairly small, the cases in this section show that under the right circumstances they can grow into significant commercial ventures that generate substantial income. Comunidad Indígena de Nuevo San Juan Parangaricutiro in Mexico, for example, produces US\$11 million in annual sales of its timber and non-timber products, and generates some 900 permanent and 300 seasonal jobs in a community numbering around 7,500 members. Similarly, the ecotourism lodge operated by Comunidad Nativa Ese’jeja de Infierno in the Peruvian rainforest netted over US\$660,000 between 1997 and 2007 for the isolated 500-member community.

Given their importance, it is not surprising that much of the work of community-based forest initiatives goes into building these local enterprises, especially in the beginning. This involves selecting an appropriate activity, acquiring the technical and business skills necessary, and tackling the many obstacles that rural enterprises face, including physical isolation, lack of market connections, inadequate finance, and unfriendly government regulations. The help of partners and support organizations is often essential in meeting these challenges. In Honduras, MOPAWI provides the technical training needed to isolate and harvest non-timber forest products such as batana oil—an ingredient in high-quality hair products. Just as essential, MOPAWI has established market supply chains that enable communities to export the oil to cosmetic companies in the United States and Canada. Establishing such value chains has allowed local batana oil production to increase from 1,000 liters per year to 88,000 liters per year, and to raise the price obtained per liter from US\$1.50 to US\$7.

**“Forest enterprises are a prominent feature of most community forest management efforts—the vehicle for generating local employment and turning forest resources into local income.”**

The work of MOPAWI is just one example of the vital role of partners in developing local enterprise capacities. In reality, community forest initiatives often make use of several different sources of support and training. These include government extension services, universities, and local and international NGOs that can provide select training, business planning, market development, and certification for products that are organically or sustainably produced. It also includes organizations such as cooperatives, producer groups, and federations that allow community groups to network, share information, and jointly market their products.

Private sector partners can also be beneficial—even essential—as several of the cases demonstrate. The timber enterprises of Nuevo San Juan Parangaricutiro, the ecolodge of the Ese’jeja, and the Yawanawá’s lucrative contract to produce urukum fruit were

all dependent on the participation of commercial partners, at least in the early stages. These partners brought their technical and business expertise, as well as their markets, and invested considerable time to develop the capacities of their rural clients. In the case of Aveda (the commercial partner of the Yawanawá), it took six years and thousands of dollars of investment for the commercial relationship to become profitable.

Overall, these cases show us that when local enterprises are chosen carefully, supported consistently as they develop their capacities, and encouraged to link together in cooperatives, federations, and knowledge networks, they can become the foundation for a more sustainable and equitable rural economy—the leading edge of the ‘green economy’ that generates jobs and income while it supports local indigenous culture and maintains the community’s ecosystem assets. This kind of local green economy is vital if forest communities are to prosper when they choose climate-friendly forest management.

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**Local Knowledge and Innovation:** Local knowledge is a key resource that makes community-based initiatives relevant and adaptive, while local innovation bridges between traditional practices and modern forest management.

The reservoir of traditional knowledge and local expertise that communities bring to bear on their forest management is a key asset in their success. Forest communities draw on generations of experience in managing local ecosystems, often codified into well-defined traditional resource management regimes with long records of sustainability. Although many of these indigenous practices have been modified or abandoned as communities have faced demographic and political changes in recent decades, this knowledge remains as a potent resource that can be called on to adapt community forest initiatives to local conditions and ground them in local culture.

Many Equator Prize-winning forest initiatives are dedicated to applying this indigenous expertise. FITEMA in Madagascar provides an example of where the initiative has made the most of its sustainable forestry traditions within a modern context. Prior to the French colonization of Madagascar in 1895, the

use of natural resources in the Manambolo area was governed by a system of rules known as *dina*, which determined how and by whom forest resources could be harvested. FITEMA reintroduced this indigenous land use system as a way to reign in uncontrolled forest use and reestablish a sense of local control and a method for social acceptance of the new forest use rules. While the forest is still officially under state control, the local *dina* system has been formally recognized by the government, making it legally binding. The system is administered by elders working through forestry management associations in each participating community, and strives for equitable and sustainable sharing of resources by local residents.

**“Forest communities draw on generations of experience in managing local ecosystems, often codified into well-defined traditional resource management regimes with long records of sustainability.”**

Indigenous knowledge and traditional practices are not just a link with the past, but a bridge to the future as well, and a basis for local adaptation and innovation. In most cases, local forest initiatives draw on both traditional knowledge and modern practice to construct a hybrid management system that fits the community. In Honduras, MOPAWI meshed traditional cacao production with new agroforestry practices to yield a mixed system of fruit trees, timber species, and organic cacao. Likewise, the Yawanawá in Brazil used their knowledge of local plants to help solve their river transport problems. They are now processing seeds of the abundant *jatropha* plant, long used for medicinal purposes within the community, into biodiesel that can be used to power their boats—the most important transport link to the world outside their indigenous reserve. In another cross-over between traditional practice and modern culture, the Yawanawá have launched a line of clothing for sale in Brazil using traditional drawings and paintings as the design motif—an instance of adapting traditional culture to support modern Yawanawá identity and fund its social and alternative livelihood programs.

**Scalability and Sustainability:** Successful community-based forest initiatives have shown high potential to scale up through a combination of local advocacy, effective outreach, and the help of intermediary support organizations. Inclusive governance, consistent benefits, and continuous public education promote sustainability.

Community-based forest initiatives need not remain simply 'local' solutions to isolated problems. Because many of the challenges faced by rural forest communities are quite similar, Equator Prize forest initiatives often have broad appeal as models, and have shown a consistent ability to scale up, sometimes quite rapidly and at low cost. The Ekuri Initiative has expanded its activities beyond Ekuri itself to five other villages and another 10,000 people, while FITEMA has exported its model to eight other buffer zone communities within the rainforest region in which it works. Such growth often begins organically, driven by peer-to-peer communication and site visits by representatives from other communities anxious to see results firsthand. Documentation of local benefits by journalists, and dissemination of success stories by NGOs, donors, and even forestry department personnel also play an important part in promoting successful community projects and driving growth.

Scaling of community forest solutions can also be aided by intermediary support organizations that link communities through technical assistance and capacity building programs, and facilitate networking and information sharing. In Peru, Asociación para la Investigación y el Desarrollo Integral (AIDER) serves this function, working with communities in five different regions of the country to tailor alternative livelihoods to local needs, and transfer sustainable forest management methods and technologies. As it has helped community forestry efforts go to scale, AIDER has explicitly addressed climate concerns by promoting forest carbon trading in some of its projects and working to have the community forestry initiatives it works with recognized under the country's REDD+ program.

Beyond expanding their size and reach (quantitative scaling), community-based forest initiatives have also shown the ability to scale up in other crucial ways. For example, they may broaden their portfolio of activities, diversifying their forest enterprises

and expanding the services they render to the community. They also tend to broaden their influence in policy circles, and thus their impact as examples of sustainable rural development and local empowerment. This functional and political scaling is essential to the maturation and organizational development of local initiatives, allowing them to multiply the benefits they bring to local communities and at the same time elevating them to the national stage where their innovations can be shared more widely and can influence forest practices on a larger scale.

Of course, to create lasting impact, community-based forest initiatives must not only grow and expand, but must also mature and demonstrate sustainability over time. This is particularly relevant when considering climate benefits such as carbon storage, whose value lies in longevity. In this respect, Equator Prize forest initiatives have an impressive record of sustainability, with all but one of the initiatives that appear in this section established more than 20 years ago, and some much more. Comunidad Indígena de Nuevo San Juan Parangaricutiro, for example, began its sustainable timber harvesting initiative in 1982.

**“Indigenous knowledge and traditional practices are not just a link with the past, but a bridge to the future as well, and a basis for local adaptation and innovation.”**

The sustainability of these and other forest initiatives is a product of changing the incentive structure around forest use through the consistent delivery of income and social benefits. But it also depends on changing—through education and information—the local perception of what forests can provide, so that the community endorses the initiative's forest management goals and enforces its rules on forest use. The governance structure of the community initiative is also critical. An organizational structure that empowers community members and gives them a continuing voice in forest management decisions provides an incentive to remain involved in the initiative's work and claim its successes as their own.





## Theme 2

### Forest-Friendly Agriculture



**T**he cases in this section highlight the importance of agriculture to rural forest communities and the ways in which communities have incorporated agriculture into their sustainable forest management regimes. Agriculture is the most important source of income and employment in rural communities. Even in forest communities, smallholder farming is essential for food security and household income. Unfortunately, poor farming practices are a common source of forest degradation and outright forest loss.

As these cases reveal, many forest communities have found ways to reconfigure their smallholder agriculture so that it is both more productive and less harmful to local forests. This may involve applying agroforestry techniques, adopting new cropping systems and farm technology, or experimenting with alternative agricultural products. It also usually entails a serious reorientation of local thinking and a substantial investment in training in new farming practices and marketing approaches. The local benefits can be profound: increased food security and income, and reduced pressure on local forests – which translates to less forest conversion and greater climate benefits.

## CASES IN THIS SECTION

- **Association Adidy Maitso**, Madagascar
- **Programa de Campesino a Campesino, Siuna (PCaC, Farmer-to-Farmer Program, Siuna)**, Nicaragua
- **Farmers Association for Rural Upliftment (FARU)**, Philippines
- **Riba Agroforestry Resource Centre**, Cameroon
- **Muliru Farmers Conservation Group**, Kenya

## UNDERSTANDING THE CASES

To understand the relationship between local farming practice and forest health, the factors at work behind successful forest-friendly agriculture, and the implications for community forest initiatives more broadly, readers should consider the following questions as they read the cases.

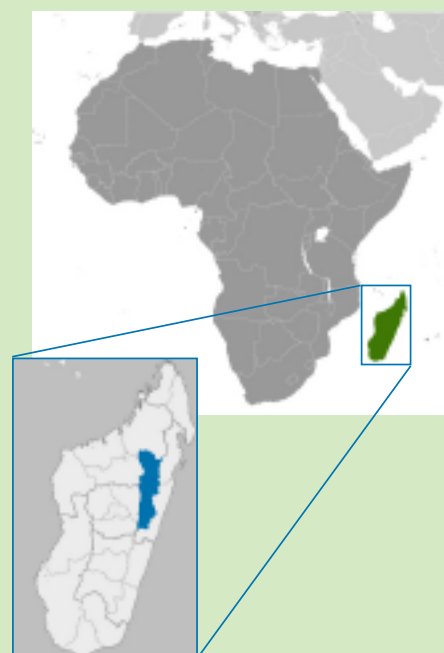
- **Food Security and Forests:** How do food security concerns affect local forest use? How much of a factor is smallholder agriculture in deforestation, forest degradation, and the loss of the climate benefits of forests?
- **Forest-Friendly Options:** What are the ways in which community forest initiatives have reduced the negative impact of local agriculture on nearby forests?
- **Education and Training:** What role does education and training play in making local agriculture more forest friendly? What kinds of training programs have been most effective?
- **Integrated Landscape Management:** How can agriculture, forestry, and other land uses be integrated across rural areas to create sustainable working landscapes?





## PROJECT SUMMARY

Association Adidy Maitso was established in 2005 with the aim of conserving the natural resources of Didy Forest – a dense moist forest in the Alaotra Mangoro region of eastern Madagascar. The forest lies within the Ankeniheny-Zahamena Corridor, which is renowned for its high species endemism and biodiversity. The Association works through 16 local community associations to manage and restore the forest corridor, educate local communities on the economic benefits of biodiversity conservation, and provide training to local farmers and women's groups on agricultural and income diversification. The group is actively engaged in maintaining an indigenous tree nursery, patrolling and surveying local forests to regulate against unsustainable forest use, radio programming, training on improved agricultural practices for greater crop yields, and running a demonstration and training center for local farmers.





## BACKGROUND AND CONTEXT

Association Adidy Maitso is a for-profit association located in the Ambatondrazaka district of the Alaotra Mangoro region of Madagascar. Founded in 2005 and formalized in 2008, the organization brings together 16 Koloharena community associations (Koloharena, literally, 'looking after the wealth,' refers to a village association that works to improve community living standards). Significant pressures were being exerted on the natural resources of the Didy Forest, a dense mid-elevation (600-1200 m) moist forest within the Ankeniheny-Zahamena Corridor, which is renowned for its endemism and recognized for connecting an important network of protected areas in the country. A shortage of arable land had driven the local population to rely too heavily on the resources of the forest, resulting in environmental pressures such as slash-and-burn farming, illegal logging, over-collection of non-timber forest products, and illegal mining.

### *Drawing up of community management plans*

After numerous lobbying efforts on the part of partners and supporting organizations, the management of natural resources in Didy was transferred to the community level in 2000.

Vondron'Olona Ifotony (VOI, community management units) were constituted to bring together volunteers to sustainably manage and protect renewable natural resources, and improve community living conditions. Each management transfer site was furnished with a management plan, which was developed in a participatory manner with stakeholders from the Ministry of Environment, communities, and local partners. The plans included three zones, based on ancestral boundaries and local knowledge: a management site, where community member are allowed to access and use natural resources; a conservation site, where community members are forbidden from using resources; and a controlled occupancy site, where communities are allowed to build homes and use land for farming.

Each community management unit also drafted a tailored management plan. One common feature to all management plans were monthly monitoring of the core conservation zone through community patrols. These patrols were done on a voluntary basis without remuneration. While some community members were motivated by a conservation ethic, there were no economic incentives in place and the local standard of living was low. Community members quickly realized that conservation could not take



place over the long-term without creation of income-generating activities and economic incentives to adhere to newly introduced restrictions on resource access and use. It was in response to this challenge that the Association Adidy Maitso was formed.

### *Roles of Adidy Maitso and its member communities*

The Association Adidy Maitso and community management units agreed to divide responsibilities, with communities retaining responsibility for the conservation of biodiversity and the sustainable management of natural resources and Adidy Maitso assuming responsibility for alternative livelihoods. Adidy Maitso also committed to improving the governance framework of each management unit, and to introducing incentive structures. The association has since ensured that each community management unit has accountable leadership with transparent decision-making mechanisms. The association also facilitates reporting to local authorities and the local forest service, providing the latter with information on how it can support community regulation of zoning restrictions, thereby harmonizing enforcement efforts.

## KEY ACTIVITIES AND INNOVATIONS

### *Mobilizing community action in forest conservation*

Adidy Maitso focuses on capacity building, awareness raising, and oversight of community patrols to safeguard local forests and forest resources. Outreach activities are primarily conducted via workshops, festivals, and through extensions services to individual communities. Awareness raising has concentrated on the importance of biodiversity and healthy ecosystems to human well-being, the economic benefits of sustainable natural resource management, and the respective roles and objectives of the *Koloharena*. The association has also been active in reforestation, planting native species in natural forests and planting eucalyptus trees outside conservation sites as a means of offsetting community fuel wood and timber needs.

### *Improving agricultural productivity*

To improve agricultural productivity, the association uses agricultural extension officers to disseminate new technologies, training in animal husbandry, and information on efficient production methods. Specific interventions include rice cultivation, crop protection



(hedging), seed distribution and diversification, and organic fertilizer use. Crop protection methods to fight parasitic plant diseases use natural materials rather than chemical inputs. The association has disseminated locally evolved insect and pest management methods, including the use of local artisanal soaps, white citron, and a liquid made from zebu bone and sisal leaf extracts. Seed diversity is encouraged, as is the use of compost and organic fertilizers.

### *Livelihoods diversification*

To promote alternative livelihoods and diversify local sources of income, Adidy Maitso is involved in identifying and linking with emerging markets and distribution channels for new products. The association has focused on several sectors to expand local income-generation opportunities, including small-scale potato and bean processing businesses, chicken rearing, apiculture, fish farming, and pig and zebu rearing. By promoting off-season crops such as onions, bananas, soybeans, cassava, corn, and white beans, the association is supplementing local diets and improving food security.





### *Awareness-raising and extension services*

Most of Adidy Maitso's resources are dedicated to outreach. One early barrier to outreach was the high rates of illiteracy amongst local farmers. This challenge was overcome through radio programming which broadcasts news and information on environmental conservation activities, sustainable livelihood opportunities, resource management rules and regulations, and improved farming techniques. The use of on-the-ground agricultural extension officers (engineers) to conduct trainings, workshops, and on-site demonstrations on environmentally responsible farming techniques has been another successful way to raise awareness in communities.

A training center has been established to carry out capacity building activities and serves as a communications outpost for participating engineers. Engineers fall into one of three categories: farming facilitators, farming extension agents, and model farmers. The farming facilitators are responsible for outreach, make courtesy calls at the request of community members, raise awareness on

conservation issues and priorities, and disseminate improved agriculture and animal husbandry techniques. Farming extension agents conduct demonstrations for farming communities throughout the corridor. The model farmers apply new techniques in their own fields, which are then used as demonstration sites for the entire community. Model farmers also mobilize people from the community to attend demonstrations, where they share their expertise, methods, and achievements. Each farming facilitator, extension agent, and model farmer is also responsible for producing one radio show each week.

In addition to the above activities, Adidy Maitso also helps its members to secure micro-finance. As one example, the association works with a network of rural microcredit agencies to connect member families with financing for off-season crop cultivation.

### *Innovative uses of communication technology*

The Association Adidy Maitso takes pride in its innovative communication methods. The local radio and farming engineer programs have been crucial to the project's success. Initially, the radio program consisted of a do-it-yourself microphone that had



a range of only 20 m, but the potential of this tool to disseminate information was quickly recognized. Working with partners, the association obtained solar panels, a more powerful transmitter, office furniture, and computers. Radio programming now reaches farmers across the entire corridor. The radio program office has an information and communications center, which houses a library, TV room, and the radio station equipment.

The farming engineer program was ultimately the outcome of a trial and error process. Unsatisfied with the sustainability of one-off training courses, the association worked with partners to train and position technicians in the roles of farming facilitators, farming extension agents, and model farmers. This train-the-trainers approach has thrived, and represents a model for effective environmental education and knowledge transfer. The Association Adidy Maitso financially compensates the engineers and communities receiving the training often provide meals.

## BIODIVERSITY IMPACTS

The Didy Forest is rich in biodiversity. The area is home to several species of lemur, including the critically endangered greater bamboo lemur, and is an important nesting site for the Madagascar ibis and other birds. Notable plants include the red stinkwood and several species of orchid.

Adidy Maitso helps protect the fragile ecosystems of the Ankeniheny-Zahamena Protected Area by supporting community management units to sustainably manage their natural resources. Since the initiative began, deforestation, poaching, and hunting of threatened species have been reduced (previously, an estimated 10 to 20 tons of the endangered rosewood tree were illegally being felled and removed from the forest every week). The association undertakes comprehensive awareness-raising campaigns to educate the local population on the importance of biodiversity and ecosystems services to human well-being. These campaigns also sensitize farmers to the irreparable damage caused by slash-and-burn agriculture. Slash-and-burn agriculture was a leading cause of forest fires, causing enormous damage to the protected area. Community patrols report a 65 percent reduction of forest fires. Adidy Maitso's reforestation efforts amount to 15 ha per year and the association's long-term goal is to expand the corridor by six times its current size.



Adidy Maitso's conservation activities have had a positive impact on species numbers and ecosystem health, as observed by the local community, local government, and local forest services. The association is currently in the process of developing an improved system of data collection based on a combination of surveillance patrols and information received via instant message.

## SOCIOECONOMIC IMPACTS

### *Improved livelihood options*

Agricultural diversification and the adoption of improved agricultural techniques have allowed the majority of families to improve their diets and livelihoods. Intensive rice growing has more than doubled rice yields (from 0.8 – 1 tons per hectare to 5-6 tons per hectare). Market garden cultivation, generally practiced by women, diversifies diets and increases financial stability, with some household selling surpluses in the local marketplace. Market operators at the provincial bazaar have signed a deal with an association member community to purchase their entire crop of beans. As a result, heads of households are no longer attracted by jobs with forestry businesses that pay a meager salary and farmers practicing slash-and-burn agriculture are now abandoning the method in favor of an improved rice-cultivation.

### Job creation and income generation

Income has increased by 25 percent in communities served by the association, with much of the gains attributable to new income-generating activities. Employment of farming engineers and radio technicians has created 48 new jobs. Temporary allowances and benefits have been shared with over 80 community associations.

## POLICY IMPACTS

Because the Didy Forest is included in the new protected area of Ankeniheny-Zahamena, the association has participated in the local consultations to draft a development plan. The association holds an advisory role in the management of this new corridor. Adidy Maitso was similarly consulted prior to implementation of the Development Plan of the Alaotra Mangoro Region. The association also holds significant sway at the municipal level. For example, the association successfully petitioned the mayor to pass a decree confronting illegal mining and logging in the town.

## SUSTAINABILITY AND REPLICATION

The Association Adidy Maitso contributes to environmental sustainability through its role in integrating communities into the management structure of the Ankeniheny-Zahamena Protected Area. The association plans to develop 12,000 ha of 'plain lands' in Didy so communities have more arable land on which to expand agricultural production instead of encroaching on the natural forest. This proposed venture will require a number of drainage and irrigation projects. The association is currently building a canal for this purpose, and is using funds exclusively gathered from member contributions. This capacity to self-finance community development projects reflects a high degree of local ownership as well as a sustainability model to meet local demands and overcome identified challenges.

Through environmental education and awareness-raising campaigns, the association has effectively instilled a conservation ethic amongst the local

population, which is essential for long-term sustainability. So too, community members are provided with up-to-date information on the laws that govern natural resources. Awareness and education have empowered community members to tackle emerging problems, encouraged reciprocal investments of time and resources by community members, and, by turn, bolstered the association's social sustainability.

Association Adidy Maitso is one of the first cooperative associations established in the Ankeniheny-Zahamena Corridor. It plays a leadership role in disseminating environment, farming, and development best practice with other communities. Demonstration sites, extension services, exchange visits, and radio programming have attracted new members to become involved in conservation and sustainable farming activities, and are replicable. To date, two cooperative associations modeled after Adidy Maitso have been formed in the Alaotra region.

## PARTNERS

Conservation International: Technical and financial support, lobbying for protected area, small grants for livelihood initiatives, radio station

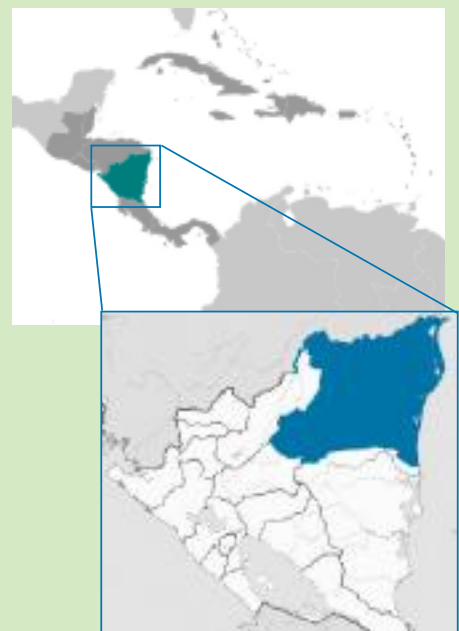
- **Conservation International:** Technical and financial support, lobbying for protected area, small grants for livelihood initiatives, radio station
- **French Global Environment Fund (FFEM):** Lobbying for protected area
- **ERI (with USAID funds):** Technical program, radio station, training center
- **UNDP/GEF Small Grants Programme:** Financial support, linkages to PES and Reducing Emissions from Deforestation and Forest Degradation (REDD+)
- **Ministry of Environment and Forests:** Monitoring and evaluation, training
- **Local government:** Local decree granting land rights, facilitation, stipends to radio staff



## PROJECT SUMMARY

Programa de Campesino a Campesino (PCaC) has operated throughout Nicaragua since the post-war period of the early 1990s, as part of the worldwide *Via Campesina* (Farmer's Way) movement, which advocates for food sovereignty, agrarian reform, and democratic governance of food production systems.

Programa de Campesino a Campesino in the northern municipality of Siuna is one of 65 such programs in Nicaragua which provide technical assistance to small-scale agricultural producers. The network began its work in 1992 with the goal of controlling the rapidly expanding agricultural frontier within the Bosawás Biosphere Reserve, while aiming to enhance food security, increase household incomes, and strengthen regional governance throughout Nicaragua's newly designated North Atlantic Autonomous Region. Three hundred volunteer extension officers work in over 80 communities and serve more than 3,000 subsistence-farming families.





## BACKGROUND AND CONTEXT

Programa de Campesino a Campesino (PCaC) operates in the municipality of Siuna, Nicaragua, through a growing network of socially driven, environmentally minded farmers. Founded in 1992, the program aims to control the rapidly expanding agricultural frontier within the Bosawás Biosphere Reserve, while enhancing food security, increasing household incomes, and strengthening regional governance. PCaC Siuna is a member of the worldwide *Via Campesina* movement, which advocates for food sovereignty, agrarian reform, and democratic governance of food production systems. To achieve its goals, PCaC employs a farmer-to-farmer information-sharing methodology to promote sustainable farming and ranching practices. Over the past two decades, PCaC Siuna and its network of volunteer ‘promoters’ have contributed to the development of an effective resource management framework for the region and the social integration and reconciliation of this war-torn rural population.

### *Civil war and its impact on agriculture*

PCaC emerged after Nicaragua’s civil war, a conflict that ravaged the country, and in particular, the Siuna region. A US- imposed trade embargo, coupled with the Sandinista government’s adoption of Soviet-style economic policies, crippled the economy and the agricultural sector. Total agricultural production fell by over 40 percent in 1980 following the onset of war, and continued to decline by five percent per year until 1993, dropping Nicaraguan farm incomes to their lowest levels in modern history.

### *Bosawás Biosphere Reserve*

Bosawás was established in 1979 and covers approximately 14 percent of Nicaragua’s territory, or 2,800,000 ha, with a core zone of one million hectares. Together with three neighboring protected areas in Honduras – Río Patuca National Park, Tawhaka Anthropological Reserve, and Río Plátano Biosphere Reserve – Bosawás constitutes the ‘Heart of the



Mesoamerican Biocorridor' and represents the largest protected tropical mountain moist forest north of the Amazon basin. Biodiversity within the reserve is extremely rich, with many rare and endangered species. Nearly 130,000 people live within the reserve. Occupation within the core zone is limited primarily to the Miskito and Mayangas indigenous groups, who number approximately 25,000. Another 100,000 Mestizos, mainly subsistence farmers, inhabit the buffer zone.

As a national reserve, Bosawás is legally the property of the Nicaraguan government. However, given the variety of ethnic groups living within and around the reserve, the diversity of interests in its many natural and mineral resources, and its geographic location, control and management of the reserve has been a complex issue. Throughout the 1980s and 1990s, the combined lack of a unified management authority, a poorly articulated management strategy, and the ongoing armed conflict in the region resulted in unchecked use of the reserve's resources. It also led to unsustainable agricultural encroachment deep into the reserve's buffer zone.

#### *UNAG and origins of farmer-to-farmer exchange, 1980s*

In 1981, the Sandinista government established the National Farmers' and Ranchers' Union (UNAG) to forge a populist agrarian front. It was from UNAG that the Nicaraguan PCaC movement emerged in 1987, as an attempt to provide relief to rural populations suffering from the effects of war. UNAG believed that a farmer-to-farmer exchange program focusing on improved agricultural methods would help Nicaraguan farmers and stabilize the agricultural sector. This was also the only concrete assistance available to small-scale farmers in an unstable political climate that favored state agricultural enterprises and inefficient rural cooperatives.

#### *PCaC initiated in Siuna, 1992*

In 1992, UNAG promoted PCaC activities in Siuna to slow the advancing agricultural frontier within the Bosawás Nature Reserve buffer zone. However, UNAG was unfamiliar with the region's humid tropical conditions and was unable to provide a resource management plan for the area. Slash-and-burn farming was widely practiced in Siuna and contributed to a variety of environmental problems; unsustainable hillside farming damaged soils and



accelerated erosion and nutrient loss, leading to decreased crop yields. This compelled farmers to penetrate further into the poorly managed Bosawás reserve where they could freely clear forestland. In response, UNAG and PCaC Siuna developed an innovative farming methodology centering on farmer-to-farmer information sharing as a way to stabilize the agricultural frontier and develop a resource management strategy attuned to the interests in the region.

#### *An approach rooted in local capacity*

The farmer-to-farmer information sharing methodology is one of the most salient features of the PCaC Siuna initiative, allowing direct, 'horizontal' sharing of information between farmers, without intervention by professional staff. The approach represents a departure from the vertical transfer of knowledge and technology that is common in Nicaragua. Many farmers have attributed their success to the fact that the transfer of *conocimiento*, or 'knowledge', is more effective when it comes from





people they can understand and whose agricultural lexicon is similar to their own. Initially, organizations and institutions were skeptical of the movement, believing that PCaC's methods signaled a reversion to 'archaic' agricultural practices, which the government had been attempting to modernize, but the group continued to promote farmer exchanges with great success.

## KEY ACTIVITIES AND INNOVATIONS

### *Use of velvet bean*

In 1993, three farmers from the Rosa Grande community of Siuna participated in an exchange program to a community where velvet bean (*Mucuna pruriens*) was used to restore soil fertility. Velvet bean is a nitrogen-fixing species that is widely used for improving degraded soils, as well as for animal consumption, crop rotation, and weed control. The bean can also be roasted and mixed with maize, rice and beans, meat, and other staples to supplement

local diets. As a fertilizer, it is incorporated into the soil at the flowering stage, decomposing rapidly and releasing nitrogen. Velvet bean also provides erosion control benefits. The dense ground cover provided by the plant reduces the erosive forces of rain, improves rainwater infiltration, and preserves soil humidity. On steep slopes, use of velvet beans can mean the difference between meager or healthy harvests.

Upon return, the inspired Rosa Grande farmers began experimenting with velvet bean, and word of their successes quickly spread to other communities in Siuna. From this beginning, PCaC in Siuna grew into a movement of more than 300 agricultural producers with expertise in farming techniques suited to the humid tropics. These farmers work voluntarily in over 80 communities to transfer their knowledge to more than 3,000 families engaged in subsistence agriculture within the municipality of Siuna. Their efforts have transformed agricultural production from traditional slash-and-burn to a sustainable system using green fertilizers and cover crops. These efforts



limit the advance of the agricultural frontier into the Bosawás Biosphere Reserve. The group has also made significant contributions towards the reconciliation of war-torn rural populations, strengthening systems of regional governance, and awakening a new campesino social awareness.

### *Additional strategies for improving agricultural productivity*

Intercropping has enabled farmers to diversify their crops with complementary species such as cacao, coffee, peppers, and soil-enhancing legumes. This technique provides habitat for insects and soil organisms that would not be present in a single-crop environment, thereby enhancing biodiversity and reducing pest outbreaks. Farmers also produce natural pest repellents made from on-farm items such as chili peppers, garlic, onion, and tobacco. With a focus on repelling instead of killing, the shift from chemical insecticides to natural insect repellants illustrates another strategy to conserve biodiversity.

Farmers are also experimenting with minimum tillage and contour farming. Minimum tillage involves tilling only specific portions of land where seeds are planted. This allows biomass covering the soil to remain intact, protecting topsoil from erosion. Contour farming is well suited to hillside farming, and is less labor intensive and more feasible for subsistence farmers than terracing. Crops are planted in concentric rows perpendicular to the slope of the hill, forming a natural barrier to run-off and soil erosion.

### *Holistic farmland management*

The success of farm management strategies has allowed farmers to intensify cattle ranching on existing ranchland without having to expand their ranches in search of more productive soils. Previously, farmers needed almost two acres of land per cow for grazing. Since the introduction of PCaC's farm management techniques, only 0.5 acres per cow is required for grazing.

### *Communication methodologies*

PCaC promoters use a variety of methods to teach others about the benefits of their farming techniques, including exchange visits, workshops, demonstrations, photographs, radio programs, folk music, and theater. The active role that the farmers have taken has created an environment of mutual respect and support and motivates others to contribute to the effort.

## BIODIVERSITY IMPACTS

Since the beginning of the project, PCaC has protected 20,000 ha of land, creating a bulwark against deforestation. Forest protection and active reforestation has increased populations of species whose habitats were previously threatened by agricultural encroachment.



PCaC reports that an estimated 3,000 farmers in 32 communities use the velvet bean cover crop on more than 5,000 ha, rather than relying on burning practices. 300 families have seen their agricultural production stabilized through farm diversification. By increasing family food security, PCaC has reduced land clearance, protected 2,500 ha of forests, and set aside 15,000 ha for restoration so far.

PCaC contributes to the management of Bosawás by reducing pressure on the reserve and promoting conservation awareness among farmers and ranchers. Tree nurseries have been established in many communities and PCaC has planted 25,000 allspice trees and an additional 10,000 trees (mixed species) in agroforestry systems. PCaC and its participants have created 1,000 ha of 'farmer biological corridors' that serve as buffer zones to Bosawás, with 300 producers putting aside 3.5 to 14 ha of forest each. Thousands of trees have been planted along regeneration corridors that link their lands to the biosphere reserve, with plans to plant a further 10,000 trees.

## SOCIOECONOMIC IMPACTS

Over 3,000 producers in 32 communities are using velvet bean fertilizer on their farms, proving that staple crops (e.g., corn, rice, beans, and plantain) can be grown without slash-and-burn techniques, while improving soil fertility. This has enabled families to improve their diets, meet basic food needs, and sell extra produce at market, even during drought years.

Another impact of the new techniques is that farmers are no longer forced to travel long distances to find fertile lands to cultivate. By enhancing the productivity of lands closer to home, they can dedicate more time to farming, safeguarding their crops, and raising animals. More efficient farms also reduce the need for family labor, giving children more time to attend school.

## SUSTAINABILITY AND REPLICATION

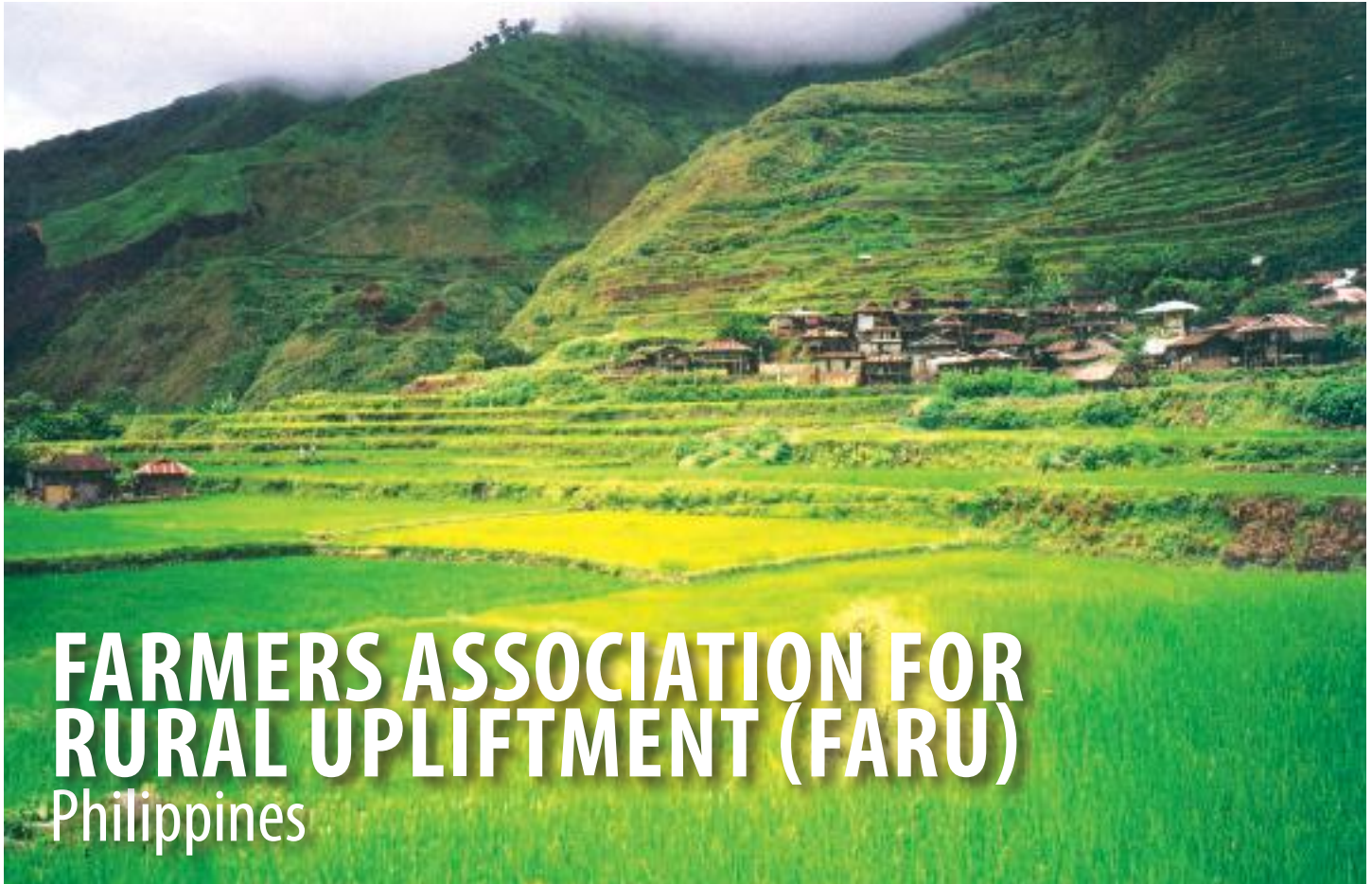
The PCaC program, which began as an experimental soil conservation project with a simple communication strategy, has scaled up to become a

holistic movement with the potential to reconfigure Nicaraguan agriculture. PCaC Siuna is a well-known rural development program in Central and South America, and the Caribbean, and has evolved from a pilot project to an alternative development initiative at the local, national, and international level. As one example of its growth, the network of PCaC promoters grew from just ten farmers in 1993 to over 3,000 farmers by 1999.

Several universities across Nicaragua have incorporated the farmer-to-farmer horizontal information sharing methodology into their graduate programs in rural development. By 1999, PCaC Siuna had expanded to over 60 percent of the communities in the municipality. Knowledge of sustainable agriculture techniques has been passed on to more than 113,000 families in over 80 communities, and 73 associations affiliated with PCaC Siuna are currently seeking legal status.

## PARTNERS

- ***Farmers and Ranchers Union of Nicaragua (UNAG)***
- ***Ford Foundation***
- ***Interchurch Organization for Development Cooperation***
- ***Oxfam Great Britain***
- ***Agricultural Frontier Program (PFA)***
- ***Central American Indigenous and Peasant Coordinator of Communal Agroforestry (ACICAFOC)***
- ***Promoción de Equidad Mediante el Crecimiento Económico (PEMCE)***
- ***Lutheran World Relief***
- ***Echanges et Solidarité 44***
- ***Brot für die Welt***
- ***MS America Central***



# FARMERS ASSOCIATION FOR RURAL UPLIFTMENT (FARU)

## Philippines

### PROJECT SUMMARY

Farmers Association for Rural Upliftment (FARU) is an initiative of the Chananaw indigenous people of Kalinga Province in the Philippines. The initiative aims to protect the environmental integrity of the Chananaw's ancestral domain through improved land management and more efficient agricultural techniques. Catalysts for the formation of the initiative included large-scale mining and geothermal projects, as well as local slash-and-burn agricultural practices. In response, FARU revived an indigenous community conserved area – the *Chananaw Ullikong* – and improved farm productivity through the introduction of locally appropriate technologies and agricultural practices. Since the initiative began, rice production has increased by 36 percent, significantly reducing poverty rates.





## BACKGROUND AND CONTEXT

For generations, the Chananaw indigenous peoples have occupied the mountain ecosystems of Kalinga Province in the north of the Cordillera mountain range of Luzon, the largest island in the Philippines. The land is a mix of tropical forests, agricultural land, and small fresh water fishing ponds, and provides the primary source of local livelihoods and community well-being. The Chananaw have a distinct culture and identity, and are practitioners of indigenous knowledge systems for sustainable land management. Their traditional livelihood is farming, and more recently, the production of handicrafts. The staple crop is rice, supplemented by assorted vegetables and root crops. Agroforestry is also practiced, with the integration of coffee, avocado, and other fruits. The surrounding old growth forests are home to a wide variety of plants and animals.

The Farmers Association for Rural Upliftment (FARU) emerged in 1990 in response to the incursion of modern development into Chananaw lands, leading to the erosion of the local environment and indigenous culture, and deepening poverty and socioeconomic marginalization of a community that depends on the land for its well-being. The practice of slash-and-burn agriculture, or *Kaingin*, also caused

destruction to the watershed, and a loss of forest cover and biodiversity. As these challenges emerged, so did the community's awareness of a need for countervailing activity and action, resulting in the creation of FARU.

### *Origin of the initiative, catalysts, and vision*

The original objective of FARU was to revive the local *Chananaw Ullikong*, or indigenous community conserved area (ICCA), as a mechanism for sustainable community development and cultural rejuvenation. FARU has evolved as a partnership between the Chananaw indigenous peoples, local government units, indigenous peoples NGOs, and the private sector. In the partnership, however, the Chananaw are the key players and architects of activities, action, and institutional development.

Foremost among the project catalysts were large-scale mining and geothermal projects planned for the region. While FARU has been successful in lobbying against these development projects, geothermal companies continue to have designs on this area, making advocacy against large-scale industrial development an ongoing priority for FARU. Exploring sustainable livelihoods options to reduce poverty is an equally high priority.



The organization is committed to five activity areas to achieve these goals, namely: agricultural development, enterprise development, environmental conservation, health and education, and ancestral domain and culture.

### *Agricultural development*

With the goal of food security and self-sufficiency, objectives in agricultural development include improving upland farm productivity; the practice of sustainable indigenous peoples' agricultural technology for poverty reduction and biodiversity conservation; the introduction of gender and culturally appropriate agricultural technologies and equipment; the sourcing and prioritization of traditional knowledge and indigenous knowledge systems in rice production; and the promotion of traditional crop varieties, including drought-resistant rice.

### *Environmental enterprise development*

With the goal of sustainable environmental enterprise development and income diversification, objectives include: building the capacity of local farmers, women, youth and children through hands-on training; provision of capital for environmentally friendly, gender-sensitive and profitable micro-enterprise businesses; the establishment of market outlets for local products in regional centers; promotion of traditional and indigenous knowledge systems in micro-enterprise businesses; and promotion of local agri-business, and community arts and handicrafts.

### *Sustainable environmental management*

With the goal of sustainable environmental management, objectives in environmental conservation include promotion of the indigenous community conserved area; definition of the ICCA, watershed areas, and local conservation areas; maintenance of community-based tree nurseries and associated reforestation activities; and Reducing Emissions from Deforestation and Forest Degradation (REDD+) to mitigate climate change.

### *Health and education*

With the goal of access to quality health and education services, objectives include promotion of an indigenous community healthcare program that combines traditional and indigenous health knowledge systems with modern health services;



promotion and reintroduction of traditional medicine and medicinal plants; and promotion of indigenous culture through holistic education programs.

### *Ancestral domain and cultural preservation*

With the goal of protecting and sustaining the indigenous territory and culture, objectives include documentation of traditional and indigenous knowledge systems; design and implementation of Ancestral Domain Sustainable Management and Protection Plan; advocacy for government recognition of indigenous ownership over their lands; and defending ancestral domain against outside threats, including mining and geothermal development.

### *Organizational Structure*

All decisions of the organization are made by a Farmers Assembly, as part of an Amung – an indigenous policy-making body that brings together various stakeholders to the initiative. It is through the Farmers Assembly that community members contribute to decision-making; they are encouraged to attend meetings to give input on implementation of project activities. Decisions on the operation and





management of the organization are also overseen by a Council of Elders, which acts as the Board of Trustees. The organization retains an executive director, cashier, and bookkeeper, as well as seven additional staff members for project management. The Amung Farmers Assembly collectively comprises all members of the organization, inclusive of the Council of Elders, management, partners and consultants. It is the highest decision-making body of the organization and handles the overall direction, planning and sustainability of the group. The Farmers Assembly also holds an annual general assembly.

## KEY ACTIVITIES AND INNOVATIONS

While the major activities of the organization are based around sustainable agriculture, enterprise development, environmental conservation, health and education, and ancestral domain and culture, these are all pursued under the larger umbrella of the *Chananaw Ullikong* indigenous community conserved area – an area of nine km<sup>2</sup> in which 40 percent is protected forest, 30 percent is reserved for rice terracing, 20 percent is for sustainable *Kaingin* farming, and ten percent is for pasture land. The target population is the approximately 34,000 indigenous inhabitants of the community conserved area, as well as the 65,700 indirect beneficiaries in Tabuk, a neighboring city.

### *Tackling the root causes of poverty*

FARU uses its financial resources to support sustainable livelihoods and micro-enterprise initiatives that reduce poverty. The initiative's Ancestral Domain Sustainable Management and Protection Plan provides for the long-term management of the *Chananaw Ullikong* in a manner that balances biodiversity conservation and the well-being of the local population. Key elements and activities undertaken through the plan include sustainable forest harvesting through enforcement of a community cut-plant customary law policy; similar enforcement of customary laws around reforestation, exclusively with endemic plants and trees; indigenous forest protection practices and watershed management; efforts to enhance nutrient cycling; the integration of fishing ponds and vegetable crops into rice fields; and enforcing customary legal governance of *Kaingin* farming, hunting practices, and pasture land expansion, issuing sanctions and penalties to violators.

### *Using customary knowledge in holistic land management*

Traditional and indigenous knowledge – and governance through customary law – is the cornerstone of FARU's work to protect and conserve the biological diversity within their ancestral domain. Enforcement of customary laws and the use of



traditional knowledge systems has increased food security and reduced poverty. The integration of fishing ponds and the diversification of agricultural crops contributes to community well-being, broadening the base of products for consumption and sale. Traditional watershed management systems – which include land management arrangements that provide for ecological rejuvenation and environmental protection – ensure a continuous supply of water to the community for consumption and irrigation.

A 2000 study reported the production of an average of 90-110 sacks of rice per farmer per hectare and poverty rates of 81 percent. In 2010, the average farmer produced between 122-150 sacks of rice and poverty rates had dropped to 54 percent.

## BIODIVERSITY IMPACTS

The main biodiversity impacts from the project have resulted from land-use planning and zoning in the indigenous community conserved area. The forest conservation area alone has led to the planting of 4,500 native trees, resulting in greater populations of birds, wild pigs, and deer, more water for human consumption and irrigation, and the conservation of over 2,700 forest species.

Prior to the initiative, incidence of forest fires and slash-and-burn for commercial bean production were high, causing local forest destruction. Over the past 25 years, forest fires have decreased by 75 percent and slash-and-burn agriculture has been reduced by 50 percent. The latter is attributable to alternative livelihood and micro-enterprise alternatives offered by FARU.

A 2009 flora and fauna inventory found an increase of plant and animal species since the initiative began. Of note are increased populations of bird species that keep rat and other pest populations in check, demonstrating the value of biodiversity conservation for local agricultural production. Flora and fauna inventories are conducted through indigenous sampling methods as well as more scientific measurement systems.

## SOCIOECONOMIC IMPACTS

Through protection of its ancestral domain, FARU has cultivated local awareness of the linkages between environmental conservation and sustainable living. By the beginning of 2010, direct beneficiaries of the project had increased rice production by 36 percent (compared

to baseline data collected in 2000). The improvements in rice yields, along with diversified crops and sources of income, have led to a reduction in those living below the national poverty line by 27 percent.

Prior to the initiative, the community's agriculture-based subsistence economy produced few surpluses to meet expenses such as school fees or medicine and healthcare. FARU has trained the local population – with a particular emphasis on youth and women – in micro-enterprise development and entrepreneurship of environmentally friendly and marketable products and handicrafts. FARU also provides producers with access to markets in centers of higher population density, expanding outlets for their commodities.

FARU has raised awareness amongst the area's indigenous peoples of the importance of giving local youth access to formal education. Before the initiative began, indigenous farmers placed little value on formal education, as support was needed from youth for farming activities and often the reality of covering school fees would necessitate selling their property or relinquishing use of their land to cover expenses. FARU provides local youth with support to attend school as well as with employment opportunities that supplement household income. In addition to changing local attitudes towards formal education, illiteracy rates have dropped by 63 percent.

FARU created a revolving fund for seed capital to support local entrepreneurs to open businesses that focus on environmentally responsible techniques to improve agricultural productivity. To date, the revolving fund has extended support to over 9,000 indigenous entrepreneurs, a majority of them women, encouraging self-employment and income diversification.

## POLICY IMPACTS

FARU and their partners were instrumental in advocating for the Filipino Indigenous Peoples Rights Act of 1997, which gives indigenous peoples rights over their ancestral lands and recognizes distinct indigenous cultural identity as expressed through traditional science, education, and other customary knowledge systems. Currently, FARU and partners are advocating for formal recognition of the *Chananaw Ullikong* as an indigenous community conserved area and as a Watershed and Local Conservation Area through relevant ordinances at the local, municipal, and provincial levels. If successful, the Chananaw will be the first indigenous peoples in the Philippines to declare their ancestral domain as an ICCA.

Chananaw leaders have gained prominence in regional planning authorities, further helping the causes of the organization. The FARU Council of Elders is a member of the Provincial Policy Advisory Board through the Kalinga-Apayao People-Oriented Development Organizations Network, a leading local policy and advocacy group. The President of FARU is a representative and member of the Municipal Planning and Development Council.

## SUSTAINABILITY AND REPLICATION

The project is largely sustainable due to the creation of the indigenous community conserved area and the forum this has given for drafting a common vision and activities for conservation and sustainable development amongst the local population. The allocation of 3.6 km<sup>2</sup> for forest and watershed protection has also helped to ensure ecological sustainability.

Advocacy for recognition of the *Chananaw Ullikong* as an indigenous community conserved area is tied to the long-term financial viability of FARU. If legally recognized, FARU intends to develop an ICCA business plan based on payments for ecosystem services. If not legally recognized, FARU plans to make their ancestral lands a site for low-impact ecotourism, environmental research, and a venue for environmental education.

Placing a high degree of ownership and decision-making power within the community has contributed to the overall sustainability of the initiative. By basing decisions and land management plans on indigenous governance systems, FARU has facilitated a socially and culturally embedded program model that resonates with the local population. At the same time, FARU emphasizes a multi-stakeholder partnership model and owes its financial sustainability and the funds needed to sustain operation of the initiative to its partners.

A neighboring indigenous community has adopted the FARU model, particularly its emphasis on reviving and mainstreaming traditional knowledge and indigenous knowledge systems. FARU has also received a number of requests to share its project model with other groups through community-to-community and farmer-to-farmer knowledge exchanges and learning programs. To date, FARU has shared its project model with more than ten communities in *Kalinga* province.

## PARTNERS

- **Barangay Local Government Units:** Technical, financial, and legal support
- **Kalinga Mission for Indigenous Communities and Youth Development, Inc:** Technical assistance, networking, support for ancestral domain
- **Kalinga Apayao People Oriented Development Organizations Network:** Technical assistance, capacity building trainings
- **Kalinga Apayao State College:** Technical assistance for forest inventories
- **National Commission on Indigenous Peoples, Kalinga:** Technical assistance, networking
- **Municipal and Provincial Governments of Kalinga:** Financial support, rice terrace rehabilitation
- **Chamber of Kalinga Producers, Inc:** Financial support, capacity building for micro-enterprises
- **Philippine Tropical Forest Conservation Foundation, Inc (PTFCF):** Financial support, support for ancestral domain

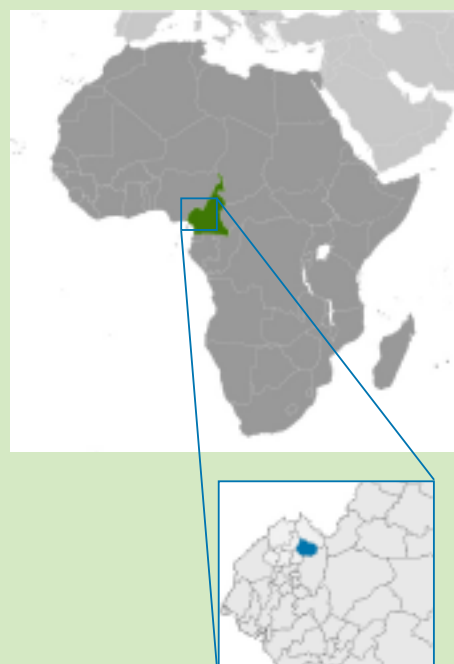


# RIBA AGROFORESTRY RESOURCE CENTRE Cameroon

## PROJECT SUMMARY

Riba Agroforestry Resource Centre is a community-based organization working in mountainous northwest Cameroon, close to Kilum-Ijim Mountain Forest. The Centre promotes sustainable tree-based farming to rehabilitate watersheds and degraded land and generate income for the local community. A rural resource center provides training in agroforestry and nursery management, watershed protection, beekeeping, microfinance, and marketing of tree seedlings and farm produce.

The initiative's tree-based farming system has successfully halted deforestation and improved soil fertility, while sales from tree nurseries and honey are supporting sustainable livelihoods. The initiative is guided by a self-help ethos, which has served to empower members of the community, promote gender equity, and instill a belief in the community's collective capacity to achieve positive change and a sustainable future.





## BACKGROUND AND CONTEXT

Riba is located in the mountainous Bui Division of Cameroon, 2,000 m above sea level. A prominent feature of the local landscape is the Kilum-Ijim Forest, the largest remaining Afromontane forest in West Africa. The forest is a vestige of regional biodiversity, and contains a wide range of unique ecosystems, flora, and fauna. The local economy is highly dependent on agriculture. The mountainsides and hills surrounding Riba have been heavily logged and degraded, resulting in a loss of soil fertility, biodiversity, and economic security for the local population.

Riba Agroforestry Resource Centre (RARC) was established to promote sustainable tree-based farming systems as a way of restoring soil fertility and improving local livelihoods. Most RARC activities are carried out through a community-funded Rural Resource Centre where trainings are provided to local farmers in agroforestry and nursery management, watershed protection, beekeeping, and marketing of tree seedlings and farm produce.

### *A local 'center of excellence' in agroforestry*

The group began as the Riba Young Foresters Club in 1995 with the aim of promoting sustainable agroforestry. In its early years, the club was assisted by several Peace Corps volunteers to become legally registered as a 'common initiative group'. This is a legal designation under Cameroon land law, which designates community leaders as official custodians of government-owned land, giving them control over its distribution and responsibility for resolving any emerging conflicts. In 2002, the architects of the initiative agreed on a project structure that would include training activities, business development, and farming. After fundraising within and outside the community, and following establishment of demonstration farms to test activities and techniques, Riba Agroforestry Resource Centre was born.

RARC now covers seven hectares of land with a woodlot, a system of tree hedges, a tree nursery, and fertile fields where wheat, beans, and potatoes are grown. The Rural Resource Centre grounds house offices, a training hall, and dormitories for visiting





farmers groups. RARC oversees 26 satellite farmer groups, all of whom are applying their agroforestry techniques. Satellite group members range on average between 10 and 45 farmers. Each group is engaged in soil fertility restoration activities, nursery management of native fruit trees, and tree domestication techniques.

## KEY ACTIVITIES AND INNOVATIONS

Riba Agroforestry Resource Centre helps farmers to improve soil fertility and, by extension, crop yields through the planting of leguminous plants (what the initiative refers to as 'fertilizer trees'), which fix atmospheric nitrogen. The group also trains farmers in contouring, alley cropping, fallows, and composting, all of which help to improve soil fertility and combat soil erosion.

In addition, RARC supports farmers to domesticate superior varieties of native fruit trees – African plum (*Dacryodes edulis*), bush mango (*Irvingia gabonensis*), and other species – thereby reducing farmer dependency on cash crops such as cacao and coffee, which are often subject to dramatic price fluctuations. Tree domestication is undertaken through a participatory approach wherein farmers work cooperatively with researchers from the World Agroforestry Centre and other technical agencies

to develop and select varieties of trees that are adapted to individual farms. In this activity area, RARC has helped to introduce non-mist propagators, marcotting, grafting, budding, and pre-treatment of seeds for quick germination.

RARC serves as a learning platform for farmers to share their experiences peer-to-peer and to receive technical guidance that is tailored to their livelihood needs. Knowledge exchanges have been carried out in the areas of tree domestication, soil fertility restoration, biodiversity conservation strategies, watershed management, organic farming, honey and wheat production, environmental education, and the cultivation and promotion of medicinal plants.

## BIODIVERSITY IMPACTS

The Bamenda Highlands of the northwest region of Cameroon are home to a wide variety of animal species, including endangered primates such as chimpanzees and Preuss's monkey, as well as the critically endangered western lowland gorilla. Endemic plant species include *Oncoba lophocarpa* and *Chassalia laikomensis*, the latter classified as critically endangered. The area is also home to a number of medicinal plants, including *Prunus africana*, *Allanblackia gabonensis*, and *Entada abyssinica*.





### Combating deforestation and species loss

The region and its ecosystems have suffered extensive deforestation. Between 1987 and 1995, 25 percent of the forest was lost in one area of the Bamenda Highlands. RARC has worked to rehabilitate degraded lands through sustainable agroforestry, which has paid dividends for local farmers in improved on-farm productivity. From 2008 to 2009 alone, over 117,000 seedlings were grown and planted on community farms. By integrating traditional forest species into farming systems, RARC has successfully reduced pressure on surrounding forests. Where timber and non-timber forest products were previously harvested in the forest – exerting pressure on an already fragile and declining ecosystem – they are now harvested on farms. The region's watersheds have also been protected. Over 16,000 trees have been planted in riparian strips to improve availability of water in the community, which is particularly important during the dry season.

Many of the trees now grown on community farms – for example *Enantia chlorantha*, *Voacanga africana*, and *Prunus africana* – are threatened Afromontane species, which also provide important habitat for

local bird and monkey species. There has been a resurgence of one native tree species (Iroko) and one monkey species (red Colobus), whose populations were reportedly dwindling. Birds have also proliferated on community farms. When the initiative began, the grey beaked Camaroptera warbler was one of the only birds seen in large numbers. By 2010, 21 different species of birds were living and thriving in the region, with an additional 19 visiting on a seasonal basis. RARC has created a biodiversity impact assessment form, which is completed by participating farmers and RARC outreach officers who travel door-to-door.

### SOCIOECONOMIC IMPACTS

The socioeconomic benefits of RARC's work are closely linked with its biodiversity benefits. Diversification of tree species and subsequent proliferation of wildlife and soil micro-biodiversity have translated to better agro-ecological functions, more productive farming systems, a reduced need for artificial fertilizers and pesticides, and improved local incomes.



For instance, RARC promotes intercropping with nitrogen-fixing tree species, such as *Calliandra*, *Sesbania* and *Tephrosia*. This has improved soil fertility and resulted in a doubling of crop yields. One farmer, who previously harvested only one 50 kg bag of maize per year now harvests up to six bags from the same plot of land. Boundary trees have also been planted on community farms to act as windbreaks. A woodlot on the hilltop provides fodder for livestock and a habitat for bees.

### *Income generation*

Sales from the RARC tree nurseries have rapidly improved local incomes over the last five years. Early on, nurseries made little in the way of profits. Once they were well established, however, they began to generate significant revenues. On average, farmers who built and maintained tree nurseries began to see a return on their investment in two years. Annual sales from RARC nurseries rose from US\$90 in 2003 to US\$1,350 in 2009. For satellite nurseries, over the same period, sales grew from US\$20 to US\$100. Similar income benefits were seen over the same period from potatoes, where revenues rose from US\$800 to US\$2,250.

A further income source has been the commercialization of honey produced from flowering trees. Total honey production during the period from 2007 to 2009 amounted to US\$13,000. Beehives have been established in and around forest reserves, and apiaries have been created using both native and non-native tree species. The honey value-chain sees local farmers producing and selling beeswax and honey-based drinks, which also supplement incomes. Other farmers are engaged in raising medicinal plants, which have improved incomes and served as a secondary healthcare system for the community, saving them money on hospital visits. Others still are engaged in livestock farming, another area of training provided by RARC. Where farmers in the past had a difficult time finding enough fodder at certain times of year for their livestock, now through the planting of *Acacia* and *Calliandra* trees, they produce enough fodder for use in the dry season.

### *Co-benefits from agricultural improvements*

In disseminating its model of agroforestry, RARC has been particularly successful in reaching women and youth: over 40 percent of RARC members are women and over 30 percent are men under the age of 35 years.



This is relevant because women traditionally handle most of the agricultural activities in the area. Agricultural improvements and greater farm productivity have reduced the drudgery of women's work, availed more time for other activities (e.g. reducing the need to travel long distances to collect wood for fuel), and improved their incomes. Revenues from biodiversity conservation activities have also been reinvested into school fees, hospitals, local infrastructure, alternative energy technology, reforestation projects, and water purification. Other socioeconomic benefits from the project include improved food security, a wider range and abundance of local food markets in the communities, greater access to micro-finance and loans, and improvements in community health.

## SUSTAINABILITY AND REPLICATION

RARC members have acquired knowledge and skills that are enabling them to scale up agroforestry activities in the community. The Rural Resource Centre is community owned and operated, and is used to cultivate crops with desirable agronomic traits such as high yields, good taste, early and regular fructification, pest and disease resistance, and climate resilience.

The centre is the heart of the project's sustainability. It serves as a training and demonstration centre to build and expand local capacity, and is where local farmers produce plant seeds, grafts, marcotts, and cuttings.

Demonstration sites are owned and managed by local farmers, who use their farms to disseminate new techniques and skills to other farmers and visiting communities. Local government staff has even visited to learn what agroforestry techniques are proving successful. As a result of demonstration farms, the adoption rate of agroforestry techniques has increased more than 50 percent at both the household and community levels.

To further its long-term sustainability, RARC is also engaged in a primary and secondary school program, where students visit demonstration farms and cultivate their own farms at their respective schools. Involving children ensures the passing of a carefully cultivated conservation ethic from one generation to the next.

The RARC model has been shared with more than 80 communities and has been actively replicated in 18. Knowledge is exchanged through trainings and workshops, field days, and exchange visits between farmer groups. The RARC model is also being promoted by the World Agroforestry Centre, which has developed training materials based on the RARC experience.

## PARTNERS

- **World Agroforestry Centre (ICRAF):** Financial support, capacity building, training, farm equipment, processing machinery, motorbikes, polythene pots and water system for nursery
- **Heifer International:** Small livestock (rabbits, sheep, poultry, pigs)
- **Grassfield Participatory and Decentralized Rural Development Project (co-funded by Cameroonian government):** Tree planting to protect water catchment areas
- **Netherlands Development Organization (SNV):** Beekeeping initiative, support for timber and non-timber forest products, identification and protection of rare and endangered species in Bamenda highlands
- **Food and Agriculture Organization (FAO):** Financial support for nurseries, training in production of high-value species
- **Ministry of Agriculture and Rural Development:** Improved seeds
- **Traditional and administration authorities:** Implementation of local regulations and the governance of agro-pastoral activities, conflict resolution



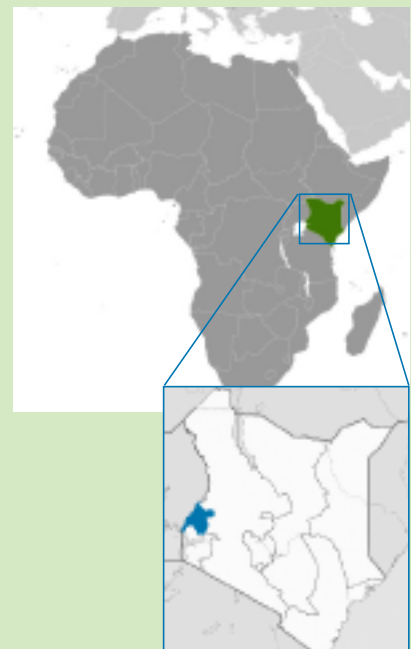
# MULIRU FARMERS CONSERVATION GROUP

Kenya

## PROJECT SUMMARY

Muliru Farmers Conservation Group is a community-based organization located near Kakamega Forest in western Kenya. The group generates income through the commercial cultivation and secondary processing of a native medicinal plant, to produce the Naturub® brand of medicinal products.

The enterprise reduces pressure on the biodiverse Kakamega Forest by offering an alternative to the exploitation of forest resources, while the commercialization of the medicinal plant has heightened local appreciation of the value of the forest's biodiversity. Over half of the project participants are women and 40 percent rely entirely on this initiative for their income. A portion of the enterprise's revenues is invested in forest conservation and biodiversity research.





## BACKGROUND AND CONTEXT

Kakamega Forest is the last surviving rainforest in Kenya. Located in the country's Western Province, it is home to a unique variety of ecosystems and diversity of flora and fauna. The forest is a cornerstone of local livelihoods, income, and well-being, providing timber, fuel wood, fodder, building materials, medicinal plants, and fresh water. Kakamega is under threat from population growth, local economic pressure, and extractive industries, and is steadily being degraded. Human population density around the forest is as high as 1,200 people per square kilometer. High levels of poverty have led the 35,000 households in adjacent communities to overexploit the forest resources, harvesting timber and non-timber forest products in an unsustainable manner.

### *Ocimum kilimandscharicum* - a potent remedy

One species native to Kakamega Forest is African blue basil (*Ocimum kilimandscharicum*). This medicinal plant has been used by the local population for generations, notably for the treatment of colds, flu, and coughs. African blue basil leaves contain essential aromatic oils and are traditionally added to boiling water to treat respiratory problems. The local population also uses the plant as a mosquito repellent, a source of nectar for bees in apiculture, as a flavoring agent, and to protect stored grains from pests.

The Muliru Farmers Conservation Group formed in 1997 with a goal of sustainably cultivating *Ocimum kilimandscharicum* as a means of conserving the Kakamega Forest and offering forest-adjacent communities an alternative income source. The initiative combines traditional knowledge with modern science and technology, and brings together a diverse partnership – rural farmers, research institutes, and the private sector – to harness the commercial value of this medicinal plant. The organization works with local farmers to develop purified extracts of *Ocimum kilimandscharicum* which are developed into products under the brand Naturub®.

## KEY ACTIVITIES AND INNOVATIONS

The Muliru Farmers Conservation Group mobilizes communities living adjacent to the Kakamega Forest to cultivate *Ocimum kilimandscharicum*. As a high-value commercial crop, the medicinal plant is an appealing livelihood option for the otherwise economically marginalized local farmers. Training is provided on domestication and processing. Processing to extract the essential oil was previously undertaken on a small-scale basis, mainly at the household level. In 2005, Muliru built a centralized processing facility. Farmers are supported to gather the *Ocimum kilimandscharicum* leaves and transport them to the processing facility. At the facility,





the leaves are dried and processed using hydro-distillation equipment, which produces purified essential oils that are used in the production of Naturub® balms and ointments.

### *Production and marketing of Naturub*

Since the processing facility opened, over 770 tons of *Ocimum kilimandscharicum* leaves have been processed, producing over 700 kg of essential oil. Over 400,000 units of Naturub® products have sold in urban and rural areas of Kenya. The products have received wide acceptance in the market and are competitive with major international brands. Total revenue from the project thus far exceeds US\$70,000. Over 360 rural households cultivate the plant on smallholder farms and the acreage under cultivation has increased by 700 percent, from 2.5 ha in 2005 to 20 ha in 2010.

In addition to producing and marketing Naturub® products, Muliru Farmers Conservation Group offers workshops on biodiversity conservation and alternative livelihood solutions based on sustainable natural resource management. It also operates several indigenous tree seedling nurseries, which are used for on-farm planting, reforestation, and agroforestry trainings.

The most noteworthy innovation of the project is a new brand of registered medicine, developed and commercialized by local communities. Not only is cultivation of the plant community-driven, the processing facility is fully owned, operated, and managed by local farmers. In addition to improved livelihoods, local community members have attained new knowledge in modern science and technology, industrial processing and production, and entrepreneurship, including in sales, marketing, and management. The initiative effectively combines traditional knowledge with modern science and technology, and is an effective model of a multi-stakeholder partnership.

### *Organizational structure*

A Management Board leads the Muliru Farmers Conservation Group, which is the highest governing authority within the management structure of the MFCG medicinal plant enterprise. The primary responsibility of the board is to protect the farmers' interests and ensure they receive a decent return on their produce. The initiative's Field Extension Department provides training to farmers and links them to the Muliru enterprise. The department guarantees the quality of the raw materials used in





processing, and ensures that farmers are paid a fair price for their produce. They also assist farmers in preparing farm plans and record keeping.

After processing, Muliru's products are marketed by the initiative's Sales and Marketing department. Through this office, the organization has partnered with private marketing companies to undertake packaging design, market surveys, advertisements, and the distribution of products to different retailers. Through these partnerships, MFCG has been able to engage with large Kenyan retail chains such as Uchumi, Nakumatt, and Tusky's Supermarkets.

Finally, the group works at the local level through three organizational units. The Farmers Cluster Representatives work on behalf of small clusters of farmers, representing their views during planning meetings. The Plant Operators are responsible for the initial distillation of essential oils from the raw plant materials. Nine youths – four female and five male – were trained in essential oil distillation using hydro-distillers. They have since acted as 'Trainers of Trainees', passing on training to others in the operation and facilitating training of schools and colleges that visit the enterprise.

## BIODIVERSITY IMPACTS

The main biodiversity impact of the project has been the conservation and sustainable use of the Kakamega Forest, protecting its estimated 380 plant and 350 bird species. As well as improving local attitudes to conservation of the forest, Muliru's sustainable harvesting of *Ocimum kilimandscharicum* has been undertaken as part of the devolved sustainable management of the forest under the Mwileshi Community Forest Association.

### *Mwileshi Community Forest Association*

The Kenya Forest Service (KFS) is in the process of developing participatory forest management plans for areas across the country in close consultation with rural communities. Communities are able to partner with KFS and local county councils in the form of Community Forest Associations (CFAs). CFAs enter into agreements with KFS for management of a forest under a management plan. Kakamega was among three forests in Kenya selected to pilot collaborative forest management between local communities and the Forest Department. This led to the formation of the Mwileshi CFA, registered in June 2009, with



an initial 25 forest user groups (these have since grown to 31). These groups include all community-based organizations around the forest involved in conservation activities, including the Muliru Farmers Conservation Group.

The association is predominantly involved in the management and conservation of Kakamega Forest, which includes maintaining a tree nursery and active afforestation; 10,000 seedlings were planted in September 2010 alone. They conduct all activities according to the Kakamega Forest Management Plan. The association is also involved in sensitizing communities on conservation, monitoring the forest condition, training groups in nursery management, developing ecotourism, and assisting in forest policing. Environmental education is largely carried out by Muliru Farmers Conservation Group in partnership with local primary and secondary schools through the Kakamega Environmental Education Program.

### *Changing attitudes to conservation*

Based on surveys conducted by Muliru, of the 360 households participating in cultivation of *Ocimum kilimandscharicum*, over 85 percent participate in forest conservation activities. The survey, broken down

by activity, finds: 49 percent promote conservation awareness among other community members; 59 percent have reduced their collection of firewood, fodder, and timber; 37 percent have deterred other community members from misusing forest resources; and 5 percent have reported poaching and illegal forest activities to local authorities.

The harvesting of *Ocimum kilimandscharicum* is environmentally friendly because it regenerates naturally and gives the community a source of income that does not damage the forest. In addition to reducing local dependence on unsustainable extractive activities, the initiative has improved local awareness about the threats to Kakamega Forest and the importance of conserving biodiversity. Naturub® products bear a distinctive message on their packaging regarding biodiversity conservation.

## SOCIOECONOMIC IMPACTS

Muliru activities cover five districts in the vicinity of the Kakamega Forest, with a total participating population of roughly 2,500 community members, or 360 households. This is an economically marginalized community with few livelihood options. Over 40 percent live below the poverty line and over 30





percent of households own less than 0.4 ha of land, an important statistic in an economy where land is tantamount to earning capacity.

The Muliru Farmers Conservation Group has positively altered the local economy, providing greater employment and income-generation opportunities. Eight community members are employed on a full time basis to operate the initiative. Day-to-day management needs include field supervision, quality control, and oil distillation. More important, on-farm cultivation of *Ocimum kilimandscharicum* has created nearly 900 jobs for smallholder farmers. Participating community members receive a share of revenues from Naturub® products three times a year, and have increased their income by an average of 300 percent since the initiative began.

### Co-benefits

There have been a number of secondary benefits of Muliru's work. Based on surveys conducted by the organization, 31 percent of participating community members have used income generated from the project to start small businesses to produce alternative sources of income. Additionally, 83.5 percent use their income for food security needs; 57 percent to

cover school fees; 26 percent to purchase clothing; 17.5 percent to buy livestock (an additional source of income); and 7.65 percent have invested in housing renovations. Hands-on training and workshops are provided by private sector and NGO partners, giving community members new and marketable skills.

The initiative is also a source of community empowerment. Smallholder farmers are directly involved in the decision-making and strategic direction of the organization. In addition to greater access to finance, information on sustainable farming, and training opportunities, the initiative provides a forum for social networking among community groups, thereby improving community cohesion, trust, and social capital – all of which are essential ingredients of the collective action needed to address common challenges.

The initiative works to ensure gender equality in all of its activities, and has aimed to ensure that women receive equal access to productive resources and equal engagement in marketing and enterprise development. Women form the majority of participating community members and have been elected to leadership positions.

## SUSTAINABILITY AND REPLICATION

Muliru has established a ten-year plan for the distribution of net profits from the sale of Naturub® products, which will be allocated as follows: ten percent for conservation activities around Kakamega Forest; ten percent for a community development fund, to be used for projects in and around Kakamega Forest; 60 percent for distribution to participating members of the Muliru Farmers Conservation Group; and 20 percent for a conservation research fund for further diversification of other natural products.

According to Muliru projections, the organization is financially sustainable and has significant room for growth. Naturub® is the first registered natural medicine by the Pharmacy and Poisons Board of Kenya. As a pioneer in this space, the organization has a competitive edge. Market intelligence has established the total market for Naturub® to be 100 million Kenyan Shillings per annum, with rural markets representing the largest market share and greatest potential for growth.

Under SWITCH Africa Green project (SAG), the Muliru Farmers Conservation Group is being supported to scale up production and commercialization of medicinal plants through capacity building and adoption of sustainable production practices. This is aimed at transforming the ongoing community-based activities into small and medium-sized green social enterprises that enhance livelihood improvement and environmental conservation. The SAG project supports the development of green businesses, eco-entrepreneurship, and use of sustainable consumption practices among micro, small, and medium enterprises (MSMEs) in the 6 pilot countries – Burkina Faso, Ghana, Kenya, Mauritius, South Africa and Uganda – creating an environment for knowledge exchange and replication of best practices.

Muliru has shared its project model and experiences with communities from Kenya, Uganda, Tanzania, South Africa, and Nigeria. Over 830 people have visited the initiative to learn first-hand about their cultivation, production, and marketing techniques. Knowledge generated from the project has been exchanged through lectures, demonstration activities, storytelling, exchange visits, radio, and television. Two communities are now actively applying the initiative model: The East Usambara Farmers Conservation Group based in Maramba, Tanzania (adjacent to the East Usambara mountain forests) and The Budongo Community Development Organization based in Masindi, Uganda (adjacent to the Budongo forest reserve).

## PARTNERS

- **International Centre for Insect Physiology and Ecology:** Technical assistance
- **University of Nairobi:** Research assistance
- **World Agroforestry Centre (ICRAF):** Technical assistance
- **Kenya Forestry Research Institute (KEFRI):** Technical assistance
- **Kenya Wildlife Service:** Technical assistance
- **GBDI:** Marketing assistance
- **UNDP/GEF Small Grants Programme:** Funding
- **Ford Foundation:** Funding
- **The BioVision Foundation:** Funding
- **The MacArthur Foundation:** Funding
- **German Development Service:** Funding
- **PACT-Kenya/USAID:** Funding
- **WHO Multilateral Initiative on Malaria/Research and Training in Tropical Diseases (TDR):** Multilateral Initiative on Malaria/Research and Training in Tropical Diseases (TDR): Funding
- **SWITCH Africa Green (SAG):** Support for upscaling commercialization of medicinal plants funded by the EU and implemented by UNEP in collaboration with UNDP and UNOPS.



# WHAT DO THE CASES TELL US?

These cases show the major impact that small-scale agriculture has on local forests, and the need to accommodate the legitimate food security needs of local communities within any sustainable forest management plan. They also indicate that improving the efficiency of local agriculture and thereby reducing pressures on local forests is not just a matter of identifying new farming practices and technical fixes, but of developing the willingness and capacity to use these new means through education, training programs and extension services. Ultimately, adopting an integrated approach to land use that recognizes the link between local farming and forest use, and accommodates them both within a larger framework of landscape management, is the best way to create a sustainable working landscape that delivers continuous climate benefits.

With the questions posed at the beginning of Theme 2 in mind, the following lessons emerge.

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**Food Security and Forests:** Smallholder agriculture is a bulwark of local food security, but also a primary source of pressure on local forests. Successful community-based forest initiatives address this essential forest-farm connection.

Agriculture is the cornerstone of the rural economy, essential to household subsistence and income. Even in communities where forest-related income is substantial, local field crops, livestock, and produce from home gardens are the basis of community food security. But small-scale agriculture is a source of forest conversion as well. While large-scale forest clearance for palm oil plantations and commodity crops is a major source of deforestation in many areas, smaller-scale forest clearance by smallholder farmers is still a significant source of forest pressure at the community level, and, along with fuelwood collection and other forest extractions, is an important driver of the degradation of local forests.

Low farm productivity, a lack of sufficient agricultural land to meet local demand, and continued reliance on traditional slash-and-burn farming techniques are common in many forest communities, driving the

clearance of local forestland for additional crops and livestock. In Siuna in northern Nicaragua, for example, declining soil fertility from poor farming practices drove local farmers to invade the Bosawás Biosphere Reserve, using slash-and-burn to open up new farm tracts deep within the Reserve. Likewise, in Kalinga Province in the Philippines, traditional use of slash-and-burn led to a loss of biodiversity and degradation of the forest and watershed within the Chananaw people's indigenous lands. Invasion of agriculture into forest areas due to population increase was also a major source of forest pressure in the Didy Forest in Madagascar, combined with illegal logging of valuable rosewood timber.

The cases in this section demonstrate the effectiveness of dealing directly with the food security needs of the local population as a first step in sustainably managing community forests. In fact, the vast majority of Equator Prize-winning forest communities have incorporated local agriculture in their community forest management plans, understanding that forest use and agriculture are inevitably intertwined, and that failing to account for local agricultural patterns and food needs will undermine their community forestry goals as farmers turn to local forests as a land bank for farm expansion.

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**Forest-Friendly Options:** Agricultural intensification, agroforestry, and adoption of alternative crops and markets can increase the productivity of local agriculture and integrate it into the forest structure, easing the pressure for forest conversion and overuse.

Successful programs to decrease the pressure from agriculture on local forests usually begin with efforts to intensify local agriculture, often with a combination of new, more suitable crop varieties, new methods and farm technology, and better water management. These strategies address the low productivity that typifies most smallholder agriculture, allowing local farmers to make better use of existing farm plots to generate more income and a better local food supply without turning to the conversion of forest lands.

**“Adopting an integrated approach to land use that recognizes the link between local farming and forest use, and accommodates them both within a larger framework of landscape management, is the best way to create a sustainable working landscape that delivers climate benefits.”**

In the Didy Forest in Madagascar, local farmers have abandoned slash-and-burn cultivation in favor of a rice intensification program that has doubled and even tripled local rice yields. This is complemented by a farm diversification program that includes potato and bean processing, livestock rearing, fish farming, and establishment of home gardens. In addition, farmers have added off-season crops such as onions, bananas, soybeans, and corn that can earn premium prices and improve the variety of foodstuffs available locally, resulting in better household nutrition.

Similarly, in Siuna in northern Nicaragua, alternative farming practices championed by the Farmer-to-Farmer program transformed local subsistence agriculture based on slash-and-burn to a sustainable system based on planting legumes as cover crops to improve soil fertility, practicing contour plowing and minimum till practices to reduce erosion, using plant-based organic pesticides, and diversifying their crop mix and livestock rearing. This has enabled families to meet their basic food needs and sell their excess produce at market, reducing the impetus for new forest clearance.

Encouraging agroforestry that utilizes tree species with high agricultural value is another strategy frequently used to address local food supply and farm income issues in a way that maintains forest canopy and strengthens the bond between farmers and the forest. The Riba Agroforestry Resource Centre in Cameroon promotes tree-based farming systems in which leguminous trees are inter-planted in local fields to fix nitrogen and improve soil fertility, typically resulting in a doubling of crop yields. The cultivation of indigenous fruit trees such as African plum and bush mango is also encouraged to diversify farmer income. Likewise, in Siuna, farmers have planted some

25,000 allspice trees and other useful species that can yield income without tree felling. In fact, many of these trees are being planted on ‘Farmer Biological Corridors’—buffer zones around the Bosawás Reserve consisting of land contributed by individual farmers to protect the park, resulting in a net increase in forest canopy.

The Muliru Farmers Conservation Group in Kenya offers another approach to supporting local farmers with forest-compatible activities. In this case, the community has commercialized the production and sale of an extract from a medicinal plant indigenous to the adjacent Kakamega Forest. The plant extract is purified and formulated into products under the brand Naturub®, and used to treat flu, colds, aches, and insect bites. Rather than collect the plant from the forest, it is cultivated by area farmers on a contract basis, providing a secure income source for nearly 900 smallholder farmers, and avoiding any forest impact. The Muliru Farmers Conservation Group works under the auspices of the Mwileshi Community Forest Association, to which the Kenyan government has devolved local forest management rights.

In all of these cases, it is critical to understand that these farm-focused programs are not undertaken on their own, but in concert with forest restoration and protection measures, so that the two activities are linked in farmers’ minds as one effort, emphasizing the supportive relationship between farm and forest.

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**Education and Training:** Locally adapted training programs, demonstration sites, extension services and community education efforts are essential to build the capacity and willingness to change local agricultural practice.

Education drives the change to forest-friendly agriculture. Changing ingrained local practices requires both a convincing rationale and an effective program to develop the technical and business skills required. For this reason, successful community forest initiatives spend a good deal of effort crafting farmer education and training programs and developing demonstration sites and model farms.

The cases in this section showcase several different models of agricultural extension, demonstration, and farmer-to-farmer learning. In Madagascar, Adidy Maitso has established a cadre of farming facilitators and extension agents who travel to local farms to conduct

trainings, workshops, and face-to-face demonstrations of farming techniques. This is complemented by a group of model farmers who act as best practice exemplars, and a training center for larger group classes and coordination of the extension agents and facilitators. To insure widespread dissemination of both farming news and environmental conservation information, Adidy Maitso also uses radio broadcasts produced by the farming facilitators, extension agents, and model farmers.

**“Local farm-focused programs are not undertaken on their own, but in concert with forest restoration and protection measures, so that the two activities are linked in farmers’ minds as one effort, emphasizing the supportive relationship between farm and forest.”**

The Riba Agroforestry Resource Centre is a community-funded training facility offering courses in agroforestry, nursery management, watershed protection, beekeeping, and marketing of farm produce and tree seedlings. The Centre itself spreads over seven hectares, with a nursery, demonstration fields, a training hall, and dormitories for visiting farmer groups. The Centre oversees 26 satellite farmer groups in the surrounding area, acting as a knowledge-exchange platform for these farmers to share their experiences on a one-to-one basis.

In Siuna, the Farmer-to-Farmer Program uses a ‘horizontal’ learning approach based on direct information sharing between participating farmers. It employs a number of different communication strategies, including workshops, photo exhibits, radio programs, folk music, and plays, and has morphed into a broad agricultural movement through which farmers are adapting to the challenges of rural forest agriculture.

All of these programs are distinguished by their direct and personal appeal to local farmers, with trainings, facilities, and communication methods tailored to the local audience, and delivered by farmers and trainers with local roots and credibility. In addition,

all are embedded in larger programs that relate the agricultural techniques and marketing approaches to broad community goals that link food security and forest care.

It should also be noted that training programs for Adidy, Riba, and Siuna were not confined to a single project or village, but were designed to create impact over a wide area, creating a network of ‘local experts’ that were linked throughout the landscape in a joint effort to upgrade the prevailing agricultural practice. As such, these programs act as a primary tool in scaling up successful community-based programs so that so that they can effect landscape-level change.

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**Integrated Landscape Management:** Sustaining local forests can be helped by adopting an integrated approach to land management that sees the interdependence of local agriculture, forestry, and other land uses.

While forests and forest management is the focus of this book, these cases show that a simple focus on forests alone is too restrictive and ultimately ineffective. In the village and community context, forests fit into a larger landscape in which several different land uses occur simultaneously that support the community’s well-being, resulting in a mosaic of forests, fields, pastures, waterways, dwellings and other physical infrastructure. Sustainably managing community forests requires integrating these sometimes-disparate activities so that they are compatible within the landscape, and sustainable when taken together. Successful community forest initiatives tend toward this whole-landscape approach to land management, and incorporate technical and educational programs that link these land uses and attempt to manage the trade-offs that may be required for them to coexist.

**“Successful community forest initiatives tend toward a whole-landscape approach to land management.”**

The Farmers Association for Rural Upliftment (FARU) in Kalinga Province in the Philippines explicitly adopted this kind of landscape approach in its program to manage the Chananaw’s indigenous lands. Their



program addresses local forest degradation by revitalizing their traditional land management system, in which protected forest areas are integrated with designated zones for rice culture, pasture land, and vegetable crops, all governed under a detailed system of customary law that specifies land uses and harvest practices. The landscape is looked upon—and governed—as a single integrated system from which many benefits are produced that, while clearly differentiated, are nonetheless interrelated and mutually supportive.

The FARU program is also a good reminder that communities themselves are part of the landscape, with their social and economic cultures as important to landscape dynamics as the ecosystems and natural processes at work on the land. The Chananaw social system provides the backdrop and platform for their successful land management scheme. Successful

forest initiatives in general reflect local cultural values and the community's social and economic context. This helps to explain the heavy emphasis that most community-based forest initiatives have on creating economic and social opportunity and empowerment, rather than just transferring technology and management systems to address technical forestry and farming issues.

**“Communities themselves are part of the landscape, with their social and economic cultures as important to landscape dynamics as the ecosystems and natural processes at work on the land.”**







## Theme 3

# Forest Restoration





**T**he cases in this section examine the motivations, means, and benefits associated with community-initiated restoration activities in local forests. These restoration efforts vary from simple augmentation plantings in standing forests, to full-scale regeneration of forests in logged-over areas and on converted agricultural lands. The fact that restoration is so common among community forest initiatives indicates the widespread nature of local forest degradation. But it also speaks to the determination of forest communities not just to halt forest decline, but to return their forests to health.

Restoration is generally driven by the goal of increasing forest productivity to better support local livelihoods, or to regain ecosystem services that forests delivered in the past, such as watershed management or coastal protection. Because of its ability to increase local forest resilience and regain ecosystem functions, forest restoration is also a common strategy for local climate change adaptation. Both of these strategies are on display in the cases that follow. No matter what the motivation, restoration efforts often become a defining activity for local forest initiatives, allowing local people to refine and act on their vision for the community forest resource.

## CASES IN THIS SECTION

- **Kijabe Environment Volunteers (KENVO)**, Kenya
- **Comité Villageois de Développement d'Ando Kpomey (Village Development Committee of Ando Kpomey)**, Togo
- **Empresa Forestal Integrada de Bayamo (Integrated Forestry Enterprise of Bayamo)**, Cuba
- **Abrha Weatsbha Community**, Ethiopia
- **Trowel Development Foundation**, Philippines

## UNDERSTANDING THE CASES

To understand the breadth of restoration efforts, the motivating factors behind them, how they contribute to climate change mitigation and adaptation, and their impact on the overall success of community forest initiatives, readers should consider the following questions as they read the cases:

- **Motivation and Results:** What do communities hope to achieve with their forest restoration work? How does it relate to other forest management activities? Is it worth the investment of time and money?
- **Mitigation Potential:** How does forest restoration affect the climate change mitigation potential of community forest initiatives? How does community-based restoration differ in impact from large-scale afforestation projects? In what situations is community-based forest restoration most advantageous?
- **Restoration and Adaptation:** How does community-based forest restoration contribute to local climate change adaptation efforts?

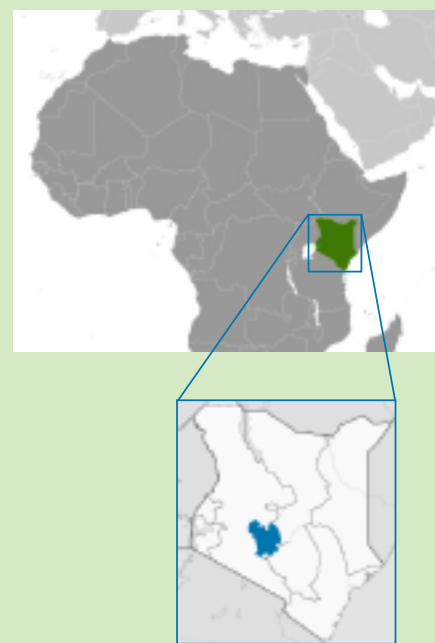


# KIJABE ENVIRONMENT VOLUNTEERS (KENVO)

## Kenya

### PROJECT SUMMARY

Kijabe Environment Volunteers (KENVO) has worked with rural communities on the Kikuyu Escarpment in Kenya since 1996, with a primary focus on forest conservation and reforestation in response to human pressures on the escarpment's forests. The organization has evolved beyond this initial focus, however, into a flexible vehicle for holistic local development. Current activities include selling affordable fuel-efficient stoves to poor farming households; distributing mosquito nets to combat increased incidence of malaria; encouraging beekeeping and aquaculture as alternative livelihood activities; facilitating conflict resolution over water access between local tribes; environmental education; and developing ecotourism.





## BACKGROUND AND CONTEXT

With the objective of conserving Kenya's natural habitats and biodiversity, Kijabe Environment Volunteers (KENVO) provides local communities with the information, education, and resources they need to advance environmentally friendly businesses. KENVO has been active in the Kijabe area of the southern slopes of the Aberdares since 1996, working to combat forest degradation on the Kikuyu Escarpment. The most common source of income in the region is from small-scale agriculture. Households grow carrots and potatoes, and typically harvest one grain – such as maize – per year.

### *Reforestation and diversifying livelihoods*

The main focus of KENVO's work is forest rehabilitation. A group of young activists with university educations in conservation united in 1996 to address deforestation in Kereita Forest. Pressure on the forest resulted from the local communities'

practices of felling for timber and charcoal burning, as well as grazing within the forested area. The education of local villages on the importance of conservation became the main aim of KENVO's work, and led to the broad portfolio of alternative environmentally friendly livelihoods they support today. Agroforestry has formed a large part of this, enabled by three partner-funded tree nurseries, which produce almost 100,000 seedlings each year.

Other significant areas of work include youth empowerment, through mentoring and involvement in the Canada World Youth international exchange program. Community outreach is continued through workshops, and KENVO's monthly bird-watching monitoring program in Kereita Forest. Training and financial support to community organizations for livelihood improvement projects such as beekeeping and aquaculture is providing alternative sources of income for local people. An ecotourism campsite is also being developed on the escarpment with the indigenous Il Parakwo Maasai tribe.





## KEY ACTIVITIES AND INNOVATIONS

### *Kereita Forest tree nursery*

Tree planting for forest rehabilitation forms the main part of KENVO's work, taking place at three tree nurseries. The first nursery, based at KENVO's Resource Center adjacent to Kereita Forest, began in 2000, when a Nairobi-based UNEP department undertook an Early Warning Assessment to calculate their paper usage. Through a Responsible Consumer Behavior paper compensation scheme they pledged that, for each ream of paper used, they would give KENVO the funds calculated to grow one native seedling to planting height. The division initially contributed around 2,000 seedlings per year; currently there are eight divisions taking part, contributing 16,000 seedlings. With other private companies involved in compensating for trees they have harvested for medicinal and cosmetic research, the Kereita nursery plants around 30,000 seedlings annually.

Most of the seedlings are croton, which has bio-diesel uses. It is also non-palatable for livestock, which is an important consideration: local communities are allowed to graze their livestock in the forest in return for a tax, as long as they follow their herds. In practice they do not, so many seedlings are destroyed before they can grow. Olive is another non-palatable, native species that is grown for restoration purposes. Greveria, a fast-growing exotic species, is grown for fodder, firewood, and timber. These seedlings are sold to households at five Kenyan Shillings (US\$0.06), below the market price of ten Kenyan Shillings, for use in agroforestry.

The Kenya Forest Service (KFS) has divided Kereita Forest into two blocks: 4,700 ha are reserved for native trees, which are protected against felling by law, while 2,600 ha are for commercial use. The nursery is divided into three sections, for use by KFS, KENVO, and communities. Communities are given initial capital by KENVO, in the form of seeds and pods, and develop their own cooperative nurseries, taking cuttings from mother trees in the forest before selling the seedlings at planting height to KENVO. Local schools also fundraise for the cost of seedlings and volunteer at the nursery. The nursery provides incentives to reduce deforestation and charcoal burning, thereby lessening forest degradation.



### *Carbacid tree nursery*

The second nursery at Carbacid was founded in 2004 and is funded by Carbacid mining company, which mines carbon dioxide in the area through their 'Greening Kereita Forest' project. The nursery is managed wholly by KENVO, with labor support from the local community. Carbacid provides 50,000 seedlings annually, of which 30,000-40,000 are native, including croton and olive.

### *Matathia tree nursery*

KENVO encourages environmental education by supporting the development of school environment clubs. At Matathia Primary School, students participate in organized clean-ups, field trips, bird watching programs, education about native and exotic trees, and KENVO's annual World Environment Day activities. Students have also learned about water and soil conservation, as well as the importance of waste management and recycling. The school's Environment Club has cultivated a tree nursery within the school compound. This nursery grows seedlings that are used for the rehabilitation of the Kikuyu Escarpment: 30,000 have been planted since 2009, with a target of 100,000. KENVO supports replication of this model with neighboring school districts, through Kikuyu Escarpment Forest Important Bird Areas.



### Alternative livelihoods

KENVO is promoting apiculture as an ecofriendly alternative livelihood activity. Since 2005, 150 beehives have been distributed to ten local youth and women's groups. KENVO has also given training and donated protective harvesting gear to the groups. Group members are expected to create the hive areas and ensure security. The Esibonia Youth Group harvested their first batch of honey in 2010. After using a portion of the honey for their own consumption, the group marketed the surplus.

KENVO supports aquaculture initiatives by providing local groups with access to microloans. Kaharlu Women's Group, a women's agricultural cooperative, was given funding to dig a fish pond. KENVO then supplied the plastic lining and 400 tilapia fish. The women harvested 900 fish in 2010, and have received marketing assistance from KENVO.

In collaboration with the indigenous Il Parakwo Maasai, KENVO is supporting an ecotourism project. The construction of lodges at Osotua Camp began in 2007 after participatory work with tribal leaders. Maasai representatives had originally been engaged by KENVO to create tree nurseries, to compensate for the effects of their livestock grazing and charcoal burning on the escarpment forests. The idea behind developing an ecolodge was to promote the rehabilitation of these forests and to generate revenue for local education projects. The lodge

and campsite will allow visitors to interact with the Maasai people, view performances of cultural dances and traditions, and to purchase locally made artisanal handicrafts. Construction work is undertaken alongside a tree-planting scheme in the surrounding area, with 2,500 trees planted by 2010. Once open, the proceeds will be reinvested in developing a tree nursery, expanding the local primary school to eight classrooms, and funding a water distribution project.

## BIODIVERSITY IMPACTS

### Measuring improvements in conservation

KENVO has observed a reduction in community dependence on escarpment forests, with corresponding increases in biodiversity. These positive impacts are seen through regular community participation in monitoring of birds, wildlife, and vegetation. Annual monitoring is conducted along one-kilometer-long transects, wherein participants measure the extent of canopy cover and tree diameters, and record incidence of human disturbances. One notable improvement in the regenerated forest areas has been higher numbers of *Prunus africana*, which is an important medicinal species used by local communities.

Birds are also counted annually along these transects, as well as in monthly birding programs. Kereita Forest is home to 120 bird species, including



hawks, turacos, cuckoos, weavers, eagles, and the endangered Abbott's starling (which led to the forest being classified as an Important Bird Area by BirdLife International). One observed trend within the regenerated portion of Kereita Forest has been the return of green-headed sunbirds and red-chested sparrowhawks from the interior of the forest to its edge, indicative of the reduction in human activities within the forest. The higher forest areas are also home to elephants. This has led to some human-wildlife conflicts (e.g., elephant encroachment on household farm plots), so KENVO worked with Kenya Wildlife Service to erect an electric fence.

The Kikuyu Escarpment is an important water catchment area and supplies Nairobi with most of its water. In light of this, growing and planting of eucalyptus trees for commercial reasons was stopped, due to their harmful effects on water sources and soil fertility. The importance of the region for local and national ecosystem services is reflected in the involvement of government partners (e.g., Kenya Forest Service and Kenya Wildlife Service) in KENVO's initiatives.

## SOCIOECONOMIC IMPACTS

Over time, apiculture and aquaculture are expected to provide participating groups with a reliable source of revenue. Similarly, the ecotourism project

will generate income for the local Maasai. The tree nurseries have given families access to timber and encouraged agroforestry, thereby enhancing their livelihood prospects.

### *Delivering health and education benefits*

KENVO has sponsored various infrastructure and health projects. Since 2005, the group has sold 300 fuel-efficient, ceramic stoves to local communities, reducing their need for fuel wood and improving health conditions in their homes. Local youths were trained in stove installation and use, and acted as pioneers in training others. Women, who are the primary firewood-gatherers in households, reap the greatest rewards from the health and social benefits provided by the stoves.

KENVO donated two water tanks to Matathia Primary School to serve their tree nursery. Access to water is a pressing concern for many of the communities on the escarpment, and has been a priority area for KENVO. At the site of the new ecolodge, KENVO helped to mediate a dispute between the uphill Kikuyu agrarians and the downhill Maasai pastoralists over water use for agriculture and livestock, respectively. By installing water tanks and pipes, KENVO has helped to resolve the problem, and ensure water security for the two tribes.





KENVO has also targeted the increased incidence of malaria on the escarpment, which has been linked to warmer temperatures and higher numbers of mosquitoes. In 2010, as part of a joint project with civil society organizations from Canada, 500 mosquito nets were distributed to local health centers and dispensaries. The scheme was an example of community-based measures to mitigate the effects of climate change, and was made possible through KENVO's participation in international youth exchanges.

## POLICY IMPACTS

KENVO's influence on policy changes has been growing steadily through their membership in national bodies such as the Kenya Wildlife Service (KWS), Kenya Forests Service (KFS), a Kenya Forests Working Group, and Nature Kenya. They also sit on the District Environment Committee and benefit from their close proximity to the local government headquarters.

Increasingly, Kenyan policies on environmental conservation are taking community interests into account. Kenya's Forestry Act of 2005 enabled the creation of Community Forest Associations (CFAs). CFAs act as umbrella organizations for various forest user groups and give communities a mandate to make their own decisions on conservation and the sustainable use of their natural resources. Five CFAs have been formed in five forest blocks in the district, with a further five being planned for the remaining blocks in the larger escarpment forest. This participatory approach has had some notable benefits: an example is when KFS planned to grant concessions for ecotourism ventures on the Kikuyu Escarpment. KENVO collected opinions from local communities and communicated these to KFS; as a result, the forest service has given first priority to communities' applications for ecotourism concessions. Participatory forest management has also allowed for the establishment of a number of usage rights within the forest for local communities.

## SUSTAINABILITY AND REPLICATION

KENVO has been able to sustain its work through the engagement of community members, partnerships with NGOs and government offices, and funding support from international organizations. Volunteer support has been crucial for the maintenance of KENVO's tree nurseries, and has involved youths, schools, and women's groups. The sale of exotic,

fast-growing tree species have helped to fund KENVO's work, as have profits from the ecolodge. A new forest rehabilitation scheme called 'Tupande Pamoja' is also being planned by KENVO, which will involve corporate sponsors buying tree seedlings from local communities to replant forests.

KENVO has been able to act as an influential model for forest conservation through its role as a Site Support Group with Nature Kenya and BirdLife International. It has hosted other groups from Important Bird Areas, who have learned from KENVO's initiatives. Together these groups form a national network for bird conservation. Youths mentored by KENVO have replicated the organization's model in other communities on the Kikuyu Escarpment, placing a particular emphasis on forest conservation and monitoring.

## PARTNERS

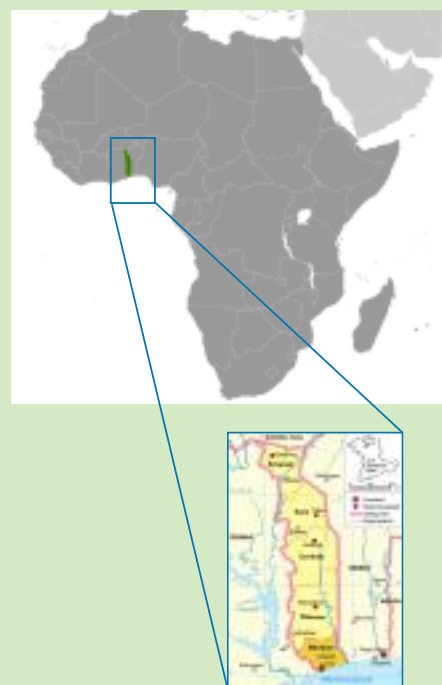
- **Nature Kenya:** Technical support
- **Kenya Forests:** Working group networking
- **Ecotourism Kenya:** Support for ecotourism
- **EU Biodiversity Conservation Programme:** Funding for beekeeping initiative
- **Carbacid Mining Company:** Funding for tree nursery
- **IUCN:** Funding for ecotourism
- **PACT Kenya (USAID):** Funding for ecotourism
- **UNDP/GEF Small Grants Programme:** Funding for ecotourism
- **Kenya Wildlife Service:** Technical support, fencing
- **Kenya Forest Service:** Technical support for forest monitoring
- **Kenya Ministry of Agriculture:** Technical support for aquaculture, beekeeping
- **United Nations Environment Programme:** Funding for tree nursery
- **Kenya Forestry Research Institute:** Technical support



## PROJECT SUMMARY

After a devastating bush fire in 1973, the village of Ando Kpomey created a 'green belt' buffer around its community that has grown into a 100 ha forest. A participatory management committee has been established to monitor the forest and its resources and to regulate its use. The community authorizes limited resource extraction to meet livelihood needs and manages revenues generated from the sale of forest-based products. Local women are authorized to enter the community forest in order to access firewood, significantly reducing the average time needed to forage for cooking fuel. Various crops are grown in the forest, including medicinal plants which contribute to local healthcare needs.

Neighboring communities have been enlisted to protect the forest, and have benefited from knowledge sharing on natural resource management, participatory planning, and forest conservation. The village hosts peer-to-peer learning exchanges to share lessons learned, and has done so with communities and organization across Togo and Burkina Faso.





## BACKGROUND AND CONTEXT

### *Persistent threats to forest cover and biodiversity*

The total forested area in Togo is around 287,000 ha, but with an annual deforestation rate of 5.75 percent, is rapidly shrinking. Decades of socio-political crises and poor resource governance have resulted in severe degradation of protected areas. Between 2005 and 2010, Togo lost an average of 20,000 ha of forest cover each year. The combination of deforestation, bush fires, and hunting has resulted in the widespread loss of forests and the biodiversity they support. Deforestation has led to land degradation, leaving towns and villages vulnerable to flooding, bush fires, and other extreme weather events.

### *A 'green belt' to buffer against bush fires*

Ando Kpomey is a small town in the Maritime Region of southwestern Togo, located 70 km northwest of the capital, Lomé. The community, with a population of 1,000, was formerly surrounded by dry shrub savanna. Bush fires were common, devastating housing, local infrastructure, crops, and food stores. On two occasions, the entire town burned to the ground.

After a devastating fire in 1973, a group of town elders proposed the establishment of a 'green belt' of forested area around the perimeter of the community to buffer against future bush fires. In addition to protecting the town from bush fires and enhancing

biodiversity, the elders reasoned that a green belt would also allow the community to diversify crops under cultivation. The proposal was accepted and since 1973 Ando Kpomey has maintained and expanded a community forest that surrounds the village.

### *The genesis of a community forest*

The initial creation of the green belt was based on local initiative and voluntary efforts. Family heads and community elders agreed on the proposal and devised a plan to move forward. Community members first cleared a 14 m fire break around the town. Beyond the fire break, they planted a 10 m strip of trees. Every year since, the community has planted an additional 10 m strip to expand the green belt. Little by little, the green belt has grown to become a community forest, which now surrounds the town to a depth of between 350 and 850 m and is a great source of pride for Ando Kpomey's residents. For 18 years, the community maintained and expanded the forest with no outside help.

### *Management, participation, and financing*

In 2003, Ando Kpomey, with assistance from partners, developed formal regulations for the management of the community forest, including the formation of management and watchdog committees to ensure accountability, transparency, and good governance. The management approach is largely participatory in







nature, and regulations over forest use and access are widely respected by local residents. The community forest in Ando Kpomey is nothing short of a regional anomaly. The majority of surrounding towns have entirely cleared their forests. As such, Ando Kpomey has reached out to neighboring villages in an effort to engage a larger population in the protection of its forest and local biodiversity. Forest visitors from outside the community are charged an entrance fee, and profits go to support local infrastructure and reforestation. Additional funding for reforestation efforts is collected from local businesses: any business that extracts resources from the forest must make a contribution towards the maintenance of the forest and other village development activities.

## KEY ACTIVITIES AND INNOVATIONS

The primary activity of the Comité Villageois de Développement d'Ando Kpomey has been the establishment and ongoing maintenance of its community forest. Thanks to the efforts of volunteers, the forest now covers an area of more than 100 ha. Rather than focusing on continued expansion, however, the organization is now concentrating its efforts on forest maintenance and monitoring to ensure that existing tree cover stays intact and is not deforested or degraded.

To feed community tree-planting and reforestation efforts, village nurseries were established in which tree saplings were cultivated and nurtured to maturity before being planted in the annual forest expansion. With the support of its partner organizations, the Village Development Committee developed a simple management plan, in open consultations with community members. The forest management committee is composed of elected community members and is responsible for enforcing forest regulations, and authorizing the felling of trees and other resource use in the forest. When a member of the community needs timber for a construction project, they must submit a request to the management committee which authorizes the type and quality of wood that should be used as well as the location in the forest where the tree(s) can be harvested. Through this process, the forest is able to meet the needs of the community without being overharvested. Women are also permitted entry to the forest to collect fallen branches for firewood. Because pressures on forest resources are not limited to inside the community, the Village Development Committee has also made efforts to involve neighboring communities in the protection of the forest. Residents of adjacent communities are invited to participate in traditional hunts and sustainable resource extraction events, with the intention of demonstrating the value of the forest and the ecosystem goods and services it provides, and to discourage people from degrading or setting fire to the forest.



The Village Development Committee has also expanded into alternative livelihood activities that aim to raise local incomes and reduce pressure on forest resources. The project has trained 25 women in snail farming and 17 men in apiculture and the marketing of honey. Project beneficiaries were provided with hives, brood snails, construction material for snaileries, and honey harvesting and storage equipment.

## BIODIVERSITY IMPACTS

Increased forest cover has improved biological diversity around the village. Many of the trees and vegetation that now thrive in the community forest are endangered or threatened in the larger region. The forest is home to various species of vines, medicinal plants, edible mushrooms, and wildlife, including birds, rodents, and deer. The livelihoods projects that are now being advanced by the Village Development Committee provide the local

population with income-generating alternatives to unsustainable extraction of forest resources and illegal logging.

The community forest contributes to healthy ecosystem functioning and, by extension, the provision of ecosystem goods and services. Among the more important services that have been restored through the planting and expansion of the community forest has been the regulation of water cycles. Other towns that have cleared their forests suffer from irregular rainfall and constant flooding, as the lack of tree roots compromises soil integrity and subsequent land degradation. Anecdotal reports from villagers claim that the community forest has also improved air quality in and around the village, and that the trees planted around Ando Kpomey have created a microclimate that produces regular rainfall, which has benefits for local freshwater access and agriculture.



## SOCIOECONOMIC IMPACTS

The initiative's greatest socioeconomic impact has come from the suppression of bush fires. By creating a fire break and green belt around the perimeter of the village, the community has effectively made bush fires a thing of the past. No bush fires have occurred in the town since the initiative began. Previously, bush fires caused a great deal of damage to local homes and village infrastructure, which had detrimental effects on the local economy, requiring large investments of time and resources to rebuild.

### *Regulating and provisioning services*

The role of the community forest in creating a local microclimate, and therefore potentially increased rainfall, has benefited local agricultural production and food security. Water for irrigation is a precious commodity in the region, which is characterized by arid land and water shortages. These climatic and geographic realities have been exacerbated by deforestation, land degradation, and climate change. In the early 1970s, when the decision was made to establish the green belt, there was sufficient tree cover in the surrounding region to maintain regular rainy seasons and sufficient rainfall. Since then, due to widespread deforestation, rainfall is now rare in the area. Ando Kpomey has been somewhat protected from this, owing in large part to the community forest. Although harvesting wood from the forest is

restricted and tightly regulated, community members no longer have a difficult time obtaining wood for construction projects and infrastructure needs. Another socioeconomic benefit of the community forest has been improvements in the diversity and abundance of the natural resources that are available for community use. The forest contains edible mushrooms, animals that are sustainably hunted (during regulated traditional hunts), and medicinal plants that help to meet basic healthcare needs.

### *Associated benefits of the community forest*

The community forest has enhanced the village's land tenure security. The forest which now encircles the village is actively monitored and maintained in a way that the surrounding bush and scrubland never was previously. The tree line created by the forest boundaries provides a clear delineation of Ando Kpomey land. This has been beneficial in strengthening community identity as it relates to the stewardship and ownership of communal land, as well as grounding the village's collective spatial sense of territory and protecting the village from 'land grabs', which are not uncommon in the region.

Entrance fees from visitors to the community forest are directed towards village development projects. The majority of the funds have been used to construct and operate a local school. The school has been recognized by the Togolese government and was allocated teachers in 2010.





The community forest has given the women of Ando Kpomey a stronger voice in community life through representation in decision-making processes and in the management of the forest. The participatory management approach employed by Ando Kpomey gives men and women equal representation in forest governance. Community members report that women are increasingly inclined to voice their opinions at town meetings and during exchange visits with other communities. Women also benefit from being given access to the forest to gather firewood, reducing the distance they would otherwise have been forced to travel to collect fuel for household consumption.

## POLICY IMPACTS

The Togolese Government has been influenced by the community-based stewardship and management model and has revised legislative text on forest management to be less top-down and more amenable to participatory, community-based action. For example, the Ministry of the Environment and Forest Resources adopted a more participatory approach to forest management that directly involves local town associations in the management of protected areas. The Ando Kpomey community forest provided proof of what local communities can achieve when empowered, and can be credited with influencing this change in policy.

## SUSTAINABILITY AND REPLICATION

Although Ando Kpomey receives some external funding, the community forest remains very much a locally driven project. All segments of the local population – men, women, and youth – are represented in each of the committees that oversee the forest. The forest is managed in such a way that all community members feel involved and have a stake in its stewardship. Community members recognize

the benefits the forest provides and willingly invest their time and energy to maintain it. Every year, residents clear the fire break and carry out any necessary planting to maintain the quality of the forest. Throughout the year, but especially during the dry season, community members monitor the forest to prevent forest fires and deforestation. All of this work is carried out voluntarily and the involvement of the entire village has helped to strengthen the solidarity and cohesion of the community.

The community forest is recognized as a best practice model in the region and has a high level of potential for replication. The Ando Kpomey forest management model has been promoted through educational visits, peer-to-peer learning and knowledge exchanges, awareness-raising workshops, radio programming, and the publication of a booklet. Visitors have included a range of farming organizations from the area, neighboring prefectures, and researchers from the University of Lomé, government agencies, and NGOs. These exchanges have led to the replication of Ando Kpomey's community forest project model in five neighboring communities.

## PARTNERS

- ***Association Togolaise pour la Promotion Humaine (ATPH)***: Technical support for watchdog committee and management committee
- ***Institut Africain pour le Développement Economique et Social (INADES)***: Technical support for forest management
- ***Department of Environment and Forest Resources***: Support for forest management
- ***UNDP/GEF Small Grants Programme***: Funding for alternative livelihoods, training, capacity building, and forest management plan



## PROJECT SUMMARY

Empresa Forestal Integrada de Bayamo is a state-run forest enterprise operating in Granma Province, Cuba. In 1999, Granma was one of two pilot sites for a *fincas forestales ecológicas* (ecological forest farms) initiative, which put reforestation of the Cauto River Basin in the hands of smallholder farmers.

The working model saw plots of land assigned to interested households for concession periods of 30 years. These households were given responsibility for managing and reforesting plots of between 12 and 25 ha, and were encouraged to plant timber-yielding trees, fruit trees, and medicinal plants. Over 3,000 ha of land along the banks of the Cauto River were reforested, improving local livelihoods and well-being. The initiative was later replicated in three hydrological regions of the country in 2004.



## BACKGROUND AND CONTEXT

The eastern Cuban province of Granma is of national importance, in terms of its cultural and historical significance, and its ecological wealth. The province is named after the yacht 'Granma', used by Che Guevara and Fidel Castro to land in Cuba in 1956. The American who sold the revolutionaries the yacht in Mexico named it 'Granma' after his grandmother, and the name of the vessel subsequently became an icon of Cuban communism. The province extends over 8,400 km<sup>2</sup> and has a population of 835,000 people, of whom more than 670,000 live in rural areas. The province contains the second largest river in Cuba, the Rio Cauto, which extends 140 km and crosses three of the five eastern provinces. This area forms the Cauto River Basin, the largest and most important water reserve in the country.

### *Land conversion and deforestation in Granma*

The Cauto River Basin has been subjected to landscape-level changes which have led to substantial environmental degradation. Large areas of forest were cleared for charcoal production, agriculture, and cattle-rearing. As a result, the province is the second largest producer of milk and rice in the country, but by 2000, only 19 percent of its land remained forested. Much of the vegetation and fauna had disappeared, canyons and hillsides had become heavily eroded, and the river valley soil had become too salinized to sustain agricultural production. During storms, the erosion of crevices and gullies from flash floods caused major landslides, devastating the landscape and human settlements. Siltation of the river from large-scale cultivation and infrastructure projects also threatened the hydrological potential of the region.

### *Government attempts to stimulate local development*

Widespread environmental degradation was exacerbated by the area's low level of economic development. In 2000, the province of Granma ranked last on the country's Human Development Index. The province is characterized by high levels of rural to urban migration, as well as outmigration from these cities to Havana. Granma, therefore, became a development priority for the Cuban government, which focused on protecting the water table, combating drought, halting soil erosion, and curbing out migration.

However, several government-led attempts to reforest the Cauto River Basin failed due to drought, inadequate site preparation, uncontrolled grazing, illegal logging,

and inadequate financing. Another significant factor in the failures was insufficient input from local communities. To address this, the provincial administration designed a program that took a more holistic approach to rehabilitating the Basin, encompassing environmental, social, and economic elements.

### *Reforestation, farm by farm*

Since 1998, the United Nations Development Programme (UNDP) has led the Local Human Development Program (Programa de Desarrollo Humano a Nivel Local – PDHL) in Cuba. The program promotes technological innovation for sustainable human development in rural communities and brings together more than 160 institutions from 11 countries worldwide to exchange knowledge and experiences on local-level development. PDHL began operating in the country in 1999 in two pilot provinces – Granma and Pinar del Río. In Granma, the program focus was a local initiative to promote sustainable natural resource management and the environmental rehabilitation of land along the Cauto River. Emphasis was placed on diversifying vegetation and creating employment opportunities for economically marginalized communities.

The program focused on a reforestation project called *fincas forestales ecológicas*, or 'ecological forest farms' that was based on a state-run forest enterprise model, Empresa Forestal Integral de Bayamo. The model divided land into plots and assigned them to interested households for a concession period of 30 years. Interested farming households were given responsibility for managing and reforesting plots of between 12 and 25 ha in size. Families planted timber-yielding trees, fruit trees and medicinal plants, and raised livestock. No restrictions were placed on whether goods produced or crops harvested were used for personal consumption or for sale to outside markets.

## KEY ACTIVITIES AND INNOVATIONS

Community members running *fincas forestales ecológicas* are empowered to manage their parcels of land as individual owners. By 2002, the project had established 55 forest farms along the Cauto River. In total, 1,300 ha of land have been reforested in the river basin. Plots were reforested with seedlings provided by the provincial forest authority. Training on farm management and environmental conservation was provided by government extension staff. To encourage families to remain on their





plots, the project also oversaw the construction of comfortable family homes, each equipped with fuel-efficient stoves, photovoltaic panels, and a television set. The project created 220 new jobs for farm families, benefiting men and women equally.

The United Nations Development Fund for Women (UNIFEM) and the Cuban Women's Federation carried out training workshops focusing on gender relationships in the context of the forest farms. The workshops produced a proposal recommending that half of newly-created forest farms be assigned to women, which was duly presented to the Granma People's Power Provincial Assembly. An agreement was subsequently drawn up that made equal allocation of land between men and women into reality. This approach was so successful that it was replicated in ecological forest farm projects in Guantanamo and Las Tunas. Gender mainstreaming was crucial to the spread of the forest farm model because it enabled a high level of community participation in project implementation and ensured the involvement of women heads of household in farm management.

The *fincas forestales ecológicas* project also served as a delivery mechanism for alternative energy technology to marginalized communities. By 2002, the program had overseen the installation of nine solar power energy stations and 79 windmills. The program was also effective in promoting the uptake

of organic farming and the use of organic fertilizers by smallholder farmers; more than 4,500 tons of fertilizer were produced and used each year, substituting for chemical inputs which had detrimental effects on the local environment. The program also improved irrigation systems, rehabilitating over 1,000 ha of farmland and strengthening food security. In total, the *fincas forestales ecológicas* project created 1,206 new jobs, 921 of which went to women.

## BIODIVERSITY IMPACTS

Empresa Forestal Integrada de Bayamo project reforested over 3,000 ha of land along the banks of the Cauto River, helping to reduce soil erosion and the degradation of the river basin. Reforestation using native timber and fruit tree species also helped restore biodiversity in the region. What has distinguished the forest farms initiative most markedly from previous reforestation attempts has been the maintenance and upkeep of the plots. The survival rate of plantations has increased due to farming families remaining on their household plots, and investing time and energy in the upkeep of their forest land.

*Fincas forestales ecológicas* have had a wide impact on reforestation across Cuba. Adopted as a nationwide program following its success in Granma, 848 forest farms had been set up in three hydrological regions of the country by early 2004, covering an area of 91,067 ha. In the province of Pinar del Río (which has the highest woodland ratio in Cuba at 39 percent), forest companies set up 111 farms to manage 74,100 ha of natural woodland, or 23 percent of the total land area of the province.

The impact of *fincas forestales ecológicas* in the eastern region has been especially notable, with some 8,573 ha planted on 11,472 ha of forest farms. Farms in the eastern region tend to be smaller on average, but have been established in greater numbers, predominantly in areas prioritized for reforestation. The eastern region has a rural population of more than 1.5 million, and over half of the country's river basins, yet water regulation remains a critical challenge. Both soil salinity and soil erosion are still high, making the eastern region a priority for reforestation efforts.

Over six years, the forest farms project was responsible for planting 13,643 ha, a figure that represents almost 35 percent of the country's annual

total. In regions with limited agricultural production due to poor soils and frequent droughts, a 95 percent survival rate has been achieved across all planted areas. This contrasts with a historical average of 36 percent for previous reforestation attempts. The incidence of illegal logging and forest fires also diminished.

## SOCIOECONOMIC IMPACTS

The forest farms project has generated social and economic benefits, primarily through direct job creation. By 2002, Empresa Forestal Integrada de Bayamo had created 220 jobs in Granma, and was providing income for 55 families. Associated benefits of the project included improved agricultural productivity and revenue generation from the sale of livestock and forest products. Growing vegetables and breeding animals for household consumption also improved food security.

Gender empowerment has been a critical component of the program's work in Granma. Through an agreement with the provincial government, 50 percent of all forest farms were titled to women heads of household. Improvements in the well-being of farming families, including the provision of televisions, photovoltaic panels, and fuel-efficient stoves, have decreased rates of migration from the countryside to urban areas.

By 2004, 75 percent of forest farm areas in the country's eastern region had been planted. The farms employed 1,025 workers, at a ratio of 8.3 ha per worker. The province of Las Tunas, once one of the most deforested in the country (with only 12 percent forest cover), led the national reforestation effort, with 146 forest farms established. Direct forestry production per 26 ha farm was 3,300 m<sup>3</sup>, worth a total of US\$75,092. Average salaries increased by an average of 17 percent, and agricultural production by 38 percent.

## SUSTAINABILITY AND REPLICATION

By involving the local community at every stage of reforestation planning and implementation, and through gender mainstreaming, the long-term success of the forest farm project has been secured. While previous attempts at large-scale reforestation had been unsustainable, this approach has ensured

that individual households have maintained their own forest plots, with a survival rate of 95 percent between 1998 and 2004. One reason for this has been the granting of land plots in usufruct to farming households for a period of 30 years. As families have a greater interest in the long term productivity of their farm plots, they have been more likely to maintain their ecological integrity. The potential for income generation has also enhanced households' interest in maintaining their individual forest plots.

The fast pace of replication of the forest farms project beyond Granma to other hydrological basins in Cuba is testament to the results achieved by the Empresa Forestal Integrada de Bayamo. The transferability of the project has made it a model for state-run reforestation programs, with its popularity in Cuba mirrored by its recognition on the international stage. Results of the forest farms project have been widely disseminated at regional and international forestry conferences, and many Latin American and Caribbean countries are now replicating the model.

## PARTNERS

- **Ecological Forest Enterprise System:** Implementation, technical assistance
- **Forestry Research Institute:** Technical assistance
- **State Forestry Service:** Technical assistance, training
- **Cuban Agriculture Ministry (MINAGRI):** Technical assistance
- **Forest Study Centre of the University of Pinar del Río:** Technical assistance
- **The Land Department of the University of Granma:** Technical assistance
- **UNDP Programa de Desarrollo Humano a Nivel Local (PDHL):** Financial and technical assistance
- **United Nations Development Fund for Women (UNIFEM):** Support for female farmers, women's empowerment
- **Oxfam Canada:** Capacity building

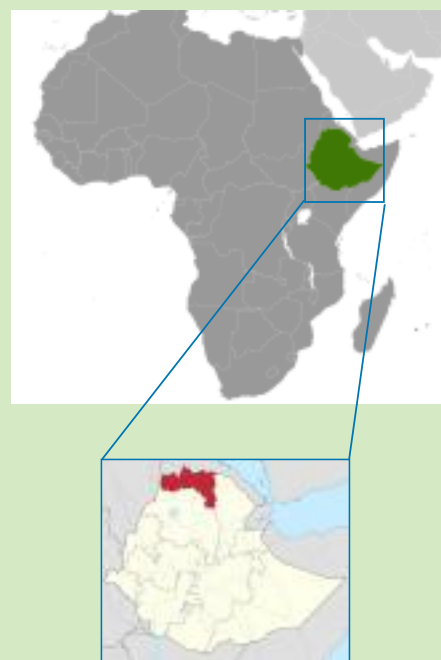


# ABRHA WEATSBHA COMMUNITY

## Ethiopia

### PROJECT SUMMARY

Once on the brink of resettlement due to desertification, soil degradation, and lack of water, the Abrha Weatsbha community in northern Ethiopia has reclaimed its land through the reforestation and sustainable management of over 224,000 ha of forest. Tree planting activities have resulted in improved soil quality, higher crop yields, increased groundwater functioning, and flood prevention. The organization has constructed small dams, created water catchment ponds, and built trenches and bunds to restore groundwater functioning. Environmental restoration has led to livelihood improvements through crop irrigation, fruit tree propagation, and apiculture. Local incomes have increased and food security and nutrition have improved through the integration of fruit trees onto farms.





## BACKGROUND AND CONTEXT

The Tigray region of Ethiopia is located in the northernmost territory of the country and borders Eritrea in the north and Sudan in the west. The region is characterized by drylands and is highly vulnerable to recurrent drought. Land degradation is one of the most serious challenges confronting the rural population, and is intensified by climate change.

### *Drought, deforestation, and land degradation*

In recent years, northern Ethiopia has experienced serious droughts and inconsistent rainfall patterns. This variability, and the impacts it has on agricultural production and local livelihoods, is aggravated by deforestation. The northern regions of Ethiopia are among the most deforested in the country. Without the protection of vegetation cover, hillside soils become easily degraded, exacerbated by free-range grazing, which prevents the re-growth of vegetation and exposes topsoil to erosion.

The village of Abrha Weatsbha is situated in a sandstone area that is particularly vulnerable to erosion and desertification. Land degradation has severely impacted the productivity of the village and surrounding agricultural lands. Short-sighted land

and water management approaches magnified the vulnerability of the community to climate impacts, and by the early 2000s, conditions were so dire that the community faced resettlement.

### *Micro-catchment ecosystem management*

In 1996, the Ministry of Agriculture and Rural Development in Ethiopia undertook a process of decentralization, working with communities to implement solutions to land, natural resource, and water issues. Among the approaches promoted was a community-based 'micro-catchment ecosystem management' model. The strategy aimed to empower communities to implement actions to stop free-range livestock grazing by: cutting and carrying grass to feed livestock, terracing hillsides to prevent erosion, damming gullies, and ensuring that any transgressors of established community by-laws were penalized. The approach was successful: soil and water conservation activities were carried out on more than 956,000 ha of land throughout the country; vegetation enclosure management was implemented on more than 1.2 million ha of land; and more than 224,000 ha of land was enrolled under sustainable forest management. The micro-catchment ecosystem management model resulted in the planting of over 40 million tree seedlings and the building of 180 wells that provided access to potable water.



Although Abrha Weatsbha was not one of the four communities selected for the piloting of the ‘micro-catchment ecosystem management’ model, Abrha Weatsbha Natural Resource Management Initiative was founded in this context and based its approach on lessons learned from the successes and obstacles faced in carrying out this strategy. One important transplant from the ‘micro-catchment ecosystem management’ model was enlisting the guidance and support of the Ministry of Agriculture’s extension system – one of the largest and most robust of its kind in the world, with development agents working in farmer training centers across all regions of the country.

### *Community-based adaptation to climate change*

The Abrha Weatsbha Natural Resource Management Initiative was formed in 2004 to address the challenges of food insecurity, land degradation, and access to fresh water. It has since emerged as a leading example of community-based adaptation to climate change. The initiative began with a community assessment of existing constraints to local health and well-being. Despite the presence of a local aquifer, one of the top priorities identified was fresh water access. The community initiated a range of actions, including a tree-planting campaign; construction of dams, wells, and water catchments ponds; and the establishment of temporary closed areas on communal land to prevent over-grazing.

### *Landscape level change*

The results of the project have transformed the landscape and rejuvenated the catchment area. Vegetation has quickly returned, soil erosion has been reduced, soil water retention has increased, and springs and streams have been replenished. The community is using hand-dug, shallow water wells to start small-scale irrigation projects. Livestock management has improved, with animal dung used for compost to improve soil fertility. Environmental restoration has improved livelihoods, diversified incomes, and strengthened food security. To date, more than 1,000 farmers have been engaged in new agronomic practices and diversified income generation activities.

The village is known for its pioneering work on ‘community-based participatory planning’, which prioritizes involvement of the local population in project design, development, and implementation. Responsiveness to local needs and the ability to

draw from local resources are strengths of the initiative and key factors underpinning its success. A community-based management system ensures local buy-in, while mechanisms ensure the inclusion of women in all aspects of community planning, project implementation, and monitoring.

## KEY ACTIVITIES AND INNOVATIONS

### *Improved water management*

Tree planting, ecosystem restoration activities, and the use of manure for compost and organic fertilizers have improved the integrity of soils and enhanced water security by facilitating groundwater recharging. Improvements in soil quality have translated to increased absorption of water into the soil, which has reduced incidence of flooding and increased resilience to drought. The community also constructs small dams, catchment ponds, and trenches and bunds to restore groundwater functioning. This has resulted in improved irrigation, making year-round agricultural production possible by allowing local farmers to grow fruits and vegetables during the dry season.

### *Changing local agricultural practices*

Abrha Weatsbha Natural Resource Management Initiative engages local farmers to improve environmental sustainability through behavior change. The group promotes crops and livestock varieties that are suitable to the carrying capacity of the land. Grazing restrictions are enforced to permit regeneration of native vegetation. The group also works with private land owners to encourage agroforestry practices, which provide the community with wood and non-timber forest products, while contributing to climate control, nitrogen fixing, soil enrichment, and water percolation.

### *Tree planting and agroforestry*

The Abrha Weatsbha community is reversing deforestation through tree planting and improving soil quality. The community focuses on temporary closures of communal areas where soils and woodlands need time to recover and rehabilitate. Communal lands are dominated by pioneer native species, which have low production function, but high environmental stabilization functions. The group also promotes the use of farm fields for agroforestry and fruit trees, which provide positive environmental and economic benefits.





### *Education, local problem-solving, and innovation*

Abrha Weatsbha dedicates significant resources to community education and training. The organization also promotes an extension program that responds to economically marginalized women by offering support and skills-training in livestock production, forestry, and agriculture.

## BIODIVERSITY IMPACTS

### *Recharging groundwater*

Rehabilitation efforts, along with reforestation, have increased retention of rainwater recharge to groundwater stores by 50 percent. Wells have been dug for irrigation of high-value crops, which are now able to be harvested two to three times per year, irrespective of rainfall patterns. The creation of enclosure areas and water conservation measures has resulted in water tables rising from nine meters depth to between two and four meters, making the digging of wells a low-labor and low-cost proposition. Natural

springs that had dried up have started to flow again, streams now flow longer distances, and pastured lowlands remain green throughout the year.

### *Abrha Weatsbha: A 'quiet revolution'*

The initiative has created a paradigm shift – locally dubbed a 'quiet revolution' – in how local farmers understand and manage the environment. One aspect of this appreciation concerns indigenous plants. Where farmers formerly removed *Faidherbia albida* – a thorny species with deep-penetrating tap roots that make it highly resistant to drought – it has become common practice to replant them. Local farmers have also voluntarily introduced seasonal land closures to facilitate plant regeneration.

### *A sustainable land management approach*

Reforestation has translated to conservation of topsoil, and watershed protection. Tree root systems strengthen soil integrity and reduce soil erosion. The sustainable land management approach promoted by the group has served as a mechanism for doing



away with local agricultural practices that were either unsustainable over the long term or, worse, were eroding the integrity of local ecosystems and damaging the environment. A focus on the interconnectedness of groundwater recharge, soil quality, and tree cover has provided a lens and rallying point through which the community has been able to address environmental challenges at the systemic level.

## SOCIOECONOMIC IMPACTS

The initiative has had a profound impact on the livelihoods of community members, improving food security to such an extent that an area once plagued by low crop yields and food shortages in the dry season now produces a food surplus. Similar improvements have come in the availability of potable water, once a serious concern which threatened local health and well-being, as well as irrigation options for off-season agriculture. Availability of water and improvements in soil quality has boosted agricultural outputs and the diversity of crops that can be grown throughout the year.

### *Access to water, irrigation, and food security*

Recharging the groundwater and local aquifer has enabled the use of shallow wells. Prior to the group's interventions, the water table was too low for communities to access it. Now, with a minimal amount of technological input, the village has been able to dig more than 180 wells, using treadle pumps to access potable water. Treadle pumps are also used by villagers to access water from ponds, springs, and the nearby river, thereby strengthening water security.

Access to water has made irrigation during the dry season and supplementary irrigation during the rainy season possible, creating reliable year-round agricultural production. The ability to produce crops, fruits, and vegetables year-round has fundamentally changed the area and the lives of the local population. For the average shallow well user, food self-sufficiency is now possible for over nine months of the year, while for 27 percent of users, food self-sufficiency is possible year round. Additionally, a total of 39 percent of shallow well users are now consuming a wider variety of vegetables at least once a week.



### *Agricultural outputs, diversified crops, and local income*

Between 2004 and 2007, the amount of irrigated land under cultivation for vegetable production increased from 32 to 68 ha. This has had positive effects on local incomes. Between 2007 and 2010, incomes from the sale of vegetables and spices nearly tripled from US\$32,500 to US\$93,750. Farmers have also been supported to grow high-value fruit trees including apple, avocado, citron, mango, and coffee. The group has also promoted apiculture as an income diversification strategy. Training in beehive management and the use of apiculture equipment led to an increase in honey production between 2007 and 2010 from 13 to 31 tons.

### *Resilience*

One of the most significant impacts of the project has been improvements in the resilience of the community to withstand environmental and economic shocks and to adapt to climate change in a region known for extreme climate variability. The initiative has improved soil quality and integrity, which has in turn reduced the susceptibility of the community to floods and land degradation. Efforts to improve food security and diversify agricultural products have created less dependency on a small number of crops, thereby strengthening income certainty for farmers who regularly confront drought.

### *Empowerment of women*

Development of potable water sources and establishment of enclosed woodlots have had important implications for women and girls. Where previously women travelled great distances to collect water, fodder, and firewood, they now are able to access these basic needs in less time. This has given women and girls more time to engage in other activities, including education. Women also play an active role in the governance of the initiative, taking a lead on decision-making and program implementation.

## POLICY IMPACTS

The organization has had an important role in shaping regional policy development and has garnered the attention of regional and national policymakers. Lessons learned from the initiative have

been channeled into regional government steering committees, which decided to adopt the use of ponds as irrigation sources and to make the technical and resource investments needed to implement the approach across the region. This initiative spurred many local farmers in the region to adopt shallow wells and other locally adapted water harvesting strategies to increase irrigation.

## SUSTAINABILITY AND REPLICATION

Abrha Weatsbha Natural Resource Management Initiative is not reliant on outside funding and its sustainability model is tied closely to its ability to foster community ownership and investment. The initiative is deeply embedded in local life, culture, and identity. All interventions are implemented using locally available resources. Self-sustainability and self-reliance are defining characteristics of the organization, as well as important aspects of local identity.

The Abrha Weatsbha Knowledge Management Center serves as a training center for farmers from across the region and facilitates knowledge sharing between Abrha Weatsbha and other villages, enabling replication by communities facing similar challenges from climate variability and land degradation. There is also a demonstration site in the village for testing new methods of natural resource management. Governmental recognition of the village as a model for community development has influenced many to visit and participate in knowledge-sharing with the community.

## PARTNERS

- **Mekelle University:** Technical assistance, support for Knowledge Management Center
- **Tigray Bureau of Agriculture:** Technical and resource support
- **Relief Society of Tigray:** Technical and resource support
- **Ministry of Agriculture:** Technical support, training





# TROWEL DEVELOPMENT FOUNDATION

## Philippines

### PROJECT SUMMARY

Trowel Development Foundation is a community-based organization employing climate-adapted aquaculture technology to replant mangroves. Mangrove reforestation efforts have focused on planting native tree species in strategic areas, resulting in restored marine biodiversity, food security, and protection of coastal areas. The initiative also works to increase local incomes and improve livelihoods through a value-chain system to market tie-crabs. The group has established five community-managed tie-crab farms that benefit 250 subsistence fishing households. This innovation has been implemented in idle fishponds, where mangrove-friendly and climate-adapted tie-crab fattening technology has been employed to double the income of fishing households.





## BACKGROUND AND CONTEXT

Northern Samar province is located on the eastern side of the Philippine Archipelago. It is composed of 24 municipalities and 569 barangays covering a total land area of 3,693 km<sup>2</sup>. The province has a complex mix of ecosystems and habitat types, which include lower mountain forests, agroforestry areas, and lowland agricultural areas, as well as freshwater ecosystems and oceanic coral reefs. Cultivated crops, which cover 38 percent of the province, include coconut, rice, root crop, banana, and corn.

### *Environmental threats to an area of 'megabiodiversity'*

The waters off the coast of Northern Samar are inhabited by commercially important fish and marine species. The province's abundance of biological diversity helps to account for the Philippines' classification as one of the world's 18 areas of 'megabiodiversity'. Thirteen of the Philippines' 40 native mangrove species are found in Northern Samar, as are dozens of fish species. Northern Samar supplies 60 percent of the country's crablets. The abundance of fish, coral, and mangrove resources, however, has been severely degraded by illegal fishing and destructive activities such as dynamite fishing, coral harvesting, and mangrove deforestation. A substantial proportion of the province's mangrove forests have

been deforested to convert land into settlement areas or for fishpond use. Hundreds of hectares of fish ponds, previously used for aquaculture, have been abandoned and now sit idle and unproductive.

### *Root causes of poverty in Northern Samar*

Northern Samar is an economically marginalized province. Fourteen of 24 municipalities in the region have poverty rates higher than 50 percent, making it the seventh poorest among the country's 79 provinces. Roughly 50,000 households in Northern Samar suffer from extreme poverty. The economy is largely dependent on the agriculture and fisheries sectors. Small-holder farmers are the norm, with the average family tending less than five hectares of farmland. Copra and root crops provide the primary sources of rural income. Small-scale production also predominates in the fishing sector, where local fishermen use traditional fishing techniques to harvest marine resources. As fishing stocks, and by association fishing incomes, have plummeted, so has the ability of fishing families to provide for basic food security, health, education, and housing. Poverty in Northern Samar has also been exacerbated by the negative impacts of climate change. It is particularly vulnerable to tropical cyclones, flooding, and coastal erosion, which introduce insecurity and uncertainty for an already vulnerable rural population.



## KEY ACTIVITIES AND INNOVATIONS

### *A community-owned initiative*

The Trowel Development Foundation focuses on the local aquaculture industry and the gap in appropriate technologies for small-scale farmers and fishermen. After the organization conducted an assessment of the coastal and marine environments, availability of raw materials, and market demand, priority was given to mud crab and seaweed cultivation. Trowel Development Foundation undertook a comprehensive inventory of local technologies and practices in mud crab and seaweed cultivation. A transferable crab-fattening technique was identified that involved tying crabs to bamboo poles. To test the approach, a two-hectare demonstration farm and training facility for tie-crab fattening was established. On- and off-site training was offered to small-scale fishermen. Mangrove restoration and reforestation were instituted as a central conservation dimension of the strategy. Supply-chain and marketing support for fattened tie-crabs was also provided, closing the loop on market access and business development.

### *Evolution of project goals based on local demand*

The initial objectives of the project were to increase the income of 200 subsistence fishing households by 50 percent, to protect and enhance local biodiversity in nine mangrove sites, and to facilitate co-management arrangements for the conservation and sustainable use of aquatic and mangrove resources. These objectives were later expanded to cover incomes and food security for 1,000 households, mangrove reforestation in idle fishponds, and market supply-chain development. Growth in the number of target beneficiaries was due to local demand from small-scale fishermen. Supply-chains were also prioritized based on a perceived gap in favorable markets for locally produced products.

### *Planning and mobilization*

The Trowel Development Foundation commenced an assessment and planning phase that included institution-building, scoping and visioning, stakeholder mobilization, and awareness-raising. During this stage, operational plans were formulated and an MOU was signed between the project's stakeholders regarding partner responsibilities and commitment of resources. This was followed

by community assessments and a crab industry scoping. Local consultations and site inspections were undertaken to identify idle fishponds that could be included in the project. A crab sector study and value-chain analysis was conducted on the linkages between the community and industry stakeholders. From this study, an industry protocol was drafted to ensure a steady supply of quality lean crabs, reliable market outlets that offer fair prices to local fishermen, and accessible business development services.

The initiative next focused on awareness-raising within the local crab industry. A protocol on crab fattening was developed and shared with project participants. Mangrove reforestation and restoration was initiated in strategic portions of abandoned fishponds. Community members were involved in documentation of mangrove species as well as replanting efforts. Five fattening sites were identified, farm areas demarcated, bamboo poles set up, a feed area and 'guard house' were established, and lean crabs were procured and distributed. The final stage in this phase included a crab parade and mangrove festival, where wholesale and retail marketing support was provided and market agreements between crab farmers and buyers negotiated.

### *Innovations in technology, value-chains, and governance*

The most noteworthy innovation of this project is the disaster-resilient tie-crab fattening technique, which involves tying individual crabs, each with a buoy, to bamboo poles. As a consequence, crabs do not easily escape during farm flooding (a regular occurrence), as they are securely fastened to the bamboo poles. In the event that the poles break during floods, the crabs can be recovered, as the buoys are easily visible from above the floodwater. When faced with severe storms, farmers can quickly collect the crabs, place them in a secure shelter, and then return them to the farms after the storm has passed. Individual feeding allows for rationing based on the consumption capacity of individual crabs, meaning that there is less wasting of food as compared to grow-out methods of crab cultivation. In addition to being an easily transferred technology, tie-crab fattening adds to ease of harvesting, greater selection in harvesting based on weight and maturity, and a higher return on investment, as profits are 50 percent higher over a 15 to 20 day period as compared with the grow-out method.

By setting standards and consolidating the tie-crab association, the Trowel Development Foundation has enhanced the group's collective bargaining capacity and improved the value-chain for crab fattening and marketing. Negotiations between crab growers and buyers are facilitated in advance of the fattened crabs being ready for market in order to establish mutually beneficial outcomes and to bolster the bargaining power of local crab growers who are otherwise in a compromised position.

Another project innovation has been the co-management approach to mangrove restoration, which brings together a range of relevant stakeholders, including civil society organizations, government, and the business sector. Cooperating organizations are equal and active participants in decision-making, and each performs a distinct developmental function that complements the work of other players. Similarly, benefit-sharing is at the heart of the co-management arrangement.

#### *Multifaceted service delivery and capacity building*

Trowel Development Foundation is actively involved in integrated service delivery to the local community, including credit and savings services. These services are provided for small-scale farmer activities and enterprise development in the forest, agriculture, and fishery sectors. The initiative also provides technical support to local farmers on organic agriculture, forest management and agroforestry techniques, and coastal resources management. Trowel Development Foundation also offers trainings to community members on project management, accounting, and gender sensitivity.

### BIODIVERSITY IMPACTS

The area where the Trowel Development Foundation is currently active has a combined coastline of 102 km, comprised mainly of coastal and marine ecosystems and wetlands that include mangrove swamps, aquaculture ponds, coral reefs, and seagrass beds. The region's mangrove forest contains thirteen mangrove species from seven families. Due to ongoing degradation, mangrove forest densities are low and increasingly shrinking, with an average of 6,000 stems per hectare.

Reforestation and restoration of 20 ha of mangrove ecosystems have resulted in the recovery, reappearance, and proliferation of fish species, as well as other native marine fauna. Targeting previously

idle fish ponds, mangrove reforestation has helped to reduce shore erosion and buffer against climate-related natural disasters. Beyond mangrove reforestation, tie-crab farming itself has a number of biodiversity benefits. The tie-crab approach allows for more targeted and less destructive harvesting, and the demarcation of tie-crab farms has created *de facto* marine sanctuaries where aquatic species are able to regenerate and thrive. Tie-crab farmers report the reappearance of native fish species in farm sites and in adjacent areas. Farmers have also released fattened and matured crabs for spawning in the wild. This has led to a dramatic increase in crablet stocks, which, according to researchers, have grown to 12 million individuals.

### SOCIOECONOMIC IMPACTS

By doubling the income of project participants, Trowel Development Foundation has improved the quality of life of over 250 families, who can now meet basic needs as a result of increased incomes and diversified livelihoods. On average, participating families have seen an increase in monthly earnings of US\$69. This revenue has been invested into school fees and meeting the subsistence needs of local households. Project activities have also resulted in greater food security, through the reliable production of crabs, as well as the reappearance of wild fish in crab sanctuaries. Whereas farmers had no reliable source of fish previously, they now report an average weekly take of three kg, or the equivalent of an additional US\$32 per month.

For tie-crab fattening, the cost of land averages US\$313 per hectare per year. Since tie-crab farmers only require about 100 m<sup>2</sup> for a functioning small-scale farm, the cost of the pond area is very low. The time required to develop a crab farm is minimal, taking between 15 and 20 days for crabs to fatten, at which point they are harvestable. There is high demand for tie-crabs, both locally and for export, and tie-crab farmers can gross US\$161 per fattening cycle.

Within the community, women have been a primary target of capacity building and support. Women are supported to engage in the buying and selling of lean and robust crabs, each earning US\$5 for a half day of trading. Trowel Development Foundation has also promoted a 'Passing on the Gifts' approach, whereby primary project beneficiaries are encouraged to transfer and share the original assistance they received with other families in need of capacity building support.





## POLICY IMPACTS

Trowel Development Foundation actively advocates at the local and provincial levels for community-rights and entitlements, and the delineation of mangrove forests for both tie-crab farming and community co-management. In 2011, a municipal tie-crab ordinance was negotiated in order to formally establish mud crab sanctuaries in the mangroves of two municipalities. At the provincial level, Trowel Development Foundation has succeeded in introducing a mud crab ordinance for all of Northern Samar. Tie-crab farmers help enforce the ordinance by reporting violators who export undersized crablets outside the province, thereby diminishing the sustainable and reproductive capacity of local crab populations and ecosystems. The organization is also in conversation with the provincial government regarding the allocation of funds for mangrove reforestation efforts.

## SUSTAINABILITY AND REPLICATION

The project is financially sustainable because capital requirements are relatively small, labor inputs are minimal, and many of the materials are locally available. Also, tie-crab fattening as a technique requires only modest levels of expertise and knowledge, which is abundant amongst the local population. The financial sustainability of the Trowel Development Foundation is provided for by partners and collaborators. The 'Passing on the Gifts' approach,

wherein original recipients of project assistance are required to transfer assistance to a selected recipient, ensures future project growth.

The Trowel Development Foundation's success has bred interest in other communities. Neighboring communities and municipalities have seen first-hand the transformative effect that the initiative's activities have had on the local economy, on household-level food security, and on mangrove and coastal ecosystems. Equally attractive to communities is how easily transferred the project model is to other sites, and the relatively low levels of capital investment and resource inputs needed. The project model is currently being promoted among local government units for integration into local development plans and was highlighted at a national aquaculture conference.

## PARTNERS

- **Bureau of Fisheries and Aquatic Resources:** Technical support
- **Philippine Social Enterprise Network:** Support for value chain development
- **The College of Science and Fisheries Department of the University of Eastern Philippines:** Technical support for coastal resources management
- **Southeast Asian Fisheries Development Center:** Technical support for aquaculture

# WHAT DO THE CASES TELL US?

Community-based approaches represent some of the most effective and sustainable avenues to genuine forest restoration and recovery. They tend to be more sustainable than large state-sponsored afforestation efforts because the benefits they deliver to communities are substantial and increase over time, rewarding continued stewardship by local people. They not only increase forest productivity and support forest livelihoods, they inspire local collective action, transforming community perceptions of what can be accomplished in local forests, especially among the young. Community-based forest restoration is effective even in areas that large-scale afforestation projects avoid, such as drylands, coastal mangrove forests, village forest patches, and other areas where forest degradation has eroded local forest benefits even though standing forest remains. Thus, it offers a way to take advantage of mitigation and adaptation opportunities that might otherwise be ignored.

With the questions posed at the beginning of Theme 3 in mind, the following lessons emerge.

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**Motivation and Results:** Communities undertake forest restoration primarily to support local livelihood goals or regain lost services such as water regulation or coastal storm protection. But restoration has social benefits and teaching value as well, inspiring local participation in forest activities.

The immediate goal of most community-based restoration is to regain forest structure, to restore ecosystem functioning and to augment productivity to support forest livelihoods, often by introducing agroforestry species. In Kenya, the Kijabe Environment Volunteers works with communities to plant some 100,000 seedlings per year of indigenous species to rehabilitate the Kereita Forest. Additional seedlings are sold below cost to local households for agroforestry use in their home plots, to reduce the need for tapping forest resources. In the Kereita Forest and in many communities, local nurseries produce the seedlings used in augmentation plantings, and often become substantial forest enterprises themselves, and another source of forest-friendly income.

In addition to boosting forest productivity, restoration activities allow degraded forests to regain healthy ecosystem functions and recover the many ecosystem services they formerly provided. Prominent among these is better watershed regulation, with local communities such as Abrha Weatsbha and the residents of Bayamo experiencing more regular water supply, and diminished erosion and flooding. In Ando Kpomey, the establishment of the community greenbelt resulted not only in better water recharge of local wells, but in suppression of bush fires, which had become common after the disappearance of the original forest. In Northern Samar in the Philippines, replanting of mangroves under the guidance of the Trowel Development Foundation brought back both the rich marine habitat and the coastal protective function associated with mangrove forests.

**“In addition to boosting forest productivity, restoration activities allow degraded forests to regain healthy ecosystem functions and recover the many ecosystem services they formerly provided.”**

Community-initiated forest restoration is rarely a stand-alone activity as it is in large state-sponsored or private sector afforestation projects. In the community context, it is part of a mutually reinforcing suite of activities that include developing alternative livelihood and agricultural options to ease forest pressures, and adopting and enforcing rules to regulate forest and land use. In Abrha Weatsbha Community in Ethiopia, Bayamo in Cuba, and even the Trowel Development Foundation in the Philippines, for example, tree planting was only part of a larger, integrated plan to restore the land and resource base and stabilize the local economy.

Whatever the original motivation for restoration activities, they often take on greater importance within the community as time goes on, acting as a

source of inspiration for what can be accomplished, and a corroboration of the community's initial vision for local forest management. In some cases, restoration can be transformative in the life of the community. In Ando Kpomey Community in Togo, original ambitions for the newly planted community greenbelt were modest. But as time went on and the greenbelt began to function as a real forest, the community's vision expanded to embrace this larger function. The forest became a community focus—a contributor to the village economy and a source of pride and recognition within the largely deforested region—and the village began to see itself as a forest community again. The transformation has been so impressive that five neighboring communities have adopted the community's forest regeneration model. Similarly, restoration activities in Abrha Weatsbha in Ethiopia have transformed the village landscape markedly, and with it, the community's view of its future.

Forest restoration is also commonly used as a teaching opportunity. Ando Kpomey encourages educational visits from other communities and regularly conducts awareness raising workshops on its methods, even going out to other villages to share their experience. Because raising and planting seedlings are jobs that even young people can do, restoration efforts frequently become a platform for environmental education and a way for local schools to contribute to the community endeavor. In the Kereita Forest, for example, the Kijabe Environment Volunteers carries out extensive mentoring work with local youth groups, including enlisting their help tending tree seedlings at local nurseries.

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**Mitigation Potential:** Community-led forest restoration has considerable global mitigation potential because it is widespread, is especially suited to restore degraded forests, is applicable in areas where large-scale reforestation projects are not, and is sustainable due to community buy-in.

Community-led forest restoration has several attributes that make it flexible and widely applicable, and that give it great potential to contribute to the global forest restoration goals of the New York Declaration on Forests and the Sustainable Development Goals. First, it is well-suited for degraded forests—areas where standing forest remains, but where forest quality is low. While

these areas are not fully deforested, their ecological functions, productivity, and carbon storage are impaired. The Kereita Forest where the Kijabe Environment Volunteers works is a good example, with selective timber felling, charcoal manufacture, and uncontrolled grazing all contributing to the forest's degraded condition. The restoration effort there was effective because it not only augmented the forest with new seedlings, but at the same time acted to discourage destructive behavior through education and development of economic alternatives.

**“Community-initiated forest restoration is rarely a stand-alone activity ... it is part of a mutually reinforcing suite of activities that include developing alternative livelihood and agricultural options to ease forest pressures.”**

This ability to reverse forest degradation is especially important today, because such degradation is increasingly common, and its significance to the global climate is increasing rapidly as well. While global greenhouse gas emissions from deforestation dropped somewhat from 2001 to 2015, emissions from forest degradation increased some 150 percent over this period, according to a recent UN Food and Agriculture Organization estimate. Thus, the need to target degraded forests in restoration efforts is urgent. Community-based initiatives are one of the only strategies proven to successfully target these areas, which are difficult to address with large-scale reforestation regimes.

In a similar vein, and for similar reasons, community-led restoration is applicable in many other situations where large-scale afforestation and reforestation programs are not well-suited. These include drylands, coastal mangrove forests, village forest patches, and areas where agriculture and other land uses are closely integrated with the local forest. The labor-intensive and locally tailored approach of community-led restoration works well in these demanding situations, as in the dryland environment of Abrha Weatsbha and the coastal swamps of Northern Samar in the Philippines.



**“Many communities are using forest restoration as a way to adapt to disruptions in seasonal rainfall patterns, increasing risks from severe weather and storm surges, and other environmental stresses related to climate change.”**

Another critical attribute of community-based forest restoration is its sustainability. The return on the investment in time and energy that the community initially makes grows significantly over time as the forest's productivity is restored, providing an inducement to stay involved and protect the restored forest from renewed degradation. In Ando Kpomey, community commitment to forest restoration gradually grew alongside the new greenbelt, first planted in 1973, and residents have evolved a system of bylaws, local taxes, and training to protect their investment and assure the forest's future. In Empresa Forestal Integrada de Bayamo in Cuba, 95 percent of the households participating in the land restoration program have maintained their forest plots—a sharp contrast with the record of previous large-scale reforestation projects in the area.

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**Restoration and Adaptation:** Forest restoration is a powerful climate adaptation tool. It rebuilds landscape resilience, supports a transition to climate-adapted livelihoods, and restores protective functions such as flood control, erosion prevention, and coastal buffering.

Forest restoration is not only a way to repair the damage of the past, but to adapt to changing conditions today and in the future due to a changing climate. Many communities are using forest restoration as a way to adapt to disruptions in seasonal rainfall patterns, increasing risks from severe weather and storm surges, and other environmental stresses related to climate change. In fact, a number of Equator-Prize winning forest initiatives have been recognized for their best practices in community-based adaptation. Trowel Development Foundation in the Philippines focuses on mangrove reforestation, replanting native mangroves in abandoned aquaculture ponds. The goal is two-fold:

restore the natural coastal protection that mangrove forests had provided before their replacement with fish farms, and regenerate the original productivity of the nearshore marine environment that is a mainstay of the local subsistence and cash economy. A critical part of the community restoration work has been establishing a new local livelihood option in the form of crab raising. The restored mangrove swamps have provided a perfect environment for a new climate-adapted system of raising and fattening crabs by tethering them to bamboo poles set among the mangroves with a buoy. The bamboo tethering system encourages rapid weight gain, makes harvest easy, and yet is storm resistant, since the crabs are easy to remove when storm-related flooding hits. The combination of mangrove restoration and high-value crab farming has revitalized the local fishing economy while better protecting vulnerable coastal villages from storm surges.

In the drought-prone Tigray region of northern Ethiopia, adaptation has taken a different form. The Abrha Weatsbha community was on the verge of abandoning their village due to severe land degradation brought on by near complete deforestation of the region and poor grazing and farming practices. Climate patterns in recent years, with even more irregular rain cycles, have only made conditions worse. In response, the community launched the Abrha Weatsbha Natural Resource Management Initiative, which combined a comprehensive tree-planting campaign with grazing and soil management restrictions, and the construction of small dams, wells, and ponds. The result was the rapid return of vegetative cover to the surrounding common lands, reduced erosion, and increased water infiltration and storage. This greatly increased the quantity and reliability of the community water supply. While return of a mature forest will require many more years of community diligence, the planting of fruit trees and other agroforestry efforts has already brought gains in income and food security.

**“Forest restoration is not only a way to repair the damage of the past, but to adapt to changing conditions today and in the future due to a changing climate.”**

The experiences of the Trowel Foundation and Arbha Weatsbha demonstrate that the adaptation benefits of forest restoration are far reaching and go beyond a response to climate change alone. Community-based forest initiatives help forest communities become 'adaptive' in the widest sense. Forest restoration and the return of ecosystem functioning translates to healthier, more resilient forests, which

in turn support forest community resilience, with more economic opportunities, and less vulnerability to natural disasters and environmental threats, whether from climate change or other sources. In this sense, it is a precursor and key ingredient of a wider landscape resilience, where both communities and natural systems can continue to thrive in the face of change.







# Theme 4

## Forest Protection

**C**ases in this section feature community initiatives that protect forests in indigenous and community conserved areas (ICCAs), sanctuaries, reserves, or other explicitly protected zones. They also examine instances where local communities take a lead role in managing state-protected areas adjacent to or encompassing the communities. These community-protected forests vary widely in size from a few hectares to many thousands of hectares, and also vary in the kinds of use permitted within their borders. Regardless of size or use, the intent for all these areas is to protect them from degradation. They therefore represent areas with both high biodiversity value and high climate mitigation potential.

As these cases reveal, communities have compiled an impressive conservation record in protected forest areas that they have designated themselves. They have also proved to be engaged and committed conservation partners in state-designated protected areas when given the chance. Indigenous and local people create community conserved areas for many different reasons—to preserve forest areas or species of special cultural relevance, to protect watershed functions, to restore ecosystem functioning, or to guard the integrity of their traditional lands, to name a few. Whatever the specific local reason, and although the efforts often produce benefits far beyond the local community, these areas represent conservation done on the communities' terms. For this reason, these local conservation efforts are often more sustainable and effective than top-down government programs.

## CASES IN THIS SECTION

- **Monks Community Forest**, Cambodia
- **Corporación Serraniagua (Serraniagua Corporation)**, Colombia
- **Wechiau Community Hippo Sanctuary (WCHS)**, Ghana
- **I-tokani nei Sisi (Sisi Initiative Site Support Group)**, Fiji
- **Nam Ha Ecotourism Project**, Lao People's Democratic Republic

## UNDERSTANDING THE CASES

To understand why and how communities protect forest areas, their conservation value, and what lessons they might hold for the conservation community and the climate change community, readers should keep in mind the following questions:

- **Conservation and Local Culture:** Why do communities establish protected forest areas? How do these ICCAs differ from state-established protected areas? What benefits do local people receive when they co-manage state protected areas? How do local cultural values and customary systems of authority affect community conservation activities?
- **Wildlife protection and forest conservation:** How does wildlife protection – particularly those efforts built around a flagship or charismatic species – motivate community forest protection? What does this tell us about the scope for win-win efforts in biodiversity conservation and forest protection and the role for communities in managing these efforts?
- **ICCA Effectiveness:** How well do community-established protected areas meet local goals? What is their conservation value?
- **Beyond the Local Level:** Do the effects of local forest protected areas extend beyond the local area? Do they overlap or expand on state-established protected areas?





# MONKS COMMUNITY FOREST

## Cambodia

### PROJECT SUMMARY

Monks Community Forest is an 18,261 ha evergreen forest in northwest Cambodia. In response to widespread deforestation, the monks of Samraong Pagoda acquired legal protection of the forest, established patrol teams, demarcated the forest's boundaries, and raised environmental awareness among local communities. The monks developed unique approaches to law enforcement based on Buddhist principles, demonstrating the power of linking conservation with traditional customs and beliefs.

A co-management committee of villagers, government authorities and NGOs has been developed to manage what is now Cambodia's largest community forest. While logging and hunting are prohibited, villagers may use traditional fishing methods, collect fallen timber for construction, and harvest non-timber forest products like bamboo, wild ginger, fruit, and mushrooms.





## BACKGROUND AND CONTEXT

In 2001, the Buddhist monk Venerable Bun Saluth initiated the protection of an 18,261-ha stretch of evergreen forest in northwest Cambodia, now referred to as the Monks Community Forest (MCF). Venerable Saluth had witnessed the continued decimation of his country's forests by economic concessions, illegal logging, and land encroachment. The unlikely conservationist soon had volunteers from his pagoda and the local community organized into patrols that regularly monitored the forest to stop illegal harvesting activities.

With few resources, the monks of the Samraong Pagoda acquired legal protection of the forest, established patrol teams, demarcated forest boundaries, raised environmental awareness among local communities, organized community patrol volunteers, developed co-management committees with local villagers, linked with government authorities and NGOs, established the country's largest community forest, attracted external support for patrolling, awareness and livelihood activities, and significantly reduced forest crime in MCF.

The main purpose of this initiative is to protect forests and conserve biodiversity for future generations, as well as to maintain access to forest resources that benefit the local people's livelihoods in six villages. Currently,

nine members comprise the Community Forestry Management Committee (CFMC), which is further supported by 44 members of village subcommittees from six villages that are working on a voluntary basis to protect the Monks Community Forest.

The two primary activities of the organization are to patrol the area under protection and to raise awareness that the forest is protected. Villagers from six villages help spread awareness within their communities about the need to protect the forest and undertake patrols to stop illegal forestry and forest-related activities. As a direct result, forest crime has been greatly reduced in much of the Monks Community Forest.

## KEY ACTIVITIES AND INNOVATIONS

### *Modern and traditional forest monitoring*

Monks Community Forest has fostered good relationships with local and provincial authorities, including the Forestry Administration, police, and district and provincial governors in their efforts to combat forest crime. As monks, the group is well-positioned to work with a wide range of partners. Monks are respected in Cambodia, where the majority of the population is Buddhist and monks are looked to for their wisdom. The general population view monks as immune from corruption and as those working for the good of the people and the forests.





As villagers living close to the Monks Community Forest have stepped forward to patrol and advance conservation efforts, it has become clear that there are also myriad benefits to local livelihoods from forest resources.

Over 3,700 people from six villages participate in and benefit from Monks Community Forest activities. The vast majority of participants and beneficiaries are poor farmers, economically marginalized with no land title. The community forest provides resources, benefits, and services such as shelter, subsistence crops, and commercial products, all of which are essential for their survival and well-being. Participants have also been empowered to have a voice in the management of the community forest through their representatives on the central management committee and on the sub-committees in each village.

### *An innovative approach*

The Monks Community Forest is the first of its kind in Cambodia. The monks approach forest protection as an important part of the spiritual path in Buddhism. The monks apply key Buddhist teachings as the basis of the

environmental action and awareness-raising, emphasizing the elimination of suffering of all living beings and living ethically by not killing or harming sentient beings. In direct work with villagers and other stakeholders, the monks highlight the intimate role nature played in Buddha's life, stressing that his birth, enlightenment, and death all occurred in the forest.

The monks have developed a unique, 'soft' approach to enforcement. The approach minimizes violence by using unarmed patrols, treating offenders without anger, adopting a 'three-strikes' policy to transgression, and using photography to deter offenders. The monks conduct a tree ordination ceremony to bless the community forest's largest and oldest trees, wrapping their trunks in saffron robes to sanctify them. To cut down a tree or hunt wildlife within the ordained forest is considered as serious as harming a monk, with negative repercussions for the next rebirth. Because of their role in Khmer society as moral and spiritual leaders, the involvement of monks in the management and patrolling of the Monks Community Forest brings legitimacy to forest protection and serves as a powerful deterrent to forest crime.





## BIODIVERSITY IMPACTS

Patrolling and awareness-raising activities have dramatically reduced logging, hunting, and land clearing inside the Monks Community Forest, greatly helping to safeguard the area's biodiversity. While few scientific surveys have been conducted in the area under protection, Monks Community Forests is known to be home to a number of threatened species, including the sun bear, gibbon, gaur, slow iris, leopard, green peafowl, greater and lesser adjutants, pangolin, and dholes. The forest provides dense evergreen and semi-evergreen canopy cover rich in plant species and diversity. Patrol teams and village sub-committees use simple measurement systems to document the increase of species through MCF patrolling activities, where villagers and monks see wildlife and tree species thriving.

Monks Community Forest is participating in a Reducing Emissions from Deforestation and Forest Degradation (REDD+) project. An important component of this work is the conservation and monitoring of biodiversity within MCF through the participation of the local community, whose members are compensated for their involvement.

## SOCIOECONOMIC IMPACTS

Although logging and hunting are prohibited within the protected area, villagers can fish using traditional methods, collect old timber for materials for their shelters, and harvest non-timber forest products such as bamboo, wild ginger, fruit, and mushrooms. Villagers collect these resources for subsistence use and for traditional medicines, as well as to sell in local markets. Mushrooms are particularly lucrative, earning as much as US\$150-\$200 per month, an important cash source in a country where the average annual income is approximately US\$700.

Villagers recognize the livelihood benefits they receive from the Monks Community Forest, which translates to enthusiasm in participating in project activities. The monks have attracted external funding to assist village patrollers who currently volunteer their time and resources. The funds have provided emergency rice supplies for poor families, assisted them to bring non-timber forest products to market in more cost-effective ways, and provided them with food in exchange for patrolling services.



The monks have actively encouraged the participation of women, although there remains some cultural hesitation for women participating in patrols due to potential dangers. There are presently seven women involved in the village sub-committees. Venerable Saluth aims to have two more women assist the management committee in the Samraong Pagoda. Women are encouraged to take part in awareness-raising activities in the village. While women collect non-timber forest products they alert MCF patrol teams of suspicious or illegal activities. In addition, amongst 889 families from six villages, 1,918 women collect wild vegetables for consumption and for sale in local markets.

## POLICY IMPACTS

Monks Community Forest has brought together two key stakeholders of the Cambodian civil society – Buddhist monks and local communities – as powerful actors for environmental protection. Environmental stewardship is an emerging role for the country's monastic community, which is looked to for moral and spiritual guidance in their communities, and which increasingly wants to actively channel Buddhist principles to help society. In turn, monks have empowered villagers living near the Monks

Community Forest to take protection of the natural resources upon which they depend into their own hands by participating in conservation efforts and by co-managing the community forest.

Support received from villagers takes the form of contributions to the pagoda and volunteer work, all despite that fact that villagers are economically marginalized and spend a good amount of their time working in rice fields to provide for their families. Villagers have been motivated by the monks' efforts to protect the forest and wish to contribute. The monks dedicate their time (as well as the modest donations collected at the pagoda) to protecting the Monks Community Forest.

Monks Community Forest is one of thirteen community forests involved in Cambodia's first REDD+ carbon offset projects, which could provide sustainable financing for protection activities and livelihood support to local communities over the long term. The monks have developed unique approaches to law enforcement based on Buddhist principles, and demonstrated the power of linking conservation with traditional and religious beliefs. The experiences of the monks offer important lessons for conservation organizations in Cambodia and beyond.



## SUSTAINABILITY AND REPLICATION

Since its establishment in 2001, Monks Community Forestry has received financial and technical support from Buddhism for Development (BFD), Community Forestry International (CFI), PACT Cambodia, and, most recently, a group of nuns in the United States. The majority of conservation and awareness-raising activities, however, are possible because of donations of time and energy from monks and villagers.

If the REDD+ project is successful, carbon revenues will provide long-term financing to the Monks Community Forest. The REDD+ project also enjoys a high profile with a good amount of visibility, which will serve to further incentivize the government and other stakeholders to conserve the area and the related revenues it generates.

Monks Community Forest is arguably more sustainable than many of the large conservation projects in Cambodia, as motivation to protect the forest is deeply rooted in the Khmer spiritual and moral belief system. The monks have been able to cultivate a conservation ethic among villagers living near the MCF by explicitly linking nature conservation to the life of the Buddha, and to Buddhist principles such as karma and living a moral life. Through awareness-raising efforts, the overwhelmingly Buddhist villagers see forest protection as benefiting them spiritually, as well as materially, by stabilizing weather patterns for farming, conserving biodiversity for future generations, and maintaining access to forest resources that benefit their livelihoods.

MCF has ambitions to establish more patrols and to conduct farther-reaching awareness-raising activities. More resources are needed for fuel, food, and motor bikes to support patrol teams, as well as micro-finance projects for local community members, educational materials, demarcation posts, and convening resources for meetings with key local and provincial officials with villagers.

The initiative has resulted in the creation of Cambodia's largest community forest, with the support of the Cambodian Forestry Administration and Community Forestry International. With national goals to expand existing community forestry to more than 2 million ha from the current 113,000 ha, this site sets a critical precedent in demonstrating that communities can successfully manage larger high-value forest areas.

The example of the MCF has also demonstrated to observers at the local, national, regional, and international level the power that belief can have in conservation, expanding the potential role of Buddhist monks and Buddhist communities in the conservation of natural resources. This project has garnered the attention of several researchers and conservationists looking to civil society for new models of successful conservation.

Venerable Saluth is often invited to share experiences and the successes of MCF with different community forest groups within the province and national government officials in the Forestry Administration.

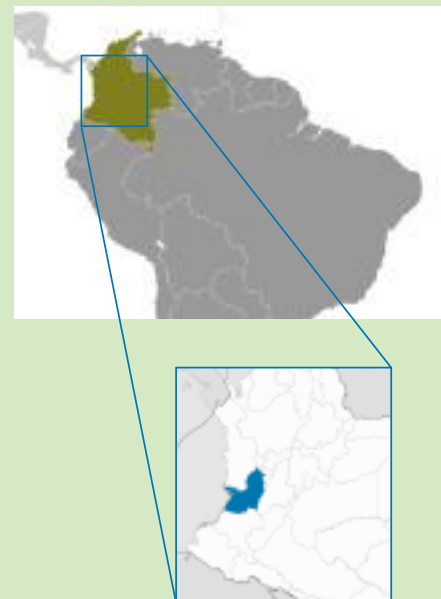
## PARTNERS

- **Buddhism for Development (BFD):** Training, in-kind support, motorbike for patrolling, petrol funds
- **Community Forestry International (CFI):** Legal and organizational assistance, forest demarcation and patrolling (US\$4,000)
- **PACT Cambodia/Forestry Administration:** REDD+ project



## PROJECT SUMMARY

Corporación Serraniagua works to ensure the connectivity of protected areas throughout Colombia's Cordillera Occidental mountain range, a key component of the Chocó-Manabí Conservation Corridor. The group connects the conservation corridors of the Tatamá National Park and Serranía de los Paraguas through a series of 60 community-managed and seven state-managed nature reserves, and encourages the participation of local and indigenous communities in environmental planning processes. Working through a broad stakeholder base, including cacao, coffee and sugar producers, ecotourism operators, environmental groups, rural schools, and women's associations, this dynamic social network protects the biodiversity of the surrounding region in a way that respects the livelihood needs of the local population.





## BACKGROUND AND CONTEXT

The Serranía de los Paraguas Mountains and the Tatamá National Park in Colombia form part of the Cordillera Occidental mountain range and the Chocó-Manabí Conservation Corridor. The high level of biodiversity, species endemism, and connectivity in these regions has made them the focus of significant conservation efforts. The Serranía de los Paraguas is covered with montane rainforest, and is an important watershed for surrounding communities. The mountains are home to 30 different nationally endangered animal and plant species, including the gold-ringed tanager, cauca guan, and the spectacled bear.

Corporación Serraniagua promotes community-managed nature reserves, encouraging the active participation of local and indigenous community stakeholders. The organization uses participatory processes to develop environmental management plans for the Serranía de los Paraguas, and its connection to the management plan of the Tatamá National Park. Primary goals of the association are the empowerment of local communities and

the effective integration of local stakeholders into decision-making processes. The ultimate objective is community input into the governance of local lands and resource management policy. Corporación Serraniagua operates at both the farm and regional levels, employing a zone-based connective strategy for land use planning, social networking, and the creation of productive agricultural systems that link local producers to equitable market supply-chains.

### *Conservation and agro-ecosystems in the Andes*

Conservation and sustainable livelihoods work takes place in the Andes where the predominant ecosystems are high Andean forests, sub-Andean forests, and secondary forests. The Tatamá Paraguas corridor contains the greatest wealth of threatened and endemic species in the continental Pacific region of Colombia. Several of these species are the focus of Corporación Serraniagua conservation efforts. The organization is also active in a number of agro-ecosystems, notably cacao and coffee, fruit orchards, and timber forest landscapes that contain palm fruit, plantains, avocados, and borojó.



### *Bridging policy and practice*

The organization was initiated by communities within the Serranía de los Paraguas mountain corridor, but benefited from the support of environmental NGOs and *corpocuenas*, state organizations with public and private involvement in watershed protection. One of the primary project catalysts was the lack of community participation in the development of land-use policies. As a result, there was little local ownership of government policies on the ground and a good deal of distance between prevailing policy frameworks and the realities of resource management activities at the grassroots. Other gaps which the association formed to address were a lack of awareness on the state of local biodiversity and ecosystems – specifically, the growing threats and drivers of biodiversity loss – and a lack of documentation on zoning. To address these issues, the organization set out to help land owners improve their spatial awareness and understanding of the territory and to better equip communities with the tools needed to form alliances, lobby government, and participate in decision-making processes.

### *Community protected area network*

Corporación Serraniagua uses a connectivity strategy to link community protected areas, local landowners, surrounding municipalities, and areas protected within Tatamá National Park. The strategy aspires to the protection of species within the wildlife corridor



by creating sustainable agricultural production networks. The association combines respect for local culture with a strong conservation ethic. Culture and conservation are linked through the promotion of customary resource management practices and land management systems.

The association has fostered a link between community practitioners, government authorities, and the science community. This multi-stakeholder model enables equitable territorial planning, and produces a strategic vision that is informed by sound science and local needs. Corporación Serraniagua emphasizes accurate data collection and the application of cartographic tools, economic needs assessments, and biological surveys in the management of the Serranía de los Paraguas region.

## KEY ACTIVITIES AND INNOVATIONS

Corporación Serraniagua connects the conservation corridors of Tatamá National Park and Serranía de los Paraguas through a series of 60 community-managed and seven state-managed nature reserves. The organization maintains a vision of sustainable development that improves the quality of life for local inhabitants while protecting regional biodiversity. The community-based network evolved in response to a need for greater stakeholder input into natural resource governance and more detailed documentation of local biodiversity.

The association oversees a diverse base of stakeholders, which includes producers of cacao, coffee, and sugar, as well as ecotourism ventures, environmental groups, rural schools, and women's associations. This dynamic social network is leveraged to protect the biodiversity and ecosystems of the surrounding region in a manner that respects the livelihood needs of the local population. A comprehensive communications program employs community radio, newsletters, environmental murals, educational videos, and community tours.

### *Protected area network*

Corporación Serraniagua brings together community-based and private protected areas to promote biodiversity conservation and the protection of threatened species in the region. Protection of the Serranía de los Paraguas wildlife corridors requires connectivity, which has necessitated the involvement of community, public, and private land owners,





as well as a number of different producer and resource user groups. The association coordinates activities between these stakeholders, developing landscape level conservation plans that connect public protected areas (e.g., municipal watersheds), community protected areas, and national protected areas. Corporación Serraniagua is also the proprietor of Cerro El Inglés, a 700 ha natural reserve that conserves the majority of threatened and endemic species within the conservation corridor. Building out from this 'conservation epicenter', the association has formed a core area of natural reserves in the region that include 60 private reserves which represent more than 2,250 ha across five municipalities and two departments.

#### *Agriculture, tailored production plans, and seed exchanges*

The conservation corridor where the association operates covers many different ecosystems, altitudinal ranges, and land uses. Corporación Serraniagua promotes harmony between these ecosystems and production landscapes to ensure the conservation of local culture, native farm species, and biodiversity. Corporación Serraniagua promotes a range of landscape management tools, including biological conservation corridors, living fences, the enrichment and diversification of coffee plantations, strategic reforestation with indigenous tree species,

the operation of community tree nurseries, and organic fruit orchards. Farmers working with the organization develop individual production plans through agroforestry systems where different species of plants are intercropped. Secondary crops (fruits, nuts, and vegetables) are added to production landscapes dominated by cacao and coffee to enhance incomes and food security. Some members of the Corporación Serraniagua also work in apiculture and honey production. The organization facilitates learning exchanges between communities, where farmers exchange seeds that form the basis of local agricultural diversity and food security.

#### *Environmental research and education*

Corporación Serraniagua works with local universities to carry out research and monitoring on ecosystem functioning, biodiversity and health, and on individual species living within the Cerro El Inglés reserve. The association carries out environmental education programs with schools throughout the region. A program called *Herederos del Planeta* ('Heirs of the Planet') targets children and youth and aims to ensure the strong foundation of a conservation ethic in future generations. The *Herederos del Planeta* group has grown into a national movement that forms part of the RED Colombiana de Reservas Naturales, a Colombian network of community-based natural reserves. In partnership with two



regional radio stations, Corporación Serraniagua oversees environmental education programming and instructional shows that promote conservation and the sustainable use of natural resources.

## BIODIVERSITY IMPACTS

The association maintains a strong commitment to biological monitoring, documentation of biodiversity, and land use planning. As a partnership of community managed nature reserves, the association has registered 343 species of birds, 56 species of mammals, 79 species of amphibians, 20 species of reptiles, three species of snails, 24 species of insects, and 646 species of plants. Over 500 community sites have received direct technical assistance on biological monitoring tools. These trainings include learning tours, reforestation and conservation workshops, and agricultural conferences. Biodiversity conservation and ecosystem management have also been mainstreamed into school curriculums through a partnership with 15 schools.

Corporación Serraniagua has created a biodiversity database for the region, filling an important void in available mechanisms to identify priority species and conservation interventions. Collected data is used to identify future conservation targets and land use plans. As a result of information gathered through the biodiversity database, the region was named an Important Bird Area by BirdLife International. Collected data has also driven expansion of the 60 participating protected areas. Greater knowledge of the status of key resources and species has translated to more effective land management strategies.

The group has steadily expanded community protected areas; since its inception, Corporación Serraniagua has integrated more than 3,000 ha of community nature reserves into the Tatamá-Paraguas regional network of private natural reserves. Cerro El Inglés, the group's proprietary natural reserve, has itself expanded by 500 ha. Ten new plant species and two new amphibian species have been found within the reserve, and ten endemic or threatened species are currently under protection within the reserve.

## SOCIOECONOMIC IMPACTS

The primary beneficiaries of this project include *campesinos* and their families, those living within the conservation corridor, private owners of the natural reserves, eco-agricultural farmers,

producers' associations, community environmental education groups, and, indirectly, those living on indigenous reservations in the region. The positive socioeconomic impacts have been varied and widespread. The association has been at the forefront of creating a market in the region for organic agriculture, providing local farmers with a reliable and steady source of income and an outlet for their produce. These networks provide otherwise economically marginalized farmers with collective bargaining power and market access benefits that were previously unattainable. Out of these producer federations and cooperatives have emerged a number of 'environmental defenders'; local leaders who champion environmental conservation and advocate for technical assistance and the provision of social services.

Corporación Serraniagua is not only an environmental network, but a social network that aims to improve communication and collaboration between land owners, eco-agricultural farmers and sustainable producer groups. Average family and producer group incomes have improved by an average of 25 percent from Corporación Serraniagua activities. While many of the benefits from the initiative are traditional socioeconomic gains (e.g., greater household incomes and more jobs), an even greater number of the benefits are nonmonetary and include the strengthening of the social fabric, a sense of pride and belonging for workers in the agricultural sector, community empowerment, and the kind of capacity development that comes from peer-to-peer learning and exchange. Three new community associations have been formed since the project began. Corporación Serraniagua itself employs 15 individuals, and has created more than 100 indirect jobs through its activities.

Among the more notable socioeconomic impacts has been official recognition of 60 community nature reserves, as well as a management plan that outlines conservation and sustainable production activities in the region. More than 500 project sites have received direct rural technical assistance, where local community members participate in learning tours, capacity building and training, agricultural conferences and workshops, and environmental and reforestation training. Through this on-site training the association has actively promoted organic agriculture and fostered markets for organic products.

## POLICY IMPACTS

Corporación Serraniagua has contributed to the formulation of a regional territorial regulation plan as well as a development plan for a local municipality. The group also contributed to the participatory environmental management plan of the Serranía de los Paraguas. The corporation has been an active participant in the debate on new protected area legislation in Colombia. The group is a founding member of La RED Colombia Verde and occupies the presidency of its board. In addition to this post, the organization participates in the Local Committee on Protected Areas for El Valle de Cauca and oversees the Technical Secretariat of the local system of protected areas in San José de Palmar—Chocó. Corporación Serraniagua provides technical support to surrounding municipalities and assists in the formulation of the management plans within these areas.

## SUSTAINABILITY AND REPLICATION

Corporación Serraniagua is in the process of creating an endowment that will ensure the sustainability of its projects. Since 1996, the scope of the association's operations has expanded. In its first two years, the association was focused geographically on the regional river basin and programmatically on protected areas, territorial mapping, and environmental planning. Since that time, Corporación Serraniagua has transitioned into the areas of social services and social networking, namely through the establishment of producer networks to support *campesinos*. It was through this evolution that a conservation organization became infused with an emphasis on sustainable production.

Corporación Serraniagua emphasizes the role of building local capacity in establishing public consensus and catalyzing collective action. Investments into local capacity building have paid dividends for community solidarity, the sense of community ownership of project activities, and participatory management. Investments in youth programs have educated and empowered the next generation of conservationists and organic farmers. Local producers of coffee, corn, cacao, and brown sugar have been empowered to connect with one another and improve their collective bargaining and marketing power. The association aims to foster an

identity among its members of communal interest and communal success. This group ethos has slowly supplanted individual profit motives, replacing it with a spirit of collaboration that has strengthened social cohesion and the sustainability of conservation interventions in the region.

Corporación Serraniagua is celebrated regionally and nationally as a model of successful community-based conservation. The association participates in regional forums for information exchange and peer-to-peer transfer of best practice, including the Inter-American Conservation Congress. Additionally, it has served as a progenitor organization, spawning other community-based activities. Replication of the group's best practices has been carried out through two national networks: RED Colombiana de Reservas Naturales, through which organizations share experiences on issues of conservation and sustainable production, and La RED Colombia Verde, which brings together community-based organizations and sustainable producers.

## PARTNERS

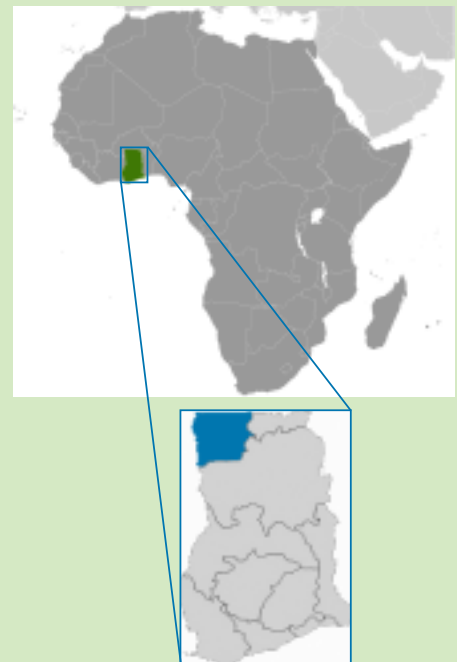
- **Tatamá National Park Administration:** Technical assistance
- **Asociación RED Colombiana de Reservas Naturales de la Sociedad:** Civil Networking
- **Herederos del Planeta:** Technical assistance
- **GREEN NETWORK de Colombia:** Networking
- **Fondo Para La Acción Ambiental y la Niñez:** Engaging local youth
- **Critical Ecosystem Partnership Fund (CEPF):** Support for private reserves, management planning, sustainable agriculture, and marketing (US\$152,945)
- **Municipality of San José Del Palmar:** Implementation support
- **Alliance for Conservation in Latin America**
- **Association of Producers of San Jose de Palmar:** Support for agricultural programs



# WECHIAU COMMUNITY HIPPO SANCTUARY (WCHS) Ghana

## PROJECT SUMMARY

This community-managed wildlife sanctuary occupies a 34-km stretch of riverine forest, floodplain, and savanna woodland along the Black Volta River, in northwestern Ghana. Created in 1998 in response to the decline of hippopotami due to high levels of hunting, the sanctuary uses revenue from ecotourism to deliver infrastructure investments to the residents of its 17 member communities. Through a balancing of ecological and social needs, the sanctuary delivers substantial conservation and socioeconomic benefits: poaching has been eliminated and the hippo population has stabilized, while investments in schools, health facilities, solar lighting, and water infrastructure have improved the well-being of approximately 10,000 local residents. The initiative was used as a model for the design of Ghana's Community Resource Management Area legislation.





## BACKGROUND AND CONTEXT

Wechiau Community Hippo Sanctuary (WCHS) is a community-based initiative in northwestern Ghana that conserves hippopotami and their riparian habitat while promoting development for indigenous local communities. Founded in 1998, the sanctuary consists of a 34-km stretch of riverine forest, floodplain, and Guinea savanna woodland along the Black Volta River, which forms Ghana's Upper West Region's boundary with Burkina Faso. Through ecotourism, substantial improvements in infrastructure, and income-diversification projects, the initiative has been able to conserve a population of 20 hippos, one of two remaining populations in Ghana.

### *Project catalysts and evolution of the sanctuary*

Situated in one of the poorer corners of Ghana, WCHS operates in a region where comparatively little land is under formal protection, wildlife serves as a primary source of protein, hippopotami are in decline, and protected areas are often devoid of animals. The sanctuary was proposed in 1998 by the Paramount

Chief of the Wechiau Traditional Area and other local leaders. These community leaders had previously rejected proposals by Ghana's Wildlife Division to establish a government-run hippopotamus reserve in the area. Instead, they opted for a community-managed sanctuary that would protect the hippopotami, restore habitat, and assure community participation.

### *Sanctuary zoning*

The sanctuary is divided into two zones: the core zone and the development zone. The core zone, which includes the Black Volta River, its islands, and a one to two km wide riparian belt, is protected by prohibitions on farming, bush-burning, hunting, the cutting of plants and trees, and vehicle access, as well as restrictions on fishing, oyster collection, grazing, and the harvest of shea nuts and locust beans. The development zone extends five to ten km to the east of the core zone boundary, and consists of wooded savanna interspersed with human settlements and farmland. There are 17 resident communities within the development zone, with a population of 10,268 people, half of whom are children.



### *The implications of ethnicity for land rights*

Communities living within the development zone comprise four ethnic groups, each with distinct languages, namely Wala (known locally as Wechiege), Birifor, Hausa, and Dagaabe. The Wala settled the area in the 17th century and have customary rights to the land, whereas the other three ethnic groups do not. Of the sanctuary communities, Wechiau and Tokali are primarily Wala. This is vital to the sanctuary's work, due to Wala land rights and tenure security. Wechiau is the seat of the Paramount Chief for the Wechiau Traditional Area and is the gateway to the sanctuary.

Both Wala and Birifor people have legends and taboos that contribute to the local conservation ethic. Wala myths maintain that their people escaped enemies in the 1800s after being carried across the river by hippopotami. The Birifor believe that the forest houses *Kontoma* (bush spirits), who hang from tree branches along the river to warn hippopotami of danger. Hippopotami also play a role in creation stories and puberty rites, and are considered children of the river spirit. Both the Birifor and the Wala have hunting taboos surrounding the species.

## KEY ACTIVITIES AND INNOVATIONS

### *Ecotourism*

The primary project focus is on developing the sanctuary as an ecotourism destination. Tourists are able to take river safaris by boat or use blinds to view the hippopotamus population. Ethnotourism, offering visitors cultural tours to Birifor communities and Wechiau village, are also included. Visitor numbers have increased steadily since 1998; in 2009, over 2,200 tourists visited the sanctuary. In 2008, the sanctuary generated US\$7,458, covering salary expenses (US\$3,061), and assuring self-sufficiency.

### *Conservation through alternative livelihoods*

The management of the hippo sanctuary has had positive biodiversity impacts and has been achieved with the cooperation and support of local communities. Sanctuary bylaws prohibit many traditional income-generating activities in the riverine forest, therefore it has been necessary to provide local residents with alternative livelihood options and conservation incentives. WCHS provides training in value-added secondary processing, which gives the local population a market premium for sustainably



harvested, organic products. The organization has also demonstrated the benefits of conservation for local well-being. Revenues from ecotourism and other conservation activities have been invested into local infrastructure and to fill service gaps in education, health, and energy.

Environmental education and outreach has helped to improve local awareness about sanctuary activities and the imperative of biodiversity conservation. WCHS has produced four publications on the sanctuary, sustainable farming, living with wildlife, and ecotourism and locally distributed 3,000 copies. Follow-up ecological awareness assessments found a high level of awareness of WCHS activities (and the benefits associated with their work) among primary and middle-school students in the sanctuary's seven schools.





### *Conservation and monitoring in the sanctuary*

The conservation priorities for WCHS include ensuring the security of wildlife in the core zone and raising conservation awareness through environmental education. Trained tour guides and research teams carry out regular biodiversity monitoring. Monitoring focuses on hippopotamus counts, but includes avifaunal species as well. A team of ten sanctuary rangers is responsible for the maintenance of the Talawona Tourist Lodge, the hippo tree platforms, and the visitor center, as well as policing the sanctuary. Introduction of motion-activated wildlife cameras has enhanced the security of wildlife within the core zone. The cameras were successful in documenting wildlife species, and collecting photographic evidence of poachers and hunters trespassing within the core zone.

## BIODIVERSITY IMPACTS

### *Hippo monitoring*

The main conservation focus at WCHS is the hippo population. Between 1995 and 1997, 11 hippopotami were killed in the area; but since the founding of the sanctuary, there have been no recorded cases. Hippo surveys are conducted by dugout canoes along river transects to establish distribution and abundance. Records are made of the numbers of hippos sighted by sex and age, in addition to exit points along the banks. Positions are marked by GPS coordinates whenever possible.

### *Biological monitoring and surveys*

WCHS also undertakes avifaunal monitoring within the sanctuary, and in comparable areas outside the sanctuary to provide a point of comparison. Surveys are



undertaken by a local bird guide along permanently marked transects: four within the sanctuary and three outside it. Transects are surveyed multiple times in dry and wet season months. The species richness per sample was similar inside and outside the sanctuary during the wet season, but was significantly higher in the sanctuary during the dry season, indicating that greater habitat quality within the conservation area provides a refuge for species when resources are scarce. Presence and distribution of medicinal plants is also monitored, although there is presently a ban on their harvest within the sanctuary.

#### *Local bylaws and the core zone*

Local bylaws have been reinforced through demarcating the core zone, employing rangers within the sanctuary area, and installing of wildlife cameras. In 2007, WCHS established a permanent boundary line between the core zone and the development zone by planting over 100 native, fire-resistant mahogany saplings. A wide fire belt was also cut along the demarcation line to prevent the spread of seasonal bushfires from farmland to the riverine forest.

## SOCIOECONOMIC IMPACTS

The social and economic benefits of WCHS's work in the sanctuary communities have been substantial, from job creation in ecotourism to alternative livelihood options. WCHS has also made significant investments in local infrastructure that have improved community well-being. All investments in community works have been undertaken with sensitivity to the local cultural context and respect for various ethnic groups. To examine the relative gains of the sanctuary model, a 2010 study compared new infrastructure acquired by the sanctuary's 17 communities between 1999 and 2007 with infrastructure acquired by all other 93 communities in the Wechiau Paramountcy during this time (the communities outside the sanctuary are ethnically and socioeconomically comparable). The study found that the number of new amenities – including health clinics, school blocks, new roads, and boreholes – per community was higher in both large and small settlements within WCHS.





### *Investments in clean water and energy access*

WCHS-funded borehole drilling projects have provided all 17 sanctuary communities with access to clean drinking water. In 2002, only seven communities had a supply of safe drinking water in their settlements. The need for water – and to access the local river – was one of the key drivers for community incursion into the core zone. The provision of boreholes has substantially reduced human pressure on the core conservation area and improved local health conditions.

The Wechiau Solar Lighting Initiative was launched in 2005 with a target of installing 550 solar-powered LED light sources in the 17 communities, and was completed in 2009. Installation required an initial tariff fee of around US\$8.25. Community contributions were paid into a local banking firm to establish a capital investment fund, which has been used to cover the maintenance costs of the program. By the end of 2006, the fund had accrued over US\$2,000.

### *Investments in health and education*

WCHS has attracted international donors to fund primary and secondary schools in the sanctuary communities, and three health clinics. Between 2005 and 2007, a primary school building with a teacher bungalow was constructed in the WCHS

development zone. The school's six teachers offer instruction in seven grades to over 200 children. A second primary school, located north of the sanctuary, was constructed in 2009, and serves three communities where children previously had to walk a great distance to attend school (and were sometimes prevented from attending due to seasonal flooding). WCHS has also established a school twinning program, which partners schools in Canada with the sanctuary schools and facilitates the transfer of school supplies and cross-cultural relationships.

Older students have also benefitted from WCHS's work. Two scholarship funds enable academically-gifted students from sanctuary communities to attend high school. These two funds, both supported by international partners, have sponsored 15 students (including one female) to attend high school. The first high school graduate returned to work for WCHS as a tour guide.

Partnerships have also improved the delivery of health services to communities in the sanctuary. A partnership with the University of Calgary has facilitated the delivery of medical supplies to three health clinics. A preliminary needs assessment was conducted, and supplies are shipped to meet these needs on a quarterly basis. These shipments recently included a movable hospital bed.



### *Sanctuary employment and shea nut harvesting*

More than 50 people receive regular income as paid sanctuary staff. Included in this group are three stipend-receiving land owners, 17 full-time employees, and 31 to 41 commissioned staff. More than 100 additional individuals have benefited from part-time or seasonal employment as masons, painters, unskilled laborers, dancers, and artisans.

One of WCHS's most successful alternative livelihood projects has been the promotion of shea nut harvesting within the sanctuary. After research indicated a significant market for shea butter, WCHS contacted the Savannah Fruits Company, a shea butter exporter, to run a training workshop. Two members from each community attended and were given advice on harvesting and storage techniques for shea nuts. To further benefit local collectors, WCHS applied for and received organic, fair trade, and conservation certification for the project. The Savannah Fruits Company agreed to pay a 5 percent 'conservation premium' on shea nuts collected by sanctuary communities.

In 2008, 716 women were registered members of the organic shea nut cooperation and their first harvest sold for US\$12,438. By 2010, an additional 309 women registered to join the cooperative, bringing

the total number of trained collectors to 1,445. The 2010 harvest value was US\$16,000, with an organic premium of US\$2,400 going directly to the collectors. The total value of organic shea nuts purchased through the cooperative to date is US\$52,000. An organic shea butter processing center and storage warehouse is under construction, which will strengthen the project's financial sustainability.

### *Benefit sharing between ethnic groups*

Despite the disproportionate authority that Wala have as chiefs and landowners, employment and infrastructure benefits span ethnic divisions, gender, and the sanctuary's geographic expanse. Ethnic representation among sanctuary employees reflects their traditional skill sets and geography. Boatmen are usually Hausa or Birifor from the riverside communities where tourists access the water for river safaris. Tour guides are Wala from Wechiau, where the visitor center is located. Rangers are distributed throughout the development zone and include a mixture of Wala and Birifor. The shea cooperative has benefited families in all 17 communities, while infrastructure improvements have also been delivered across the sanctuary. A survey of attitudes among those community members most adversely affected by sanctuary work – herbalists, anglers, oyster collectors, and farmers bordering the core zone –





found that 72 percent of respondents acknowledged benefits of WCHS work to local people, particularly infrastructure (67 percent), but also employment (9 percent), and an improved economy (4 percent). While 33 percent expressed grievances over lost farmland or access to oysters, firewood, and forest herbs, 35 percent expressed appreciation of what WCHS had done for the area.

## POLICY IMPACTS

In 2009, Wechiau Community Hippo Sanctuary was designated as one of Ghana's first community resource management areas (CREMAs). This outcome was the culmination of a three-year process of advocacy and lobbying. The initiative is implemented by the Ghana Wildlife Division of the Forestry Commission, and is a government-designated protected area scheme which recognizes and transfers authority to local people to manage their own parklands. The Wechiau Community Hippo Sanctuary was used as a model during the initial design of the CREMA designation. Since this legal category was created, a number of other community-run programs across the country have secured CREMA status.

## SUSTAINABILITY AND REPLICATION

WCHS is socially and financially sustainable. Infrastructure and community development projects (e.g., boreholes and solar lighting) have been equitably distributed across the sanctuary communities. Forty percent of women in the sanctuary are registered members of the organic shea nut cooperative, which offers a reliable source of income. This combination of alternative income-generating projects with investments in local infrastructure has fostered trust and legitimacy with the local population.

One success factor has been the governance structure of WCHS, notably because of its ties to local and traditional authorities. Chieftaincy remains integral to Ghana's electoral system. As land owners, chiefs are a respected authority that external agencies can engage directly. And in the relationship between chiefs a balance of power has been established that facilitates accountability, transparency, and the equitable distribution of benefits. Although all chiefs are Wala, the creation of a Sanctuary Management Board has allowed all ethnic groups and communities to be represented in decisions regarding the sanctuary. Consequently, the designation of the protected core zone and the establishment of conservation-related bylaws were model examples of subsidiarity in practice.

To date, the WCHS model has been replicated at three sites in Ghana. In addition, the expertise of sanctuary management personnel has been requested in conjunction with the Nature Conservation Research Centre (NCRC) for development of a comparable site in Liberia. Discussions are also currently underway for the creation of a trans-boundary protected area with Burkina Faso, with the government planning to reinvigorate forest protection along a large stretch of the Black Volta River (including areas abutting WCHS) and holding the intention to use community-based approaches to ease implementation. This potential collaboration would help secure the wider ecological integrity of the Black Volta riparian zone and the spread of lessons learned through the operation of WCHS.

## PARTNERS

- **Alice Jamieson Girls' Academy:** Support for schools
- **Calgary Zoological Society:** Financial and technical assistance, capacity building
- **Canadian Hydro Developers Incorporated:** Support for solar lighting project
- **Earthwatch Institute:** Hippo monitoring
- **Friends of Wechiau Group:** Student sponsorships
- **Ghana Tourist Board:** Support for ecotourism
- **Hope for Health:** Medical supplies
- **Light Up The World Foundation:** Support for solar lighting project
- **Nature Conservation Research Centre:** Technical support
- **Savannah Fruits Company:** Support for shea butter project
- **Stanley Jones Elementary School:** Support for schools
- **University of Calgary:** Technical support, medical supplies, lab analysis
- **USAID:** Financial and technical support
- **Wa West District Assembly:** Implementation support

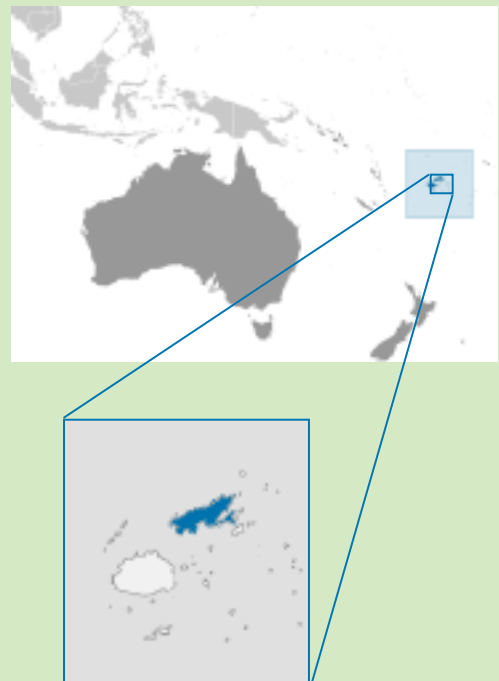


# I-TOKANI NEI SISI (SISI INITIATIVE SITE SUPPORT GROUP)

Fiji

## PROJECT SUMMARY

I-tokani nei Sisi manages natural resources around the periphery of the Natewa Tunuloa Important Bird Area. The organization has established a 6,000 ha community protected forest and developed alternative livelihood options for the area's indigenous landowners. Developed in response to illegal logging, forest fires, overgrazing, and agricultural encroachment, the organization uses an innovative incentive scheme to protect flora and fauna in Natewa Tunuloa. Communities signed a Memorandum of Understanding in which they agreed to protect the community forest and refuse logging concessions. The initiative provides alternative livelihood training and projects in beekeeping, poultry, handicraft and jewelry-making, bakery and pastry-making, and sustainable agriculture. The group's model farm and tree nursery also help to reduce deforestation. The initiative has been used as a learning model for community-based conservation and forest management across Fiji.





## BACKGROUND AND CONTEXT

I-tokani nei Sisi was established in 2005 to conserve the Natewa Tunuloa Important Bird Area (IBA) on the island of Vanua Levu in Fiji's Northern Division. It is a community-based volunteer group that works with a range of stakeholders to develop sustainable, environmentally-friendly initiatives for communities living in and around the IBA. These initiatives, including sustainable agriculture projects, ecotourism, and beekeeping, have given community members a way live in greater harmony with their natural environment, thereby reducing threats to the forest.

### *Natewa Tunuloa Important Bird Area*

The Natewa Tunuloa IBA contains most of the remaining tracts of native forest on the Natewa Tunuloa Peninsula. Communities living in the vicinity of the IBA have traditionally depended on agriculture for their subsistence and income, and rely heavily on the forest for firewood, game, timber, and medicinal plants. In recent years, the rate at which resources have been extracted from the forest grew to unsustainable levels, threatening the integrity and biodiversity of the forest.

Several threatened and endemic species are found on the Natewa Peninsula, including seven of the nine subspecies endemic to the island of Vanua Levu. The peninsula is home to the shy ground dove, classified as a vulnerable species, and the near-threatened silktail. The Natewa Peninsula was designated a 'Key Biodiversity Area' by Conservation International, a 'Site of National Significance' in Fiji's National Biodiversity Action Plan, and an 'Important Bird Area' by BirdLife International. Despite recognition of the area's natural significance, the peninsula does not have official protected area status and remains vulnerable to the threat of deforestation.

### *Deforestation, soil erosion, and water insecurity*

Excessive logging, human-induced fires, and overgrazing have historically been the main threats to the peninsula's forests. Large areas of native forests were cleared for mahogany plantations, and unsustainable logging continues to degrade the forests adjacent to the IBA. Since much of the flat land on the peninsula has been converted to coconut plantations, agriculture tends to encroach on the forest. The resulting deforestation leads to soil erosion and water insecurity, as well as threatening wider biodiversity.







The IBA is the source of the peninsula's rivers and of the drinking water for all of its villages. Logging has impacted water quality, disrupting ecosystem functioning and the water cycle, which in turn has affected the availability of drinking water and even the health of marine resources in Natewa Bay. The impact on drinking water quality in particular was severely affecting the living conditions of peninsula communities, and this provided the main impetus for community leaders to partner with BirdLife International in developing a forest conservation program.

### *Genesis of I-tokani nei Sisi*

Following designation of the 17,600 ha IBA in 2005, local communities developed a plan to conserve the forest. Local workshops were convened, community representatives were nominated to form a 'Site Support Group,' and the I-tokani nei Sisi was formally established as a community-based volunteer group charged with leading efforts to conserve the IBA. The initiative is named after the Sisi, a Fijian silktail bird and a characteristic species of the area found only on the peninsula and one neighboring island.

The primary challenge faced by the I-tokani nei Sisi was commercial logging, the second greatest source of income (after agriculture) for local residents and a cornerstone of the local economy. For conservation efforts to succeed, it would be necessary to develop economic alternatives to logging. After consultations with local communities and government officials, the initiative arrived at a Memorandum of Understanding (MOU) wherein 11 landowning clans agreed to conserve the forest and avoid logging for 10 years in exchange for village-based alternative livelihood projects. The MOU—the innovation at the heart of the I-tokani nei Sisi—was signed in 2009 and resulted in the declaration of a 6,000 ha community-managed forest.

The Site Support Group ensures equal representation of each of the landowning clans in the conservation program. Fiji is particularly well-suited to this land management approach, as over 80 percent of the country's territory is owned by indigenous peoples who are organized in land-owning units or clans known as mataqalis. In consultation with the statutory body that oversees management of land tenure issues in Fiji, these clans ultimately decide how land is used. The Site Support Group is made up of 15 members who are nominated by their respective mataqalis. Membership is limited to clans that own land within the community-managed area, meaning that forest management decisions rest solely with the landowners.

## KEY ACTIVITIES AND INNOVATIONS

### *From logging concessions to alternative livelihoods*

I-tokani nei Sisi's alternative livelihood projects provide the local community with sustainable sources of income that do not rely on felling local forests, thereby enhancing forest conservation. Model farms have been established to bolster food security and reduce encroachment of agricultural land into the forest. The farms are designed to mitigate soil erosion and enhance water and nutrient retention functions of the soil. Vetiver grass is planted to prevent erosion, while pineapples and vegetables provide supplemental food. Sandalwood farming increases local incomes and improves sustainable agricultural practices. A poultry farm has generated significant income for the local population; within two months, the community earned over US\$550 from the sale of chicks – a significant amount in a community where the average daily income is US\$11. Beekeeping is also



generating income through the sale of honey. The beekeeping project provides additional value as a reminder of the value of standing forests, as the bees source their nectar from forest plants. The initiative has also tailored livelihood projects specifically for women, including handicrafts and basket weaving (using raw materials gathered from the forest), jewelry-making and screen printing, baking, and pastry-making. These projects aim to revive local culture through traditional dance and weaving, with the intention of incorporating them into an ecotourism strategy for the IBA.

#### *Community-based forest management*

The most innovative dimension of initiative is its community-based forest management strategy, which is based on the MOU to sustainably manage 6,000 ha of forest. Under this agreement, participating communities agree to refuse logging concessions within the demarcated area, and to restore the remaining tracts of native forest. With technical guidance from Fiji's Department of Forestry, a forest resources management plan has been developed

that guides forest rehabilitation activities and the sustainable management of forest resources, which is a key factor in maintaining water quality, ensuring adequate food supplies, and mitigating flood risks. The management plan addresses the impact of deforestation on fish populations, and allows for connectivity for migratory fish species to move between the saltwater and freshwater habitats of the peninsula.

### BIODIVERSITY IMPACTS

Prior to establishment of the I-tokani nei Sisi, excessive logging in the Natewa peninsula forests was having deleterious effects on soil quality, biodiversity, and drinking water. So disruptive was the logging to ecosystem functioning and environmental health that dramatic changes were noted in marine and terrestrial species. The clearance of native forests was destroying the habitats of endemic bird species, which were also under pressure from the incursion of invasive alien species.

### *Forest conservation through local management*

By forging a MOU with local stakeholders and creating a 6,000 ha community-managed forest area, the I-tokani nei Sisi has effectively created a safe haven for birds and other biodiversity on the peninsula. The Department of Forestry has demarcated the forest boundaries of the IBA as an area that needs to be protected for the survival of the silktail. This has resulted in a decrease in the number of logging licenses issued for the IBA. Communities now report sightings of the silktail in forest areas closer to the village, a stark difference to few years ago where one had to walk deep into the forest to see the bird.

Alternative livelihood activities reduce pressure on the forest by creating options for community members who previously relied on unsustainable agriculture or logging for income. Model farms and tree nurseries reduce the need to clear forest for agriculture. Government researchers report significant improvements in sustainable agricultural practices in the communities due to the I-tokani nei Sisi. The handicraft, jewelry-making, and beekeeping projects all provide income that depends on non-timber forest products, and so provide additional incentives for the forest's conservation.

### *Building conservation capacity*

I-tokani nei Sisi has strengthened the capacity of the local population to sustainably manage and monitor their forest, empowering them to take responsibility for its conservation. Community members have been trained in bird identification and monitoring techniques so that they can take part in the monitoring of the IBA, while younger community members have been involved in reforestation and forest enhancement measures through community youth groups. By building community pride in the forest and highlighting the greater long-term value of standing forest over cleared land, the activities of the initiative ensure the conservation of the forest for generations to come.

## **SOCIOECONOMIC IMPACTS**

### *Alternative livelihoods and improved food security*

Several initiative activities are enhancing the short- and long-term food security of local communities. In the short term, pineapples, yams, and taro are being grown in nurseries and model farms, which provide

household-level food security without requiring that the local population resort to unsustainable agricultural practices. In the long term, the shift to sustainable agriculture will ensure the preservation of the forest and the services it provides. Beekeeping and poultry-rearing projects are also improving food security and providing new sources of income in place of logging concessions. The local communities now use honey from the beekeeping project to replace more expensive sugar in their diets, while the sale of surplus honey provides additional revenue.

### *Education, health, and capacity building*

Income generated through alternative livelihood projects is managed by village committees and is invested in community infrastructure, social services, and meeting collective needs. Community profits are invested in children's education and health services, while those profits accruing to individuals have also been directed towards school fees and medical costs. The training provided through the initiative is equipping local people with the skills needed to diversify their income base and sustainably manage natural resources, including the community forest.

### *Traditional knowledge and local culture*

In the management of the community forest, the I-tokani nei Sisi draws upon traditional knowledge and local customs. The MOU which established the community-managed forest is based on *mataqali*, or traditional clan land management authority, where each *mataqali* has the right to decide how their land is used. This system of customary rights is the cornerstone of both the initiative and the forest management plan. Similarly, many of the alternative livelihood strategies promoted by I-tokani nei Sisi are also based on traditional knowledge and local skills. The group has made concerted efforts to revive traditional cultural practices, such as *mekes*, or traditional Fijian dances.

## **POLICY IMPACTS**

Although Fiji has a number of protected areas and nature reserves, their effectiveness is compromised by the absence of national protected area legislation. The existing legal framework in Fiji offers a range of mechanisms with the potential to support the establishment and management of terrestrial protected areas. However, community conserved areas, and those established purely on the basis



of customary law – such as the Natewa Tunuloa community-managed area – are subject to certain limitations. In this respect, the I-tokani nei Sisi has brought needed attention to the importance of community-managed areas and the legal obstacles they confront.

The approach used by I-tokani nei Sisi has been brought to the attention of the National Protected Area Committee, which oversees the development of protected area legislation for Fiji. Since community-conserved areas do not retain legal powers of enforcement, the I-tokani nei Sisi has shown that ongoing conservation efforts, partnership with government entities, and alternative livelihood options are required to support land-owning communities. The mechanism by which the I-tokani nei Sisi established its community-managed forest area – through an agreement with local landowners – provides a powerful demonstration of how this can be achieved.

BirdLife International, in collaboration with relevant government entities, highlighted the approach employed by I-tokani nei Sisi at a 2010 forum of the Protected Area Committee. Consequently, Natewa Tunuloa has been recognized as a site where an informal protected area exists, and where legal endorsement of this status would cement community achievements. I-tokani nei Sisi is regarded as a leading example of what bottom-up action can achieve. Its experiences are now being taken into account in the development of a national policy and legal framework for protected areas.

## SUSTAINABILITY AND REPLICATION

All components of the I-tokani nei Sisi are deeply rooted in community ownership. From its inception, the I-tokani nei Sisi has prioritized a high degree of community involvement and engagement. The authority on which the forest management plan is based is derived from the communities themselves because it recognizes the ownership and control of *mataqalis* over their land. Respect for traditional authority and customs is a bedrock of the initiative

and the continued involvement of the *mataqalis* is a testament to the value they see in continuing to conserve the remaining tracts of native forest.

The I-tokani nei Sisi approach has great potential for replication in Fiji and across the region. The project model has already been replicated in five communities in Fiji. BirdLife International has facilitated peer-to-peer exchange programs whereby representatives of the I-tokani nei Sisi share their expertise and experiences with counterparts in other local conservation groups. Outputs from these learning exchanges will be deployed in the establishment of Fiji's Permanent Forest Estate Framework, which is designed to empower local landowners to become part of Fiji's sustainable forest program.

## PARTNERS

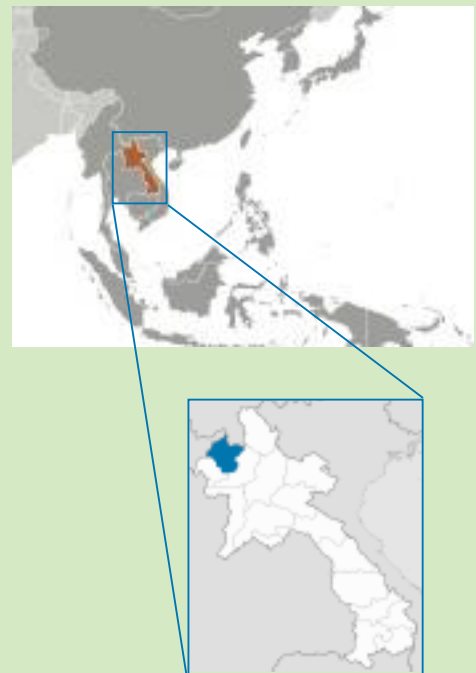
- **BirdLife International:** Financial and technical support
- **NatureFiji-MaregetiViti:** Livelihoods project
- **UNDP/GEF Small Grants Programme:** Financial support (US\$48,000)
- **Critical Ecosystem Partnership Fund (CEPF):** Financial support
- **Department of Environment:** Technical support
- **Department of Forestry and Agriculture:** Technical support
- **Cakaudrove Provincial Council**
- **I Taukei Land Trust:** Technical assistance
- **UK Darwin Initiative:** Financial support
- **Australian Government Regional Natural Heritage Programme:** Financial support
- **United States Embassy's Small Grants Program:** Financial support

# NAM HA ECOTOURISM PROJECT

## Lao People's Democratic Republic

### PROJECT SUMMARY

Located in the remote northern province of Luang Namtha on Lao PDR's border with China, the 222,400 ha Nam Ha National Protected Area includes some of the country's most significant wilderness areas. Altitudes range from river valleys and plains to highland peaks, supporting a broad range of habitats and biodiversity. Since 1999, conservation efforts in the area have been linked to improving ecotourism, which now underpins the economy for the area's 57 villages and 3,451 households. Community members are trained as eco-guides and operate village-based lodges and forest camps. They are also trained to monitor threats to biodiversity in the protected area, supporting the work of the under-resourced Protected Area Management Unit. The project has provided a model for co-management of Laos's protected areas.





## BACKGROUND AND CONTEXT

The Nam Ha Ecotourism Project is a model for the development of community-based ecotourism in Lao PDR's National Protected Areas (NPAs), and is based on the principles of local ownership and stewardship of natural resources. The Nam Ha project provides alternative employment and income-generating opportunities to communities living in and around the 222,400 ha Nam Ha NPA, thereby reducing pressure on natural resources in the NPA.

Nam Ha is located in the remote northern province of Luang Namtha on Lao PDR's border with China, and includes some of the country's most significant wilderness areas. Altitudes range from river valleys and plains to highland peaks, supporting a broad range of habitats and biodiversity. The majority of the NPA is covered by mixed secondary deciduous forest, while the mountain range is home to patches of primary evergreen forest mixed with secondary forest

and large patches of *Imperata* grassland. Mammals of significance include clouded leopard, leopard, tiger, gaur, Asian elephant, and Muntjac deer. The area is home to an estimated 288 bird species and great botanical biodiversity.

Initially established in 1993 with an area of 697 km<sup>2</sup>, the NPA was extended to 2,224 km<sup>2</sup> in 1999. As an area of natural beauty and cultural interest, the site was incorporated in the Association for Southeast Asian Nations Heritage Park Program. Local threats to the NPA include slash-and-burn agriculture, harvesting of non-timber forest products for sale and consumption, and hunting of wildlife for consumption. Other harmful activities, predominantly carried out by outsiders, include the hunting of wildlife for sale and free-ranging domestic animals that disturb wildlife populations through competition for habitat and the spread of diseases. Timber harvesting is also rife, while road construction has reduced habitats and simultaneously improved access for hunting and harvesting by outsiders.







Nam Ha guides and village-based tourism providers are drawn mostly from ethnic minority communities in Luang Namtha, one of the poorest provinces in Lao PDR. The project includes 57 villages, home to 3,451 families - a total target population of 21,227 people. Community members are trained as eco-guides and operate village-based lodges and forest camps. They are also trained to monitor threats to biodiversity in the protected area, supporting the work of the critically under-staffed and under-resourced Protected Area Management Unit.

Extensive ecotourism information and education campaigns targeted at policymakers, the private sector and local communities enhance the sustainability of tourism in Luang Namtha. As a result of these campaigns, the NPA's various stakeholders have been able to work collaboratively to create regulations that protect the cultural and natural resources that underpin the province's growing and profitable ecotourism sector. Local communities work closely with the NPA management authorities to create cooperative agreements that define stakeholder responsibilities in protecting the natural and cultural resources upon which the ecotourism ventures are based; provide guidance on the harvesting of forest resources; prohibit the unlicensed hunting or sale of wildlife; and set aside tracts of village-managed forests as bird, wildlife, and plant sanctuaries.

#### *The National Protected Area Management Unit*

The Lao National Tourism Administration is the main government agency concerned with guiding the development and regulation of Lao PDR's tourism industry. Each of the country's 16 provinces has a

Provincial Tourism Department; in Luang Namtha, the provincial office works in close conjunction with the Nam Ha NPA Management Unit. The NPA Management Unit receives a limited annual budget to carry out its duties and has little capacity to enforce regulations. Although trekking fees are dedicated to NPA management, a lack of human resources constrains the management unit's ability to effectively utilize these funds. Such institutional limitations have made the organization receptive to the implementation of protected area co-management with local communities.

## KEY ACTIVITIES AND INNOVATIONS

The management of individual cultural and natural tourist sites is performed directly by local communities that have historically acted as stewards of these resources, and their day-to-day operation is in the hands of village authorities. Since the introduction of the first community-based ecotourism programs by the Nam Ha Project in 1999, there have been 30 established tour circuits, involving some 50 communities. Participating communities have been prepared to provide services to tourists with support from the Nam Ha project management, using a community-based ecotourism methodology that includes extensive community awareness-raising and skills training.

### *Integrated Conservation and Development Initiatives*

Individual community-based projects are classified as 'Integrated Conservation and Development Initiatives' and typically comprise community agreements or concessions that provide revenue sharing mechanisms, income and infrastructure for the villages in exchange for improved local environmental management, or forest conservation schemes. These agreements, approved by the Provincial Tourism Department, village and candidate tour operator, assign a single operator exclusive access to a community-based tour circuit or host village based on rules governing the maximum size and frequency of tour groups. The agreement also sets a schedule of fees that the operator must pay the village for food, lodging, village-based guides, transportation, handicrafts, and trail maintenance. The agreement defines areas that are off-limits to tourism activities and also sets fees for permits and taxes.

Four examples of integrated conservation and development initiatives include: a scenic waterfall, critical to local water supply, that is protected and



managed in exchange for infrastructure development (e.g., walkways, bridges, parking lots, and toilets); a sacred spring, preserved in its natural state, but also offering a functional bathing pool for villagers through construction of a dam; a famous cave, now outfitted with a lighting system, generator, and walkway; and a popular tourist village that has received a clean water system.

All visitors to Luang Namtha are targeted by education campaigns that encourage cultural and environmental sensitivity. The use of posters, a handbook, and written behavioral guidelines displayed in tourist information centers, sends a clear message regarding appropriate behavior, in a number of languages. Much of the material used to develop these messages was contributed by host communities.

### *Monitoring*

Regular contact between officials and communities to monitor the economic, cultural, and environmental impacts of tourism is a critical aspect of the NPA's management. Monitoring activities measure impacts at the provincial, village, and individual household

level. NPA staff members and local guides are trained in data collection and analysis. Monitoring activities provide community leaders with information that allows them to identify and remedy problems in tourism management when they arise.

### *Generating funds and sharing revenue*

The Nam Ha Ecotourism Project has institutionalized tourism revenue-sharing schemes, generating public funds for tourism management, village development, and conservation activities. The Eco-Guide Service Unit, for instance, has a strict revenue-sharing arrangement that is based on a 'trekking and NPA user permit' fee system, the first of its kind in Lao PDR (and currently replicated in 20 NPAs across the country). These percentages cannot be changed without agreement between the Provincial Tourism Department, the Governor's Office, and the Provincial Agriculture and Forestry Office.

Tourism revenues in Luang Namtha contribute to various public funds, including: a five percent tourism tax that supports marketing activities and district tourism offices; an eight percent village development levy, used to repair infrastructure and purchase

equipment; a ten percent provincial tax to support the provincial treasury; and a US\$1 per person per day trekking permit that funds patrolling and maintenance of the Nam Ha NPA.

According to the local tourism department, visitor numbers and annual ecotourism revenue grew steadily between 2001 and 2008. Of this income (totaling almost US\$1,000,000 between 2001 and 2009), 27 percent has been directed to public funds. Other than local taxes and Provincial Tourism Department expenses, the remainder has been paid to villages, local guides, transport operators, and for local food and handicrafts. If all tourism spending in the province is taken into account, the total revenue from tourism accruing to local communities is much higher.

## BIODIVERSITY IMPACTS

### *Adding value to natural resources*

Through the development of alternative livelihoods, the Nam Ha Ecotourism Project has reduced pressure on the natural resources of the NPA, with over 100 local people now employed as guides, cooks, handicraft producers, and accommodation and transportation providers in the 17 villages that host community-based ecotours. The development of community-based tourism projects has conferred a new type of value on the ecosystems and biodiversity of the NPA in the following way: a community member who would formerly have harvested wild orchids for US\$0.20 per kg can now lead a group of tourists to these orchids as a village-based guide for US\$5 per day. The same principle holds for former hunters who are now employed as guides leading tourists to birds and wildlife. Local communities now see a direct link between the economic benefits of forest-based ecotourism and conservation.

### *Generating funds for conservation*

Ecotourism generates funds for protected area management and conservation extension work. The guide service alone has generated over US\$8,000 since 1999, while tour operators in Luang Namtha have paid over US\$10,000 to the NPA Management Unit since 2002, helping to support this otherwise underfunded office.

Another benefit for the NPA management has been the devolution of monitoring, reporting, and enforcement for conservation to the Provincial Tourism Department, the eco-guides, and the

tourism network throughout the NPA. This wide network of stakeholders has reinforced conservation messages; villages are more likely to report poaching or forest clearance transgressions to local authorities. The NPA management staff is also permanently stationed within the project area, helping to integrate conservation principles into tourism activities. The establishment of village-based sanctuaries, meanwhile, has created small 'islands' within the forest to allow for the regeneration of non-timber forest products and small wildlife species. The enforcement of cooperative agreements between communities, tour operators, and NPA managers has ensured that communities are invested in maintaining their local heritage and receive a fair share of the economic benefits that this generates. Village-based monitoring teams and the presence of tourists on walking trails in the NPA has discouraged illegal activities such as poaching and illegal harvesting of non-timber forest products.

## SOCIOECONOMIC IMPACTS

### *Alternative livelihoods and job creation*

Ecotourism generates significant revenue for the local economy. Community eco-guides and associated service providers have received over US\$600,000 since 1999 from the Eco-Guide Service Unit treks alone. This represents significant additional income for a province where the GDP per capita is US\$389, versus US\$678 nationally.

Tourism has supplemented traditional village livelihoods, allowing community members to diversify their occupations, save money, purchase household items, and pay school fees. For instance, women used to spend one or two days collecting bamboo or rattan shoots and then another day transporting them to market by foot, and could expect to earn US\$1 to US\$2 per day. These women can now spend two to three hours preparing a meal for tourists, and earn US\$3 to US\$6, without having to venture far from the village. The remaining time can be used for childcare or other activities.

The accommodation sector employs more than 300 people in Luang Namtha, while there are 172 full and part-time guides active in the province. Several hundred additional community members derive part-time employment from community-based tourism activities as food and accommodation providers. Throughout the province, the tourism sector is





estimated to support more than a thousand additional job opportunities in restaurants, on tourism-related construction projects, in tour agencies, transportation, producing handicrafts, or supplying agricultural products to the tourism supply chain.

### *Channeling revenue to communities*

In 2006, the Nam Ha Eco-Guide Service Unit sold 359 tours to a total of 1,787 persons, and generated US\$56,940 in gross revenue. This revenue accrued to participating communities in the following ways: meals and water (US\$1.5 per meal per person); village-based accommodation (US\$2 per night); transportation (US\$2 to US\$6, depending on trip); guide fees (from US\$5 to US\$15 per day); handicrafts (set price of US\$1.75 per piece); and village service fees (variable pricing for potable water, picnic sites, and services). These categories represented US\$41,069 in total trip expenses paid to local service providers in 2006. The balance of US\$15,871 was paid into public funds, including Village Development Funds and taxes, and covered the Nam Ha Eco-Guide Service Unit operations and maintenance costs.

### *Including women and minorities*

The Luang Namtha Provincial Tourism Department and Nam Ha Eco-Guide Service mainstream gender issues into all tourism-related activities. Monitoring data is disaggregated by gender and ethnicity, and shows that women and ethnic minorities have a high

rate of participation in community-based tourism ventures run by the Nam Ha Eco-Guides. In 2006, 20 percent of the guides were women and 95 percent were members of ethnic minorities. Luang Namtha province has over 20 ethnic groups, representing about 66 percent of the provincial population. The Nam Ha Guide Service has played a leading role in the provincial Ethnic Minority Participation Program, which encourages the inclusion of ethnic minorities in conservation and development activities.

## POLICY IMPACTS

The Nam Ha Ecotourism Project has formed a core part of national policies for both rural development and Lao PDR's tourism sector. The latter is guided by the Lao PDR National Tourism Strategy for 2006-2020 which emphasizes the development of tourism products and services based on the country's cultural, natural, and historic attractions. This strategy is exemplified by the Nam Ha Ecotourism Project's approach - it is estimated that more than half of the value of the Lao tourism industry is derived from nature- and culture-based activities. The positive role tourism plays in national development, with specific reference to sustainable tourism models in Luang Namtha, has been included in legislation such as the National Growth and Poverty Eradication Strategy, the National Tourism Strategy, and the National Biodiversity Strategy to 2020.

## SUSTAINABILITY AND REPLICATION

Many of the Nam Ha Ecotourism Project's programs are self-sustaining. For example, the Nam Ha Eco-Guide Service Unit has been financially sustainable since 2002. Ecological sustainability has been ensured through practices that minimize waste generation (such as use of biodegradable packaging and non-imported foods), and through renewable solar and micro-hydro energy in village lodges. Tour groups are usually kept to less than ten people to minimize environmental and cultural impacts on villages and trails.

Local communities' strong support for the project relies on several institutional factors, including organizational transparency, participation of women and ethnic minorities in service provision and management, reinvestment of a substantial portion of revenue into village development activities, and ability to cover all operations and maintenance costs without subsidies or external funding.

Based on the success of Nam Ha's Eco-Guide Service model, guide services have been initiated in two additional districts in Luang Namtha Province, while three services were opened in three separate provinces in Lao PDR between 2004 and 2006. In part, replication of the Nam Ha model has been facilitated by tourism awareness seminars held in the province for policymakers and tourism service providers.

## PARTNERS

- **Government of New Zealand:** Financial support
- **Government of Lao PDR:** Financial support
- **Lao National Tourism Administration:** Ecotourism training and certification, marketing
- **Luang Namtha Provincial Tourism Office:** Office space, equipment, ecotourism support
- **Nam Ha National Protected Area Management Unit:** Technical support (research), awareness-raising, ecotourism support
- **UNESCO:** Technical and financial assistance (US\$408,992 with funds from New Zealand Government)
- **Private sector tour operators:** Ecotourism support
- **Provincial Governor's Office:** Ecotourism laws and management
- **Provincial Tourism Department/District Tourism Offices:** Licenses and regulations for ecotourism, training, marketing, and promotion
- **Land Use Planning and Management Office:** Land use planning
- **Provincial Agriculture and Forestry Offices:** Support for NPA and ecotourism
- **District Agriculture and Forestry Offices:** Agricultural extension, forest policy enforcement
- **Department of Information and Culture:** Support for cultural tourism
- **Science, Technology, and Environment Office:** Environmental impact assessments, monitoring, solid waste disposal
- **Department of Planning and Investment:** Licensing, promotion of investment in ecotourism
- **Department of Communication, Transport, Post, and Construction:** Support for construction of ecotourism infrastructure
- **Tourist Police:** Enforcement of rules and regulations, security
- **Women's Union:** Support for gender mainstreaming
- **Department of Industry and Commerce:** Promotion of handicrafts and local products
- **Lao Front for National Construction:** Promotion of conservation, conflict resolution
- **Lao Youth Union:** Awareness-building, youth outreach

# WHAT DO THE CASES TELL US?

Indigenous and community conserved areas (ICCAs) and other forms of community protected forests demonstrate that forest conservation is not just the province of state-designated protected areas, but is an important community priority as well. Forests under community protection represent some of the highest-quality forests in the world. Their successful protection is important not just to community conservation goals, but to global biodiversity and climate goals.

Community-protected forests offer a powerful complement to state-designated protected areas. They are flexible and responsive to local demand, and often occur in otherwise neglected areas that state protected areas would pass by. They are rooted in local culture and demonstrate the power of linking conservation with traditional knowledge and customs. Also, they are frequently built around the protection of a rare or threatened species, offering an effective platform for both wildlife conservation and forest protection. Local participation in community-protected forest management is often high, increasing their effectiveness and sustainability. Emerging experience shows that linking these vital community protected spaces together in dynamic networks across the landscape can magnify their local benefits and scale up their biodiversity and climate benefits.

Considering the questions posed at the beginning of Theme 4, the following lessons emerge.

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**Conservation and Local Culture:** Community-protected forests reflect local cultures, express local conservation values, and respond to local environmental pressures. Their strength is that they are designed with the community in mind, making them more effective and sustainable than state-managed protected areas.

Community-protected forests reflect the cultural values and conditions on the ground in individual communities. As such, conservation efforts are tailored to meet local needs and express local autonomy, rather than to follow the conservation mandates of others. Levels of protection differ among community-protected forests, with some precluding local access altogether,

but many others allowing some forms of local use, such as collection of non-timber forest products. Often, they are part of larger land management systems and strategies, in which areas with stricter protection are interposed with other forest areas where greater community use is permitted, including agriculture.

**“Community-protected forests offer a powerful complement to state-designated protected areas. They are flexible and responsive to local demand, and often occur in otherwise neglected areas that state protected areas would pass by.”**

The immediate reasons for creating community-protected forests differ widely among communities and may include protecting sacred sites, preserving or restoring watershed functions, supporting traditional cultures, and contributing to the local economy. In the Monks Community Forest in Cambodia, a Buddhist monk disturbed by the rampant destruction of forests in his region inspired residents of six villages to declare the surrounding forest a protected area. With the goal of preserving the forest for future generations and retaining access for local people for sustainable forest use, volunteers from the six communities marked forest boundaries, formed forest patrols, conducted local awareness campaigns, and ultimately convinced the government to grant the forest legal protection. Similarly, I-tokani nei Sisi in Fiji was formed when continued forest degradation threatened a local forest area known for its bird diversity. A group of local residents on the Natewa Tunulua Peninsula convinced the 11 indigenous clans that are the principal land owners in the area to prioritize conservation and put in place a ten-year moratorium on logging in exchange for a suite of alternative livelihood projects directed at local villagers.

Many community-protected forests are also initiated to offer greater protection for rare or threatened wildlife species. Indeed, most if not all



of the community-protected forests presented here were initiated (or have been sustained) based on campaigns to protect dwindling populations of rare bird, mammal or plant species. Wildlife protection can, and often does, provide a proverbial rallying post around which the community mobilizes the collective will to undertake broader conservation activities in forest ecosystems. Charismatic species provide a face for conservation campaigns, a totem that represents the embodiment of community will.

In some cases, certain species have specific cultural or endemic value that motivates conservation and protection efforts. In the Wechiau Community Hippo Sanctuary in Ghana, for instance, the continued decline of the local hippo population due to poaching and habitat degradation inspired the creation of the sanctuary, which sharply restricts resource use along a 34-km stretch of the Black Volta River. Hippos are highly respected in local culture, and are featured in local creation stories and puberty rites. Similarly, in the case of I-tokani nei Sisi, the protection of several threatened species of birds endemic to the island of Vanua Levu – including the shy ground dove and the silktaill – was the motivation for designation of the 17,600-hectare Important Bird Area (IBA) in 2005 and the subsequent protection of surrounding forests.

In other cases, it is the territorial range of threatened species that motivates and sustains forest conservation efforts. In the case of Corporación Serraniagua, the Cordillera Occidental mountain range and the Chocó-Manabí Conservation Corridor is home to thirty different nationally endangered animal and plant species, including the gold-ringed tanager, cauca guan, and the spectacled bear. The Tatamá Paraguas corridor contains the greatest wealth of threatened and endemic species in the continental Pacific region of Colombia and many of these species have become the focus of Corporación Serraniagua conservation efforts. Protection of these species requires the protection of entire wildlife corridors and ensuring connectivity between community-run, public and private forests.

Further still, wildlife protection is often an investment in the local economy. Charismatic species can be the centerpiece of community-based ecotourism ventures, which provide an important source of jobs and income for local people. The Nam Ha Ecotourism Project in Laos, for instance, has become the engine of the local economy. The project includes 57 villages, home to 3,451 families - a total target population of 21,227 people. Community members are trained as eco-guides and operate village-based lodges and

forest camps. Protection of threatened populations of clouded leopard, leopard, tiger, gaur, Asian elephant, and the Muntjac deer has motivated and sustained protection of the surrounding forests. Forest and wildlife protection.

In each of these cases, the community-protected forest sprang organically from the community, reflected the desires of the local stakeholders, and was carried out and enforced by community members themselves. Because each of these initiatives evolved from and fit within local culture and practice, local adherence to any use restrictions imposed was high. These cases provide convincing examples of the power of practicing conservation on local terms, for local purposes. Their strength is that they are locally adapted and designed with the community in mind, and therefore more effective and sustainable than many state-managed protected areas, whose primary aim is preserving habitat and conserving biodiversity, with community concerns often only a secondary consideration.

**“Most of these conservation efforts exist within traditional social systems where local customs carry great weight, respect for traditional authority runs deep, and social enforcement of community norms is common.”**

Designating and managing a community-protected forest is not the only way in which communities participate in forest conservation. The Nam Ha Ecotourism Project in Laos is an example of how communities can become effective forest stewards when authority for state protected-area management is devolved to the local level. In this instance, 57 communities surrounding the 240,000 ha Nam Ha National Protected Area have been given responsibility for providing ecotourism services such as tour guides and local lodging, and for managing and protecting natural and cultural tourist sites within the protected area. Additionally, communities have established small village-based sanctuaries to encourage regeneration of forest wildlife and plants within the village environment. Although this is a state-managed project,

it effectively mobilizes the local knowledge and pride of the surrounding forest communities, and increases the financial resources of these communities through revenue sharing and employment.

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**Community-Protected Forest Effectiveness:** The conservation value of community-protected forests is high. Because they are embedded in local culture and arise from local demand, community compliance with use restrictions within these forests is good and local participation in their management is strong, boosting results.

Community protected areas can be highly effective in conserving local biodiversity and protecting resources of high value to the community, even though they often permit some forest use by local people. In the Wechiau Hippo Sanctuary, there have been no recorded cases of hippo poaching since the sanctuary's founding in 1998, and the hippo population has stabilized. Logging in the 6,000 ha community-declared protected area associated with Fiji's I-tokani nei Sisi has ended and restoration activities have begun. In the Monks Community Forest in Cambodia, illegal forest activities have declined sharply and the forest has now become Cambodia's largest community forest.

One obvious factor in the effectiveness of these community conservation efforts is the level of community participation in the formation and governance of the community-protected forests. Typically, land use plans are derived through community consultation and protection activities such as monitoring and enforcement fall to community members to perform. The result is that the majority of local people are aware of the initiatives and accept their basic philosophy and practice.

Another important factor is that most of these conservation efforts exist within traditional social systems where local customs carry great weight, respect for traditional authority runs deep, and social enforcement of community norms is common. In the Monks Community Forest, for example, the fact that Buddhist monks are leading the effort is a principal reason why the community has thrown its support behind the project, and one reason why local compliance is so high. Monks are greatly respected by the local populace, and acting against them is considered improper. As part of the forest protection process, the monks conduct tree ordination ceremonies

to bless the oldest trees, wrapping their trunks in saffron robes. For the local Buddhist population, cutting trees or killing wildlife within these ordained forests is seen as morally wrong. In close-knit rural communities, such social enforcement can be a powerful force.

**“Community conservation efforts don’t need to remain disconnected islands of local forest protection. Emerging efforts to create connectivity among the ever-growing number of community-protected forests have already demonstrated that large-scale coordination is possible.”**

The effectiveness of community-protected forests has also benefitted from the decision by many communities to not apply too rigid a standard of ‘protection’. Rather than set a standard of not allowing any community use of protected forests, the community-protected forests in these cases generally accommodate some level of local use. In Monks Community Forest, for example, community members are allowed to collect fallen timber as building materials, fish using traditional methods, and harvest mushrooms, wild ginger, and other non-timber forest products. This flexibility stands in sharp contrast to many state-managed protected areas where local use is forbidden. The looser standard applied in many ICCAs is another example of maintaining community members as allies rather than alienating them, in order to sustain their involvement in protection efforts.

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**Beyond the Local Level:** Community-protected forests can create wider impact through community-based networks that increase connectivity and state protected areas across the landscape.

Community conservation efforts don’t need to remain disconnected islands of local forest protection. Emerging efforts to create connectivity among the ever-growing number of community-protected forests and to link them to state-managed protected areas in order to create regional networks have already

demonstrated that large-scale coordination is possible. The work of Corporación Serraniagua in Colombia is indicative of what can be done. This consortium of community-based organizations promotes community nature reserves and links them together through coordinated land management plans to ensure habitat connectivity and to support sustainable livelihoods. The group's main work is to connect the conservation corridors of Colombia's Tatamá National Park and the Serraniagua de los Paraguas National Park through a series of 60 community-managed protected areas and seven state-managed nature reserves. In addition, Corporación Serraniagua also works with cacao, coffee, and sugar producer groups to harmonize the activities of these land users with the management plans of the community and state protected areas. The intent is to create a fully interlinked network of land users who can increase the functionality and

sustainability of the larger conservation corridor while supporting the rural economy. . The intent is to create a fully interlinked network of land users who can increase the functionality and sustainability of the larger conservation corridor while supporting the rural economy.

Creation of this kind of dynamic and interconnected network of community-protected forests and other land uses is one way to magnify the conservation value and climate mitigation potential of community-protected forests. As with community-based forest management in general, the strength of community-protected forests is their flexibility and grounding in local needs and experience. Connectivity across the landscape between local groups, however, is a way to add value and create synergies that are necessary for sustainability and resilience to emerge at a landscape scale.









# Findings



# A COMMUNITY ROUTE TO GLOBAL CLIMATE GOALS

The case studies presented in this book provide an overview of both the local development benefits and the global climate benefits of community-based forest management, and the link between the two. The case materials also reveal what motivates communities to organize and carry out local forest initiatives, and the enabling conditions—the resource rights, partnerships, and capacities—that help them to succeed. Below is a summary of the principal findings that emerge from the cases and the associated analysis.

## 1. The scope of potential climate benefits from community-based forest management is large

These global climate benefits are significant for several reasons.

**Significant in area.** Community-based forest initiatives work at a local level, village-by-village and forest-by-forest, but their aggregate impacts are much larger in scope. The territories and communal lands claimed by indigenous peoples and local communities encompass millions of hectares of forest. Community-based forest management already extends over larger areas than many realize, and could expand considerably under the right policy conditions and with adequate support. The case studies represented in this book alone cover well over one million hectares. Community-based forest management goes well beyond the local, and is unmistakably a global phenomenon with high potential for scaling up.

**Significant in the quality and sustainability of benefits.** Successful community-based forest management generates mitigation and adaptation benefits of high quality. In terms of mitigation, these efforts maintain and increase carbon storage in local forests by reducing forest pressures and deforestation rates, preserving intact canopy, regenerating cleared areas, and restoring degraded forests to a healthier state. What distinguishes this mitigation is its sustainability—reinforced by a

continuing stream of benefits and accumulated experience—amid the rural environment where conversion and extraction pressures on forests can be intense. But mitigation accounts for only half the climate benefits that community forest initiatives deliver. Adaptation benefits are also substantial. By increasing forest productivity and restoring ecosystem functions to degraded forests, community-based forest management generates new economic opportunities for climate-impacted communities and reduces their vulnerability to drought, flooding, storm surges, fire, and other natural disasters. At a systemic level, these initiatives increase the baseline resilience of the landscape, making communities and ecosystems more adaptive and climate-proof.

**“Community-based forest management already extends over larger areas than many realize, and could expand considerably under the right policy conditions and with adequate support.”**

**Significant for where the model can be applied.** Community-based forest management is marked by its wide applicability and ability to generate local benefits and regenerate vital forests in situations where other interventions fail. It can be effective in areas where the pressure to overuse or clear local forests is strong, as well as in areas of degraded forest. It can be applied in drylands and mangrove swamps, in hilly regions, and in forest patches adjacent to settlements—all areas where the success of top-down reforestation programs has been limited. In other words, it can protect forests that would not otherwise be protected, restore forests that would otherwise remain degraded, and



reduce the vulnerability of rural communities that would otherwise be ignored. It is a mechanism to generate climate benefits that would otherwise be missed.

**Significant for the change in local thinking it engenders.** Community-based interventions work because they change the way communities think about and act toward their local forest resources. This change in thinking is often a prerequisite for community willingness to adopt and even innovate new management practices, establish new sustainable forest enterprises, and persist through obstacles and over years. The evolution in local norms about what is appropriate forest use and what forest management should aspire to is a hidden but important climate benefit that community forest initiatives are uniquely able to deliver. Community education and outreach is key to producing this change in attitude, and it is no coincidence that successful community forest initiatives invariably have a strong education and communication component to stress the link between sustainable forest management and local well-being.

## **2. To generate global climate benefits, community-based forest initiatives must deliver on local development priorities**

Communities are motivated more by local benefits and empowerment than by global climate goals. Indeed, respecting these local motivations is the only route to achieving the significant climate benefits associated with community-based forest initiatives. A central lesson of the cases is that achieving climate benefits through community-based forestry is not simply a matter of quickly increasing the acreage under forest canopy. It is about creating a convincing rationale for sustainable forest use and protection that brings forests into the active lives of community members and provides economic, social, cultural, and spiritual benefits substantial enough to motivate the community over the long term. Community-based forest initiatives do this by supporting local livelihoods, generating new enterprise opportunities, and changing the economic circumstances of the community. They also do this by empowering the community, increasing its sense of self-determination, pride, and cohesion, and

revitalizing local culture. For indigenous and local communities, this empowerment aspect is often pre-eminent and determines whether the community will embrace the forest initiative.

## **3. Empowerment begins with secure land rights**

No single factor provides a greater motivating force for the adoption of community-based forest management, or plays a more central role in its success, than secure land rights. Many of the cases show the power of combining advocacy for indigenous lands rights with sustainable forest management practices to safeguard community forests. Such rights are needed to effectively control illegal forest incursions, fend off land grabs, and provide the stability and incentive to make long-term investments in the recovery and maintenance of forest health. Since the quest for land rights and the self-determination and control such rights allow is a defining issue for so many communities, providing such rights is a key strategy for enabling local forest action and capturing global climate benefits.

## **4. An integrated, landscape approach maximizes the potential of local forest management**

In the rural setting, forests, farm fields, pastures, protected areas, and villages all coexist in a mosaic of land uses. When these land uses clash, the forest often suffers, as when forests are cleared for agricultural expansion. Adopting an integrated approach to forest management, in which a variety of forest uses—including agriculture—may take place within the forest landscape, is the best way to maximize local forest benefits, including climate benefits. This is part of a larger ‘landscape approach’ to land use management, in which these land uses are planned together in ways meant to reinforce each other and manage inherent trade-offs, as part of a sustainable landscape. Many of the communities profiled here already practice this kind of integrated land use management, incorporating forest-friendly farm practices, buffers zones, and community protected areas into their management schemes. Emerging experience from Equator Prize-winning communities and others shows that adopting this kind of landscape approach can help tackle the governance challenges inherent in rural ecosystem management, with its many stakeholders and land uses. It can also help scale up local successes

beyond a single community by encouraging communities to coordinate their efforts within a given landscape to take advantage of synergies and to share resources and support.

## 5. Partnerships are vital

Partnerships—both at the community level and beyond—have been vital to the success of every Equator Prize-winning forest initiative. Partners provide an array of support services—from technical and business consultations to political organizing and outreach—that communities may not be able to provide for themselves. The absence of these management, business, and organizational capacities is one of the fundamental obstacles that local groups must overcome. Most community initiatives have many partners, including NGOs, universities and research institutes, private sector companies, and government agencies. These alliances give isolated communities access to specialized skills, appropriate technology, and networks for communication and learning. They are also essential to the development and growth of the forest enterprises at the heart of many forest initiatives, giving communities access to markets beyond their normal outlets, and sometimes developing into significant business relationships with private sector companies. The cases clearly indicate that such partnerships, if thoughtfully designed, do not diminish the autonomy of the community groups undertaking forest initiatives. Rather, they connect these groups to a wider audience and expand their reach beyond the community borders. Supporting and participating in dynamic partnerships with communities is one of the best ways that donors, governments, NGOs, and the private sector can encourage the expansion of local forest initiatives and the scaling up of the climate benefits they produce.

## SUPPORTING COMMUNITY-BASED FOREST MANAGEMENT IN THE NEW GLOBAL CLIMATE REGIME

Indigenous and local communities already play an important, although under-acknowledged, role in safeguarding the global climate through the sustainable management of communal forest land. But they can and should be supported to play a more prominent role in the new international climate framework. Community-based forest initiatives undertaken by indigenous peoples and local communities can make unique and timely contributions to the goals of reducing deforestation and restoring degraded forests, thereby reducing emissions. This lends credence to the international endorsement of Reducing Emissions from Deforestation and Forest Degradation (REDD+) and continuing efforts to create national programs to support community based forest initiatives. Indeed, with an enabling policy environment and the right partnerships, community efforts can be an important factor in achieving the international goal of keeping global warming to 2°C or below. Likewise, community-based forest management, because of its effectiveness in restoring degraded forests, could undoubtedly play a substantive role in achieving the forest restoration targets set by the Sustainable Development Goals and the New York Declaration on Forests. Just as importantly, community-based efforts have a record of sustainability that will be critical to maintaining progress on climate change and poverty reduction targets in decades to come.

But to play this role, forest communities require the recognition, land rights, resource access, and support to enable them to scale up their efforts and create landscape-level change. To facilitate this, the international community and national authorities should take action in three areas.

### 1. Acknowledge the contributions and potential of community-based forest initiatives

A first step in advancing the forest management efforts of indigenous peoples and local communities is for the international community to acknowledge their current achievements and potential contributions to mitigating forest emissions and adapting to climate change. The widespread implementation of REDD+ offers an opportunity to take advantage of community-level contributions. Another tangible expression of this would be to integrate community-based mitigation potential in the national climate commitments that countries make—their Intended Nationally Determined Contributions. In addition, it is essential for policymakers to acknowledge the importance of the ‘non-carbon’ benefits that community-based forest management yields. This involves embracing the fundamental logic of community approaches—that achieving climate benefits involves delivering community development benefits.

### 2. Recognize and respect indigenous territories and community land rights

There is a huge unmet demand for secure land rights in forest communities. According to a recent analysis, indigenous peoples and local communities lack legal title to nearly three-quarters of their traditional lands. This mismatch between the forest area that communities claim and the area to which they hold legal title represents one of the prime obstacles to scaling up the development and climate benefits of community-based forest management. Action to greatly expand the official recognition and legal protection of community land rights is thus a first order of business. Such tenure reform can capitalize on technological improvements in mapping, demarcation, and land titling that have recently decreased the costs of legally recognizing indigenous and community lands.

### 3. Provide an enabling environment of policies and support

Land and resource rights by themselves are not enough. They are only one element of an enabling environment that can facilitate local empowerment, develop local governance and business capacities, provide financial and capacity support where needed, and remove regulatory obstacles to the growth of small rural enterprises. Providing these enabling factors requires the participation of many actors. Governments can build local capacities through extension services, revise tax policies to encourage forest enterprises, encourage the formation of cooperatives and associations to link communities and foster knowledge transfer, and help with forest monitoring and enforcement. NGOs and other service providers can partner with communities to transfer technical skills, help draft forest management plans, conduct research, and help with education and advocacy. Private sector companies can contribute product development and market research, and donors can provide reliable and well-timed finance, strategic thinking, and communication of local achievements. While forest communities are resourceful and self-reliant, these contributions from outside actors can be vital in removing obstacles, and promoting up-scaling of local successes.

Addressing global climate change in a way that honors the sustainable development aspirations of indigenous and local communities is one of the defining rural development challenges of our day. Community-based forest management offers a route to meeting this challenge by generating a range of economic and social benefits for local people while reducing global carbon emissions, and helping rural communities to adapt to the unavoidable impacts of climate change. Embracing this approach requires listening to forest communities, empowering them with sufficient land rights and political access to carry out their management plans, and partnering with them to provide the technical, financial, and political support they need. The results can go well beyond climate mitigation and adaptation benefits and lead to more resilient rural landscapes where communities, local economies and natural systems can thrive.



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The Equator Initiative brings together the United Nations, governments, civil society, businesses and grassroots organizations to recognize and advance local sustainable development solutions for people, nature and resilient communities.



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